Advanced Data Analysis

Lecture Notes (Ch. 1–13)

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Chapter 1. Prediction Problems and Evaluation Design

Predictive Modeling, Lecture 1 (Is the score trustworthy? Splits, CV, leakage, OLS, logistic regression)

Roadmap of the predictive-modeling part

Lecture	Key question	Models and concepts
1	Can the score be trusted?	splits, CV, leakage, OLS, logistic regression
2	Many correlated variables?	Ridge, Lasso, Elastic Net
3	How to learn nonlinear relations?	decision trees, bagging, random forest
4	Strong models for tabular data?	GBM, XGBoost, LightGBM, CatBoost
5	When are neural nets useful?	MLP, minibatch, Adam, regularization, TabPFN/TabICL
6	How to combine many models?	AutoGluon, OOF, bagging, stacking

The guiding principle of the whole part: **models may change, but the prediction time, the split, the pipeline and the final-test rule must not.**

1 Prediction as a Rule for New Cases

1.1 The prediction problem

Definition: prediction rule

From observed data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$ with $x_i \in \mathbb{R}^p$ we build a rule

$$\hat{f} : \mathcal{X} \to \mathcal{Y}, \qquad \hat{y}_{\text{new}} = \hat{f}(x_{\text{new}}).$$

x_i is the feature vector already observed at the **prediction time**, y_i is the target observed later, n is the number of cases and p the number of features. The object of interest is the **prediction error on new cases**, not the fit to $\mathcal{D}$.

A predictive model does not try to explain every relation in the data; it is a rule that reduces future uncertainty *using only information available at prediction time.*

1.2 The prediction time decides which variables are usable

Every prediction problem must state *when* the prediction is made. If we predict next month's churn at the end of this month, nothing that happens after the end of the month may be a feature.

Role of a column	Question at prediction time	Churn example
Feature X_j	Is it already known at that moment?	recent usage, contract length, # complaints
Target Y	What do we predict?	churn next month
Identifier (ID)	Does it carry anything beyond row identity?	customer ID, account serial number
Post-event variable	Does it arise only after the outcome?	reason for cancellation, refund after cancellation

Post-event variables may raise the score dramatically, but they do not exist at the real prediction time and must not be used.

1.3 Regression and classification

Task	Target	Direct model output	Focus of evaluation
Regression	continuous Y	$\hat{y}(x)$, conditional mean	average error and large errors
Binary classification	$Y \in \{0, 1\}$	$\hat{p}(x) = P(Y = 1 \mid X = x)$	ranking, probability, threshold errors
Multiclass	$Y \in \{1, \ldots, C\}$	$(\hat{p}_1, \ldots, \hat{p}_C)$	per-class errors and probability quality

In classification the model first produces a **probability, not a label**; turning that probability into an action is a separate decision (§7).

1.4 Prediction versus causal inference

Prediction: $\hat{y} = \hat{f}(x)$	Causal inference: $\mathbb{E}\{Y(1) - Y(0)\}$
Who is likely to churn?	How much did a coupon reduce churn?
Small loss on new cases is the goal.	Identification assumptions and bias control are central.
Correlated (non-causal) variables can be useful features.	Confounders and post-treatment variables need care.

A good predictive coefficient alone never justifies a causal or policy claim.

2 Loss, Risk and Generalization

Motivation. A model is judged on data it has not seen. Risk is the expected loss on new data; training loss underestimates it, and the gap is what validation and test sets are designed to measure.

2.1 Loss translates decision costs into numbers

Definition: loss and risk

A loss $L(y, f(x)) \geq 0$ measures the cost of predicting $f(x)$ when the truth is y. The **risk** in the deployment environment is

$$R(f) = \mathbb{E}_{(X,Y)\sim P_{\text{deploy}}}[L\{Y, f(X)\}].$$

$R(f)$ cannot be observed directly. The **training risk** is $\widehat{R}_{\text{train}}(f) = \frac{1}{n}\sum_{i=1}^{n} L\{y_i, f(x_i)\}$, and the **generalization gap** is $\text{Gap}(f) = R(f) - \widehat{R}_{\text{train}}(f)$.

The loss should be fixed *before* the model: overstock and stock-out have different costs; missing a defaulter and rejecting a good customer have different costs; if bids are set from probabilities then probability quality matters more than labels. Accuracy is just one particular loss, not the default goal of every problem.

Theorem: what each loss targets

Let $f^\star = \arg\min_f R(f)$ over all measurable f. Pointwise in x:

1. squared loss $(y - f)^2$: $f^\star(x) = \mathbb{E}[Y \mid X = x]$ (conditional mean);

2. absolute loss $|y - f|$: $f^\star(x) = \text{median}(Y \mid X = x)$;
3. 0–1 loss for $Y \in \{0, 1\}$: $f^\star(x) = \mathbb{I}\{P(Y = 1 \mid x) \geq 1/2\}$ (Bayes classifier);
4. log loss $-[y \log q + (1 - y) \log(1 - q)]$ over $q \in (0, 1)$: $q^\star(x) = P(Y = 1 \mid X = x)$ (a *proper* scoring rule).

Proof [Supplement]. By the tower property $R(f) = \mathbb{E}_X[\mathbb{E}\{L(Y, f(X)) \mid X\}]$, so it suffices to minimize $g(c) = \mathbb{E}[L(Y, c) \mid X = x]$ in c for each x.

(1) $g(c) = \text{Var}(Y \mid x) + (\mathbb{E}[Y \mid x] - c)^2$.

(2) $g'(c) = 2P(Y \leq c \mid x) - 1$ (right derivative), which is < 0 for c below the conditional median and ≥ 0 from the median on, so g is minimized at the median.

(3) $g(1) = 1 - \pi(x)$, $g(0) = \pi(x)$ with $\pi(x) = P(Y = 1 \mid x)$; choose 1 iff $\pi \geq \frac{1}{2}$.

(4) $g(q) = -\pi \log q - (1 - \pi) \log(1 - q)$, $g'(q) = -\pi/q + (1 - \pi)/(1 - q) = 0 \iff q = \pi$, and $g'' > 0$. ■

2.2 Training score versus deployment risk

Learning algorithms fit by minimizing $\widehat{R}_{\text{train}}$, and complex models can drive it near zero. What we want is $R(f)$, the average loss on unseen cases, which the training loss cannot reveal. Hence an **independent validation procedure** is needed. **Generalization** is the property that a rule learned on training data keeps working on new cases.

Typical causes of an unstable generalization gap: searching over too many candidate models and hyperparameters; random splits that ignore time or group structure; preprocessing and variable selection done on the whole data; distribution shift between training and deployment.

Theorem: bias–variance decomposition

Let $Y = f(x) + \varepsilon$ with $\mathbb{E}\varepsilon = 0$, $\text{Var}\varepsilon = \sigma^2$, and let $\hat{f}$ be fitted on a random training set independent of the new (x, Y). Then

$$\mathbb{E}[(Y - \hat{f}(x))^2] = \underbrace{\left(f(x) - \mathbb{E}\hat{f}(x)\right)^2}_{\text{Bias}^2} + \underbrace{\text{Var}\hat{f}(x)}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{irreducible}}.$$

Proof [Supplement]. Write $Y - \hat{f} = \varepsilon + (f - \mathbb{E}\hat{f}) + (\mathbb{E}\hat{f} - \hat{f})$. The three terms are pairwise uncorrelated: ε is independent of $\hat{f}$ and has mean zero; $f - \mathbb{E}\hat{f}$ is a constant and $\mathbb{E}(\mathbb{E}\hat{f} - \hat{f}) = 0$. Squaring and taking expectations gives the result. ■

Model complexity moves the two curves differently: training error decreases monotonically, test error is U-shaped. **Underfitting** misses the signal (high bias), the right complexity balances signal and noise, **overfitting** memorizes accidental wiggles (high variance).

3 Splitting the Data

3.1 Three roles

Part	Used for	Never used for
Training part	fitting model parameters	—
Internal validation (holdout or CV)	choosing candidates and tuning	—
Test set	reporting final generalization once	choosing models

The test set is not material for the next choice; it is the *conclusion* of this experiment.

3.2 The split must preserve the data-generating structure

Data structure	Recommended split	Danger of a random split
independent rows, class imbalance	stratified split	minority-class proportion may differ by chance
clustered (customers, hospitals, households)	group split	the same group appears in training and test
time order	forward (time-ordered) split	future information leaks into training
duplicates / near-duplicates	deduplicate, then split	nearly identical rows on both sides

If we train on 2024 and predict 2025, the order past → future must be kept. Several transactions of one customer must be split at the customer level. Unless the *unit of a new case* is defined, the correct validation split cannot be defined either.

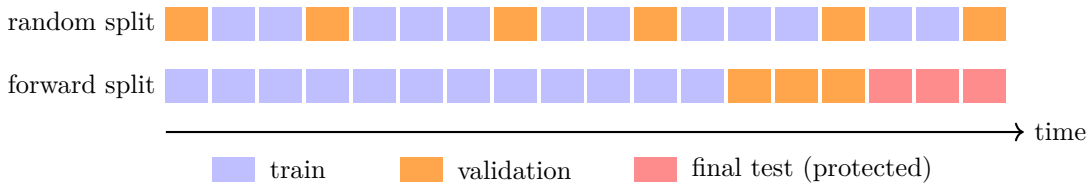


Figure 1: With time-ordered data a random split lets the model train on the future of its validation cases; a forward split trains on the past, validates on the next period and keeps the latest period untouched as the final test. Group splits follow the same idea with customers or hospitals in place of time blocks.

3.3 Reproducibility

Fix: split rule and random seed; feature list and target definition; handling of missing values, categories, scaling; code computing metrics and thresholds.

Record: extraction time, data version, number of rows; class ratios and number of groups in each set; candidate models and hyperparameter grids; fold-by-fold variability, not only the CV mean.

Reproducibility means not only getting the same numbers again but being able to *trace why numbers changed*.

4 Information Leakage

Definition: leakage

Leakage is any channel through which information unavailable at prediction time, or information from the validation/test part, enters the fitting procedure.

Type	Example	Why it is a problem
Target leakage	cancellation reason or default-handling result used as a feature	does not exist at prediction time
Preprocessing leakage	mean imputation / standardization on the full data	peeks at the validation/test distribution
Temporal leakage	future averages, after-the-fact aggregates	cannot reproduce a past decision
Duplicate / group leakage	same customer, document or product on both sides	the "new" case was effectively already seen

When performance is suspiciously high, suspect leakage before praising the algorithm.

Example: why preprocessing must be learned inside each fold

Standardizing with the full-data mean $\bar{x}$ and standard deviation s uses validation rows to build the training features. With p large and screening ("keep the 20 features most correlated with y") done on the full data, a model trained on pure noise can show a high CV score, because the screening step has already looked at the validation targets. The fix is to put every data-dependent step (imputation, scaling, screening, encoding) inside the pipeline that is re-fitted on the training part of each fold.

5 K-fold Cross-Validation

Definition: K-fold CV

Split the development data into folds $V_1, \ldots, V_K$. For each k fit $\hat{f}^{(-k)}$ on the data without V_k and compute

$$\widehat{R}_k(f) = \frac{1}{|V_k|} \sum_{i \in V_k} L\{y_i, \hat{f}^{(-k)}(x_i)\}, \qquad \widehat{R}_{\mathrm{CV}}(f) = \frac{1}{K} \sum_{k=1}^{K} \widehat{R}_k(f).$$

Each observation is used once for validation and $K-1$ times for training. Comparison of candidates is more stable than with a single validation set.

Remark: fold variability and its standard error

Report $\widehat{R}_{\mathrm{CV}}$ with $\mathrm{SE} = \mathrm{sd}(\widehat{R}_1, \ldots, \widehat{R}_K)/\sqrt{K}$. The folds share training data, so this SE is only a rough guide (it usually underestimates the true uncertainty), but it is what the 1-SE rule of Ch. 3 uses.

Algorithm: selection with CV, reporting with the test set

1. Split off the test set first and lock it away.
2. On the remaining development data, define the pipeline (preprocessing + candidate model).
3. Choose candidates and hyperparameters using CV scores and fold variability.
4. Refit the chosen pipeline on the whole development data.
5. Report final performance, error patterns and operating cost on the test set.

6 Metrics are the Language of Decisions

A good metric moves in the same direction as operational loss, is used consistently across candidates, does not hide imbalance or threshold effects, and can be explained to stakeholders. One model needs several metrics: for regression, average error, large errors, errors by segment; for classification, ranking, probability quality, errors at a threshold; and the final choice weighs metric, stability, computing time and explainability. Ask first *which errors are actually more expensive.*

6.1 Regression metrics

Metric	Definition	Interpretation
MAE	$\frac{1}{n}\sum_i \lvert y_i - \hat{y}_i\rvert$	all errors linearly; linked to median prediction
MSE	$\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2$	squared penalty on large errors; linked to mean prediction
RMSE	$\sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}$	same unit as y; emphasizes large errors
R^2	$1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$	improvement in squared error over the mean baseline

Theorem: $\mathrm{MAE} \le \mathrm{RMSE} \le \sqrt{n}\,\mathrm{MAE}$

For residuals $e_i = y_i - \hat{y}_i$, $\frac{1}{n}\sum \lvert e_i\rvert \le \left(\frac{1}{n}\sum e_i^2\right)^{1/2}$, with equality iff all $\lvert e_i\rvert$ are equal.

Proof [Supplement]. Cauchy–Schwarz: $\left(\sum \lvert e_i\rvert \cdot 1\right)^2 \le n\sum e_i^2$. The upper bound follows from $\sum e_i^2 \le (\sum \lvert e_i\rvert)^2$. ■

The gap RMSE − MAE grows with the dispersion of $\lvert e_i\rvert$: a smaller RMSE means that large errors in particular were reduced. Prefer RMSE when rare large errors are especially costly or when models minimizing squared loss are compared; prefer MAE when the influence of outlying targets should be limited or the typical error should be communicated. If the two metrics rank two models differently, the models reduced errors of different *size and distribution.*

7 Classification: Probabilities, Thresholds, Metrics

7.1 Probability first, action second

A classifier first outputs $\hat{p}(x) = P(Y = 1 \mid X = x)$, an estimate of a conditional probability. The same probability can lead to different actions under different business rules (churn probability $\hat{p} = 0.18$: early contact if coupons are cheap; only the top 5% if counselling staff is limited; a lower

threshold if losing the customer is very costly). One can also check whether the group predicted at 0.18 actually churns at about 18% (calibration).

7.2 Threshold and confusion matrix

$$\hat{y}_t(x) = \mathbb{I}\{\hat{p}(x) \geq t\},$$

	actual +	actual −
predicted +	TP	FP
predicted −	FN	TN

$$\text{precision} = \frac{TP}{TP+FP}, \qquad \text{recall} = \frac{TP}{TP+FN}, \qquad \text{accuracy} = \frac{TP+TN}{n}.$$

Lowering t produces more positive predictions: recall usually rises but false positives rise too.

Example: accuracy under imbalance

Of 1,000 customers only 50 churn. Predicting "no churn" for everyone gives accuracy = 95%, yet finds no churner (recall 0). With imbalance, look at precision (quality of the selected group), recall (share of positives found) and the costs together.

7.3 Three questions about one probability

Ranking	Reliability of the probability	Action
Whom to look at first? ROC-AUC, PR-AUC	Does 0.30 mean about 30%? Brier score, log loss, calibration plot	Where to place the threshold? precision/recall at t, cost

A good ranking does not imply well-calibrated probabilities, nor fitness for the current operating rule.

Theorem: ROC-AUC is a ranking probability

For a score s, $\text{AUC} = P\big(s(X^+) > s(X^-)\big) + \frac{1}{2}P\big(s(X^+) = s(X^-)\big)$, where X^+, X^- are independent draws from the positive and negative classes. Empirically AUC equals the proportion of (positive, negative) pairs ranked correctly (Mann–Whitney statistic).

Proof [Supplement]. The ROC curve plots $\text{TPR}(t) = P(s^+ > t)$ against $\text{FPR}(t) = P(s^- > t)$. Parametrize by t and integrate: $\int \text{TPR}\, d\text{FPR} = \int P(s^+ > t)\, dF^-(t) = P(s^+ > s^-)$ for continuous scores (ties contribute half). ■

PR-AUC is preferred when positives are rare, since it focuses on precision and recall. Brier score $\frac{1}{n}\sum(y_i - \hat{p}_i)^2$ is the mean squared probability error; log loss heavily penalizes confident wrong probabilities. AUC is invariant to any monotone transformation of the scores, so it says nothing about calibration.

7.4 Choosing the threshold from costs

Theorem: cost-optimal threshold

Let a false negative cost C_{FN} and a false positive cost C_{FP} (correct decisions cost 0). If $\hat{p}(x)$

is calibrated, acting positive minimizes expected cost iff

$$\hat{p}(x)\, C_{FN} > \{1 - \hat{p}(x)\}\, C_{FP} \iff \hat{p}(x) > t^\star = \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

Proof [Supplement]. Acting positive costs C_{FP} with probability $1-\hat{p}$; acting negative costs C_{FN} with probability $\hat{p}$. Compare the two expected costs and rearrange. ■

As C_{FN} grows the threshold for acting positive *decreases*; $t^\star = 1/2$ only when the costs are equal. The prediction of a probability and the action rule should be thought of separately.

8 The Linear Baseline: OLS

8.1 Least squares

$$\hat{y}_i = \beta_0 + x_i^\top \beta, \qquad \hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - \beta_0 - x_i^\top \beta)^2.$$

With the design matrix X including the intercept column,

Derivation: normal equations

$Q(\beta) = \|y - X\beta\|^2 = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta$, so $\nabla Q = -2X^\top y + 2X^\top X\beta = 0$ gives $X^\top X\hat{\beta} = X^\top y$. If $X^\top X$ is invertible, $\hat{\beta} = (X^\top X)^{-1} X^\top y$; the Hessian $2X^\top X \succeq 0$ shows it is a global minimum. Geometrically $\hat{y} = X\hat{\beta} = Hy$, $H = X(X^\top X)^{-1}X^\top$, is the orthogonal projection of y onto the column space, and the residual is orthogonal to every column: $X^\top(y - \hat{y}) = 0$.

If $X^\top X$ is unstable (nearly singular) the coefficient estimates are unstable too; regularization (Ch. 3) mitigates this.

8.2 Predictive versus causal reading of coefficients

Predictively, β_j is the change of the linear prediction when X_j increases by one unit with other features held fixed. Under multicollinearity the sign and size of coefficients can wobble, and coefficients of variables in different units are not comparable. A causal reading additionally requires control of confounders and a correct causal structure, exclusion of post-treatment variables, and checks for selection bias and measurement error.

8.3 Residual diagnostics point to the next model

Observed phenomenon	Meaning	Next candidates
curved residual pattern	missed nonlinearity / interaction	transformations, splines, trees (Ch. 4)
variance changing with $\hat{y}$	heteroscedasticity, scale issue	transformations, robust loss, segment-wise evaluation
a few huge residuals	outliers, label errors	Huber loss, data checks (Ch. 2)
unstable coefficients with correlated features	multicollinearity, high dimension	Ridge, Lasso, Elastic Net (Ch. 3)

Seeing where the linear baseline fails shows *why* the models of the following lectures are needed.

9 Logistic Regression

9.1 From linear score to probability

$$\eta(x) = \beta_0 + x^\top \beta, \qquad \hat{p}(x) = \sigma\{\eta(x)\} = \frac{1}{1+\exp\{-\eta(x)\}}, \qquad \log\frac{p(x)}{1-p(x)} = \beta_0 + x^\top\beta.$$

The linear score lives on the whole real line; the sigmoid squashes it into $(0,1)$. β_j is the change of the **log-odds** when X_j increases by one unit with the others fixed, $\exp(\beta_j)$ is the **odds ratio**, and the change in the probability itself depends on the current x: $\partial p/\partial x_j = \beta_j p(1-p)$.

9.2 Log loss

$$\ell(y,p) = -\big[y\log p + (1-y)\log(1-p)\big], \qquad \hat{\beta} = \arg\min_\beta \sum_{i=1}^n \ell\{y_i, \sigma(\beta_0 + x_i^\top\beta)\}.$$

A positive case given $p \approx 0$, or a negative case given $p \approx 1$, receives a huge penalty: log loss evaluates the *quality of probabilities*, not just right/wrong labels.

Theorem: gradient and convexity of the logistic log loss

With $p_i = \sigma(x_i^\top\beta)$ (intercept absorbed), $J(\beta) = \sum_i \ell(y_i, p_i)$ satisfies

$$\nabla J(\beta) = X^\top(p-y), \qquad \nabla^2 J(\beta) = X^\top W X, \quad W = \mathrm{diag}\{p_i(1-p_i)\} \succeq 0,$$

so J is convex and minimizing log loss equals maximum likelihood for the Bernoulli model. Newton's method is iteratively reweighted least squares.

Proof [Supplement]. $\sigma'(\eta) = \sigma(\eta)\{1-\sigma(\eta)\}$, so $\partial\ell/\partial\eta = -y(1-p) + (1-y)p = p - y$. The chain rule gives $\nabla J = \sum_i (p_i - y_i)x_i$, and differentiating once more gives $\sum_i p_i(1-p_i)x_i x_i^\top$. The Bernoulli log-likelihood is $\sum_i\{y_i\log p_i + (1-y_i)\log(1-p_i)\} = -J$. ■

Remark: separation

If a hyperplane separates the two classes perfectly, J keeps decreasing as $\|\beta\| \to \infty$ and the MLE does not exist; penalization (Ch. 3) restores a finite solution.

10 Comparing Candidates Fairly

Criterion	Conditions to control
same split	all candidates use the same training/CV/test structure
same feature definition	same prediction time, missing-value and category handling
same metric	primary and secondary metrics for selection fixed in advance
same compute budget	record search range, time and number of iterations
same reporting	show CV mean, variability, test score, complexity and operating constraints together

Model comparison is not a popularity contest between algorithms but a **controlled experiment**.

Summary

- Define the target and the prediction time first.
- The goal is low generalization risk in deployment, not low training error.
- The split must preserve the data-generating structure: time, groups, duplicates.
- CV is for selection, the test set for the final report; preprocessing is learned inside each training fold.
- Metrics, thresholds and baselines are chosen from decision costs before algorithms.

Chapter 2. Outliers and Robust Prediction

Predictive Modeling, Section 2 (MCD, LOF, Isolation Forest, leverage, Cook's distance, Huber, RANSAC, Theil–Sen)

1 An Outlier Is Not the Same as a Data Error

What the observation is	Example	First response
input / measurement error	wrong unit, mis-recorded target	correct from the raw source; remove only if uncorrectable
a genuine rare case	a very large transaction, a rare high-risk patient	keep it, and check that group's prediction error separately
a case from a new environment	a policy or market condition absent in training	check distribution shift and the need to retrain

An outlier is a *statement relative to a reference*, and the right action depends on why it is unusual.

1.1 Three kinds of outliers

Kind	Meaning	Simple diagnostic
univariate	one variable is far from most observations	histogram, boxplot, IQR rule
multivariate	each variable is ordinary but the *combination* is rare	scatterplot, multivariate distance, density
conditional	given x_i, the target y_i differs greatly from what is expected	residual $e_i = y_i - \hat{y}_i$

The IQR rule flags $x < Q_1 - 1.5\,\mathrm{IQR}$ or $x > Q_3 + 1.5\,\mathrm{IQR}$, $\mathrm{IQR} = Q_3 - Q_1$. Univariate rules alone miss outliers that appear only in combinations of variables.

Method	What it checks	Observations it looks for
MCD	global distance in X	multivariate outliers far from the central distribution
LOF	local density in X	points less dense than their neighbors
Isolation Forest	isolation path in X	points isolated by few random splits
studentised residual	y given X	cases far from the regression prediction
leverage, Cook's distance	influence on the regression fit	cases that change coefficients and predictions a lot

"X is unusual" and "y is unusual given X" are different problems.

2 MCD: Robust Mahalanobis Distance

2.1 Classical Mahalanobis distance and masking

With the sample mean $\bar{x}$ and sample covariance S,

$$D_i^2 = (x_i - \bar{x})^\top S^{-1} (x_i - \bar{x}).$$

Theorem: null distribution of the squared distance

If $X \sim N_p(\mu, \Sigma)$ then $(X-\mu)^\top \Sigma^{-1}(X-\mu) \sim \chi^2_p$. Hence, approximately, a point with $D_i^2 > \chi^2_{p,1-\alpha}$ is an outlier candidate.

Proof [Supplement]. $Z = \Sigma^{-1/2}(X-\mu) \sim N_p(0, I_p)$, and $(X-\mu)^\top \Sigma^{-1}(X-\mu) = Z^\top Z = \sum_{j=1}^p Z_j^2$, a sum of p independent squared standard normals. ■

Masking. Outliers pull $\bar{x}$ toward themselves and inflate S, so their own distances shrink and they may no longer be flagged. The remedy is to estimate location and scatter from the bulk of the data only.

2.2 Minimum Covariance Determinant

Definition: MCD

For a fixed h with $n/2 \le h < n$ (typically $h \approx 0.75n$), find the subset H of h observations whose sample covariance has the smallest determinant:

$$H^\star = \arg\min_{|H|=h} \det S_H, \qquad \hat{\mu}_{\text{MCD}} = \bar{x}_{H^\star}, \quad \widehat{\Sigma}_{\text{MCD}} = c_{h,n}\, S_{H^\star},$$

($c_{h,n}$ is a consistency factor), and use the robust distance

$$D_i^2 = (x_i - \hat{\mu}_{\text{MCD}})^\top \widehat{\Sigma}_{\text{MCD}}^{-1} (x_i - \hat{\mu}_{\text{MCD}}), \qquad \text{flag if } D_i^2 > \chi^2_{p,1-\alpha}.$$

$\det S_H$ is proportional to the squared volume of the data ellipsoid of H, so MCD looks for the *tightest* h points; outliers enlarge the volume and are excluded. Up to $n-h$ outliers cannot drive the estimate arbitrarily far (breakdown point $\approx (n-h)/n$). In practice the FAST-MCD algorithm iterates "C-steps": given H, compute distances and replace H by the h points with smallest distances; each step does not increase $\det S_H$.

Remark: when MCD fits

MCD suits continuous data with $p < n$ and a single elliptical bulk. With multimodal or nonlinear clouds, local methods (LOF) or Isolation Forest are more appropriate.

3 LOF: Lower Density Than the Neighbors

Definition: local outlier factor

Let $N_k(i)$ be the k nearest neighbors of i, $d_k(j)$ the distance from j to its kth neighbor, and the reachability distance $\text{rd}_k(i,j) = \max\{d_k(j), d(i,j)\}$. The local reachability density and LOF are

$$\text{lrd}_k(i) = \Big(\frac{1}{|N_k(i)|} \sum_{j \in N_k(i)} \text{rd}_k(i,j)\Big)^{-1}, \qquad \text{LOF}_k(i) = \frac{1}{|N_k(i)|} \sum_{j \in N_k(i)} \frac{\text{lrd}_k(j)}{\text{lrd}_k(i)}.$$

$\text{lrd}_k(i)$ grows as the distances to the neighbors shrink. LOF ≈ 1: density similar to the neighbors; LOF > 1: lower density than the neighbors; very large LOF: a local outlier separated from its surrounding group. Because the comparison is *relative to the neighborhood*, a point in a sparse but genuine cluster is not penalized, unlike a global distance rule. The reachability distance smooths $d(i,j)$ for very close pairs, which stabilizes the ratio.

4 Isolation Forest: Quickly Isolated Points

Algorithm: Isolation Forest

1. On a subsample, choose a feature uniformly at random.
2. Choose a split point uniformly between the feature's minimum and maximum.
3. Split recursively until each observation is alone (or a depth limit).
4. Grow many such trees and average the path lengths.

Anomaly score:

$$s(x) = 2^{-\mathbb{E}[h(x)]/c(n)}, \qquad c(n) = 2H(n-1) - \frac{2(n-1)}{n}, \quad H(m) \approx \ln m + 0.5772,$$

where $h(x)$ is the path length until x is isolated and $c(n)$ is the average path length of an unsuccessful search in a binary search tree of n points (normalization by sample size).

Short paths give large scores: $s \to 1$ means anomalous, $s \approx 0.5$ for all points means no clear anomaly, $s \ll 0.5$ means normal. No distributional shape is assumed, and it scales to large data. The parameter `contamination` only sets the share of points to be labeled anomalous. Isolation Forest gives a score, but it does not tell a data error from a valid rare case.

5 Regression: Residual, Leverage and Influence

Motivation. A single unusual point can move a regression fit a long way. Residuals detect points the model fits badly, leverage detects unusual inputs, and influence measures combine both to show which points actually change the answer.

5.1 Three different notions

Suppose most data follow $y \approx 2x$.

Observation	Feature x	Residual	Interpretation
$(3, 20)$	ordinary range	large	outlier in the target
$(10, 20)$	far from the rest	small	high-leverage point that follows the trend
$(10, 0)$	far from the rest	large	influential point likely to change the line

The residual is the gap between truth and prediction; leverage measures how unusual x_i is in feature space; an influential point is one whose removal changes the fit a lot.

5.2 Hat matrix and leverage

With $X \in \mathbb{R}^{n\times(p+1)}$ (intercept included) and $H = X(X^\top X)^{-1}X^\top$, $\hat{y} = Hy$ and $e = (I - H)y$. The leverage is $h_{ii} = x_i^\top (X^\top X)^{-1} x_i$.

Theorem: properties of leverage and residuals

1. $0 \le h_{ii} \le 1$ and $\sum_i h_{ii} = p + 1$, so the average leverage is $(p + 1)/n$.

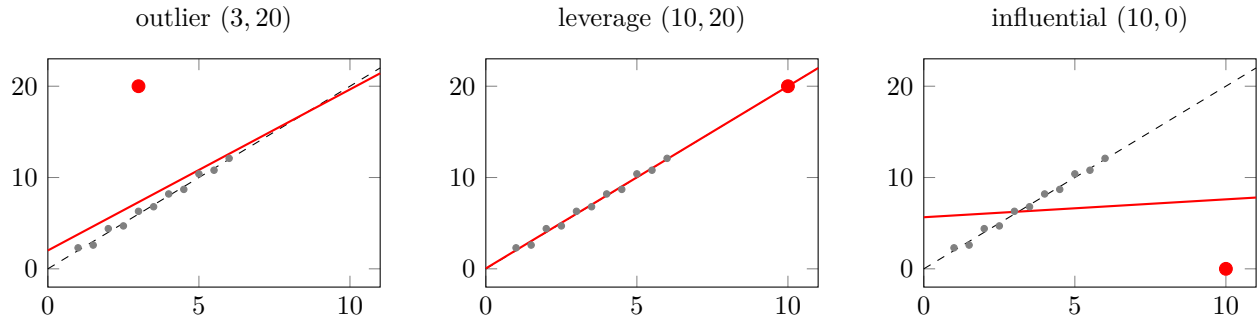


Figure 2: Dashed: OLS line on the gray points ($y \approx 2x$); red: OLS line after adding the red point (exact refits). A target outlier in the middle of the x-range mostly shifts the line upward; a high-leverage point on the trend changes nothing; a high-leverage point off the trend drags the slope — large residual *and* large leverage, hence large Cook's distance.

2. $\hat{y}_i = \sum_j h_{ij} y_j$, so $h_{ii} = \partial \hat{y}_i / \partial y_i$.
3. Under $\mathrm{Var}(\varepsilon) = \sigma^2 I$, $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$.

Proof [Supplement]. H is symmetric idempotent, so $h_{ii} = (H^2)_{ii} = \sum_j h_{ij}^2 \geq h_{ii}^2$, giving $0 \leq h_{ii} \leq 1$; $\sum_i h_{ii} = \mathrm{tr} H = \mathrm{tr}\{(X^\top X)^{-1} X^\top X\} = p + 1$. Item 2 is $\hat{y} = Hy$. For 3, $e = (I - H)\varepsilon$ because $(I - H)X = 0$, so $\mathrm{Var}(e) = \sigma^2 (I - H)(I - H)^\top = \sigma^2 (I - H)$. ■

A high-leverage point pulls the fitted line toward itself, so its *raw* residual tends to be small. Item 3 is why residuals must be standardized.

5.3 Studentised residual and Cook's distance

Diagnostic	Computation	Cases it reveals
studentised residual	$r_i = \dfrac{e_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}$	y_i far from what x_i predicts
leverage	h_{ii}, diagonal of H	far from the other observations in feature space
Cook's distance	$D_i = \dfrac{r_i^2}{p+1} \dfrac{h_{ii}}{1 - h_{ii}}$	removing it changes coefficients and predictions much

Common screening rules are $|r_i| > 2$, $h_{ii} > 2(p+1)/n$, $D_i > 4/n$. Exceeding a rule makes a case *an object of inspection*, not an automatic deletion.

Theorem: leave-one-out update and Cook's formula

Let $\hat{\beta}_{(i)}$ be the OLS estimate without case i. Then

$$\hat{\beta} - \hat{\beta}_{(i)} = \frac{(X^\top X)^{-1} x_i e_i}{1 - h_{ii}},$$

and Cook's distance, defined as $D_i = \dfrac{\|X\hat{\beta} - X\hat{\beta}_{(i)}\|^2}{(p+1)\hat{\sigma}^2}$, equals $\dfrac{r_i^2}{p+1} \dfrac{h_{ii}}{1 - h_{ii}}$.

Proof [Supplement]. Let $A = X^\top X$. Removing row i gives $A_{(i)} = A - x_i x_i^\top$ and $X_{(i)}^\top y_{(i)} =$

$X^\top y - x_i y_i$. Sherman–Morrison: $A_{(i)}^{-1} = A^{-1} + \frac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}$. Then

$$\hat{\beta}_{(i)} = \left(A^{-1} + \frac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}\right)(X^\top y - x_i y_i) = \hat{\beta} - A^{-1}x_i y_i + \frac{A^{-1}x_i(\hat{y}_i - h_{ii}y_i)}{1-h_{ii}} = \hat{\beta} - \frac{A^{-1}x_i(y_i - \hat{y}_i)}{1-h_{ii}}.$$

Hence $\|X(\hat{\beta} - \hat{\beta}_{(i)})\|^2 = \frac{e_i^2}{(1-h_{ii})^2} x_i^\top A^{-1} X^\top X A^{-1} x_i = \frac{e_i^2 h_{ii}}{(1-h_{ii})^2}$. Divide by $(p+1)\hat{\sigma}^2$ and substitute $e_i^2 = r_i^2 \hat{\sigma}^2 (1-h_{ii})$. ■

So Cook's distance is large only when *both* the standardized residual and the leverage factor $h_{ii}/(1-h_{ii})$ are sizeable, which is exactly the case $(10, 0)$ above.

Remark: leave-one-out residuals for free

The same identity gives $y_i - x_i^\top \hat{\beta}_{(i)} = e_i/(1-h_{ii})$, so the leave-one-out CV error of OLS is $\frac{1}{n}\sum_i \{e_i/(1-h_{ii})\}^2$ without refitting.

6 What to Do With Detected Cases

Finding	Action	Caveat
input / measurement error	correct or remove in raw data	remove only if it cannot be corrected and cannot be used
heavily skewed variable	log or Box–Cox transform	keeps the order, compresses gaps among large values
extremes must be capped	winsorisation	cap both tails at quantiles *estimated on training data*
valid rare case	keep	use robust learning and segment-wise evaluation

Never delete or alter observations on the basis of a detection score alone.

7 Robust Learning

Method	Core principle	Suitable when
Huber regression	caps the growth rate of loss for large residuals	a few large target outliers
RANSAC	fits many small subsamples; keeps the model supported by most points	clear inlier vs. contaminated groups
Theil–Sen	median of pairwise slopes	linear regression with few features

7.1 Huber loss

With residual $r = y - \hat{y}$:

Loss	Definition	Large residual	Target of prediction
squared	r^2	grows quadratically	conditional mean
absolute	$\lvert r\rvert$	grows linearly	conditional median
Huber	quadratic for small, linear for large $\lvert r\rvert$	linear beyond δ	compromise between center fit and tail robustness

$$\rho_\delta(r) = \begin{cases} \tfrac{1}{2}r^2, & |r| \le \delta, \\ \delta(|r| - \tfrac{1}{2}\delta), & |r| > \delta, \end{cases} \qquad \psi_\delta(r) = \rho'_\delta(r) = \begin{cases} r, & |r| \le \delta, \\ \delta\,\mathrm{sign}(r), & |r| > \delta. \end{cases}$$

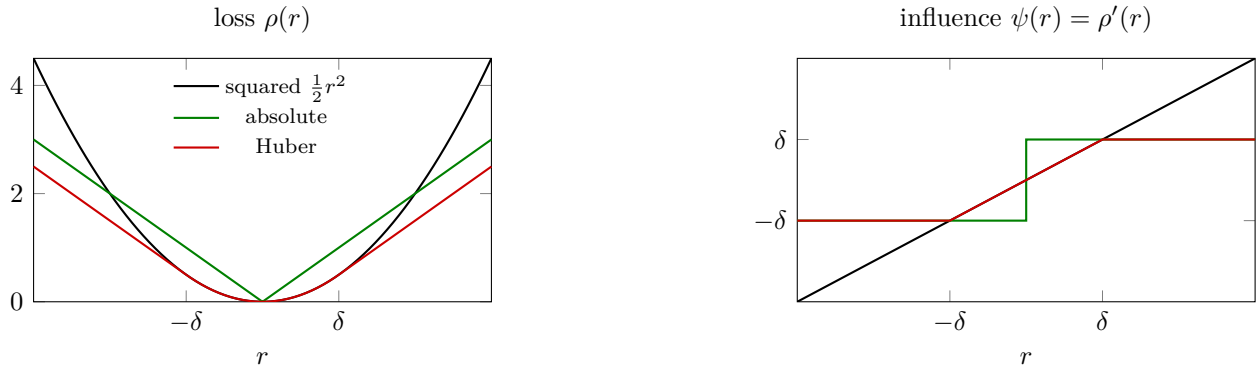


Figure 3: Squared, absolute and Huber losses (left) and their derivatives (right). The influence of a residual on the estimating equation grows without bound under squared loss but is capped at $\pm\delta$ under Huber loss.

Derivation: why Huber limits outliers — IRLS view

The estimating equation is $\sum_i \psi_\delta(y_i - x_i^\top\beta)x_i = 0$. Writing $\psi_\delta(r) = w(r)\,r$ with $w(r) = \min\{1, \delta/|r|\}$ turns it into weighted least squares $\sum_i w_i(y_i - x_i^\top\beta)x_i = 0$: iterate $\beta \leftarrow (X^\top W X)^{-1}X^\top W y$ with $W = \mathrm{diag}\{w(r_i)\}$. A case with residual 10δ gets weight 0.1; its pull on the fit (the influence ψ) is bounded by δ, whereas under squared loss it grows without bound. The function ρ_δ is convex and continuously differentiable, so the minimizer is well defined. A smaller δ limits large residuals more strongly; as $\delta \to \infty$ Huber becomes OLS, as $\delta \to 0$ it behaves like LAD (rescaled).

Since δ is in units of y, it is usually set as $\delta = 1.345\hat{s}$ with a robust scale $\hat{s}$ (e.g. MAD/0.6745); 1.345 gives 95% efficiency under Gaussian errors.

7.2 RANSAC

Algorithm: RANSAC

1. Fit the regression on a small random subsample (minimal size s).
2. Classify observations with residual below a threshold as inliers.
3. Repeat many times.
4. Refit the model on the largest inlier set.

If a fraction w of points are inliers, a subsample of size s is all-inlier with probability w^s, so to succeed at least once with probability q one needs $N \ge \log(1-q)/\log(1-w^s)$ trials. The residual threshold and the number of trials affect the result.

7.3 Theil–Sen

In simple regression compute all pairwise slopes:

$$\hat{\beta}_1 = \underset{i<j}{\mathrm{median}}\,\frac{y_j - y_i}{x_j - x_i}, \qquad \hat{\beta}_0 = \underset{i}{\mathrm{median}}(y_i - \hat{\beta}_1 x_i).$$

A few extreme slopes cannot dominate the estimate; the breakdown point is about $1 - 1/\sqrt{2} \approx 29\%$. RANSAC separates inliers and outliers sharply; Theil–Sen robustifies continuously through the median of slopes.

8 Robust Prediction Is Also Chosen by CV

Algorithm: protocol

1. Split off the final test set first.
2. Inside each CV training fold, fit scaling and outlier-detection steps.
3. Set the detection threshold, LOF's k, Huber's δ, RANSAC's residual threshold from training folds only.
4. Compare OLS and robust regressions on the same validation folds.
5. Check errors of outlier candidates and of rare groups along with the overall error.

Result to check	Purpose
MAE and RMSE	evaluate typical errors and rare large errors together
sensitivity to the detection rule	does model selection change with the threshold?
error of rare groups	did robustification ignore important cases?

Summary

- Outliers are relative to a reference; errors, valid rare cases and new environments call for different actions.
- In X: MCD (global, elliptical), LOF (local density), Isolation Forest (isolation depth).
- In regression: residual (y given x), leverage (x unusual), Cook's distance (both together).
- Robust losses (Huber, LAD) and robust fitting (RANSAC, Theil–Sen) limit the dominance of a few points; their tuning constants are chosen inside CV.

9 Selected exercises

Motivation. Three facts used above without proof: Mahalanobis-type distances do not depend on the units of the features, a C-step of FAST-MCD can only shrink the determinant, and Huber regression is least squares with case-wise outlier shifts.

Exercise 2.1 (Affine invariance of Mahalanobis distances)

Let $x_1, \ldots, x_n \in \mathbb{R}^p$ have sample covariance $S \succ 0$, and let $y_i = Ax_i + b$ with A an invertible $p \times p$ matrix and $b \in \mathbb{R}^p$.
(a) Show that the squared Mahalanobis distances D_i^2 computed from the y_i equal those computed from the x_i.
(b) Show that the MCD subset $H^\star$ is the same for both data sets, and so are the robust distances.

Solution [Supplement].

(a)

Step 1. $\bar{y} = A\bar{x} + b$ and $y_i - \bar{y} = A(x_i - \bar{x})$, so $S_y = AS_xA^\top$, which is invertible with $S_y^{-1} = A^{-\top}S_x^{-1}A^{-1}$.

Step 2. Hence

$$(y_i - \bar{y})^\top S_y^{-1}(y_i - \bar{y}) = (x_i - \bar{x})^\top A^\top A^{-\top} S_x^{-1} A^{-1} A(x_i - \bar{x}) = (x_i - \bar{x})^\top S_x^{-1}(x_i - \bar{x}).$$

(b)

Step 3. Step 1 applied to any subset H gives $S_H^{(y)} = AS_H^{(x)}A^\top$, so $\det S_H^{(y)} = (\det A)^2 \det S_H^{(x)}$. The factor $(\det A)^2 > 0$ is the same for every H, so both determinants are minimized by the same subsets.

Step 4. For that subset, $\hat{\mu}^{(y)} = A\hat{\mu}^{(x)} + b$ and $\widehat{\Sigma}^{(y)} = A\widehat{\Sigma}^{(x)}A^\top$ (the factor $c_{h,n}$ is common), and the computation of Step 2 shows that the robust distances coincide.

Changing units or rotating the features therefore never changes which points MCD flags. LOF and Isolation Forest, which use Euclidean distances or single coordinates, lack this property, so their inputs must be scaled deliberately. ■

Exercise 2.2 (A C-step never increases the determinant)

For a subset H with $|H| = h$ put $T_H = \frac{1}{h}\sum_{i\in H} x_i$ and $S_H = \frac{1}{h}\sum_{i\in H}(x_i - T_H)(x_i - T_H)^\top$. Let $|H_1| = h$ with $S_1 = S_{H_1} \succ 0$, $T_1 = T_{H_1}$, and $d_1(i) = (x_i - T_1)^\top S_1^{-1}(x_i - T_1)$. Let H_2 consist of the h indices with the smallest $d_1(i)$, and $S_2 = S_{H_2}$. Show that $\det S_2 \le \det S_1$.

Solution [Supplement].

Step 1. Using $a^\top Ma = \operatorname{tr}(Maa^\top)$,

$$\sum_{i\in H_1} d_1(i) = \operatorname{tr}\Big(S_1^{-1}\sum_{i\in H_1}(x_i - T_1)(x_i - T_1)^\top\Big) = \operatorname{tr}(S_1^{-1}hS_1) = hp.$$

Step 2. H_2 collects the h smallest values of d_1, so $\sum_{i\in H_2} d_1(i) \le \sum_{i\in H_1} d_1(i) = hp$.

Step 3. For $M \succ 0$ and any c, since $\sum_{i\in H_2}(x_i - T_2) = 0$ with $T_2 = T_{H_2}$,

$$\sum_{i\in H_2}(x_i - c)^\top M(x_i - c) = \sum_{i\in H_2}(x_i - T_2)^\top M(x_i - T_2) + h(T_2 - c)^\top M(T_2 - c).$$

With $M = S_1^{-1}$ and $c = T_1$, the first term on the right equals $h\operatorname{tr}(S_1^{-1}S_2)$, and the left side is $\sum_{i\in H_2} d_1(i)$. Hence $h\operatorname{tr}(S_1^{-1}S_2) \le hp$ by Step 2.

Step 4. The matrix $S_1^{-1/2}S_2S_1^{-1/2}$ is symmetric positive semidefinite with eigenvalues $\lambda_1, \dots, \lambda_p \ge 0$, trace $\operatorname{tr}(S_1^{-1}S_2) \le p$ and determinant $\det S_2 / \det S_1$. By the AM–GM inequality,

$$\Big(\frac{\det S_2}{\det S_1}\Big)^{1/p} = \Big(\prod_j \lambda_j\Big)^{1/p} \le \frac{1}{p}\sum_j \lambda_j \le 1.$$

So the determinants along the C-steps are non-increasing. There are finitely many h-subsets, so the sequence becomes constant after finitely many steps, which is where FAST-MCD stops. (Normalizing by $h - 1$ instead of h multiplies every determinant by the same constant.) ■

Exercise 2.3 (Huber regression is least squares with outlier shifts)

(a) For $r \in \mathbb{R}$ and $\delta > 0$ show that

$$\rho_\delta(r) = \min_{\gamma \in \mathbb{R}} \left\{ \tfrac{1}{2}(r-\gamma)^2 + \delta|\gamma| \right\},$$

with the unique minimizer $\gamma^\star(r) = \text{sign}(r)(|r| - \delta)_+$.
(b) Let $F(\boldsymbol{\beta}, \gamma) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta} - \gamma\|_2^2 + \delta\|\gamma\|_1$ with $\gamma \in \mathbb{R}^n$. Show that $(\hat{\boldsymbol{\beta}}, \hat{\gamma})$ minimizes F if and only if $\hat{\boldsymbol{\beta}}$ minimizes $\sum_i \rho_\delta(y_i - x_i^\top \boldsymbol{\beta})$ and $\hat{\gamma}_i = \gamma^\star(y_i - x_i^\top \hat{\boldsymbol{\beta}})$ for all i.

Solution [Supplement].

(a)

Step 1. $\phi(\gamma) = \frac{1}{2}(r-\gamma)^2 + \delta|\gamma|$ is strictly convex and tends to ∞ as $|\gamma| \to \infty$, so it has a unique minimizer, characterized by $0 \in \gamma - r + \delta\, \partial|\gamma|$, where $\partial|\gamma| = \{\text{sign}\gamma\}$ for $\gamma \neq 0$ and $[-1, 1]$ for $\gamma = 0$.

Step 2. If $|r| \leq \delta$, then $\gamma = 0$ works ($r \in \delta[-1, 1]$) and $\phi(0) = \frac{1}{2}r^2$. If $|r| > \delta$, then $\gamma = r - \delta\text{sign}(r)$ has the sign of r and satisfies $\gamma - r + \delta\text{sign}\gamma = 0$; its value is $\frac{1}{2}\delta^2 + \delta(|r| - \delta) = \delta(|r| - \frac{1}{2}\delta)$. In both cases the minimum is $\rho_\delta(r)$ and the minimizer is $\gamma^\star(r)$.

(b)

Step 3. For fixed $\boldsymbol{\beta}$, with $r_i = y_i - x_i^\top \boldsymbol{\beta}$, $F(\boldsymbol{\beta}, \gamma) = \sum_i \{\frac{1}{2}(r_i - \gamma_i)^2 + \delta|\gamma_i|\}$ separates over i. By (a), $\min_\gamma F(\boldsymbol{\beta}, \gamma) = \sum_i \rho_\delta(r_i) =: \mathcal{H}(\boldsymbol{\beta})$, attained only at $\gamma_i = \gamma^\star(r_i)$.

Step 4. If $(\hat{\boldsymbol{\beta}}, \hat{\gamma})$ minimizes F, then for every $\boldsymbol{\beta}$

$$\mathcal{H}(\hat{\boldsymbol{\beta}}) \leq F(\hat{\boldsymbol{\beta}}, \hat{\gamma}) \leq \min_\gamma F(\boldsymbol{\beta}, \gamma) = \mathcal{H}(\boldsymbol{\beta}),$$

so $\hat{\boldsymbol{\beta}}$ minimizes $\mathcal{H}$. Taking $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ gives $F(\hat{\boldsymbol{\beta}}, \hat{\gamma}) = \mathcal{H}(\hat{\boldsymbol{\beta}})$, and uniqueness in Step 3 forces $\hat{\gamma}_i = \gamma^\star(y_i - x_i^\top \hat{\boldsymbol{\beta}})$. Conversely, if $\hat{\boldsymbol{\beta}}$ minimizes $\mathcal{H}$ and $\hat{\gamma}$ is as stated, then $F(\hat{\boldsymbol{\beta}}, \hat{\gamma}) = \mathcal{H}(\hat{\boldsymbol{\beta}}) \leq \mathcal{H}(\boldsymbol{\beta}) \leq F(\boldsymbol{\beta}, \gamma)$ for all $(\boldsymbol{\beta}, \gamma)$.

Step 5. Given $\hat{\gamma}$, F is a smooth quadratic in $\boldsymbol{\beta}$, so $X^\top(\mathbf{y} - \hat{\gamma} - X\hat{\boldsymbol{\beta}}) = 0$: $\hat{\boldsymbol{\beta}}$ is the OLS fit to the shifted response $\mathbf{y} - \hat{\gamma}$.

Each case thus receives its own mean shift γ_i at the price $\delta|\gamma_i|$. Cases with $|r_i| \leq \delta$ get no shift; a case with a large residual is shifted by $|r_i| - \delta$, so only $\pm\delta$ of its residual reaches the least-squares fit, which is the bounded influence of the IRLS view. ■

Chapter 3. High-Dimensional Data and Regularization

Predictive Modeling, Lecture 2 (OLS instability, Ridge, Lasso, Elastic Net, choosing λ)

1 Linear Prediction and Why It Becomes Unstable

1.1 Setting

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad X \in \mathbb{R}^{n\times p}, \qquad \hat{y} = x_{\text{new}}^{\top}\hat{\boldsymbol{\beta}}.$$

n is the number of observations, p the number of candidate variables, β_j the linear contribution of variable j. Regularization selects, among all $\hat{\boldsymbol{\beta}}$ that fit the observed $\mathbf{y}$ well, a solution that is *less sensitive* on new x. That a coefficient is stable does not mean a causal effect has been identified.

1.2 High dimension is about p relative to n

Situation	Example	Problem for prediction
$p \ll n$	100,000 customers, 20 variables	fairly stable, but collinearity is still possible
p close to n	500 patients, 400 variables	small data changes move coefficients and selection a lot
$p \geq n$	1,000 documents, 50,000 words	OLS is not unique or cannot be computed
many derived features	raw variables, interactions, one-hot encodings	search degrees of freedom grow faster than signal

1.3 OLS and what it assumes

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 = (X^{\top}X)^{-1}X^{\top}\mathbf{y} \quad \text{(if } X^{\top}X \text{ is invertible)}.$$

The formula requires $X^{\top}X$ to be well conditioned, i.e. the variables must provide reasonably independent directions of information. From a predictive point of view, OLS is not a "wrong model" but a baseline that *can become unstable.*

1.4 When $p \geq n$: non-uniqueness

Theorem: interpolation is not unique

If $\text{rank}(X) < p$ (always the case when $p > n$), there is $v \neq 0$ with $Xv = 0$, and then $X(\hat{\boldsymbol{\beta}} + v) = X\hat{\boldsymbol{\beta}}$ for every solution $\hat{\boldsymbol{\beta}}$. In particular, when $\text{rank}X = n$, infinitely many $\boldsymbol{\beta}$ give $X\boldsymbol{\beta} = \mathbf{y}$ with zero training error.

The direction v does not change training predictions, but on a new case $x_{\text{new}}^{\top}v$ need not be 0: zero training error says nothing about which solution predicts well. Regularization deliberately gives up some of these degrees of freedom. (Among all interpolators, the minimum-norm one $X^{\top}(XX^{\top})^{-1}\mathbf{y}$ is the $\lambda \to 0$ limit of Ridge.)

1.5 Collinearity and the variance of coefficients

Income, credit limit and monthly card spending may all measure the same customer dimension: one coefficient may become very positive and another very negative; the combined prediction is similar

but the individual coefficients wobble and their roles can be reallocated in a new sample.

Theorem: variance along eigen-directions

Under $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I$, $\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^\top X)^{-1}$. With the SVD $X = UDV^\top$, $D = \mathrm{diag}(d_1, \ldots, d_p)$,

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 \sum_{j=1}^{p} \frac{v_j v_j^\top}{d_j^2}, \qquad \mathrm{Var}(v_j^\top \hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{d_j^2}.$$

Proof [Supplement]. $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}$, so the covariance is $\sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}$. Since $X^\top X = V D^2 V^\top$, its inverse is $V D^{-2} V^\top$. ■

If some direction of $X^\top X$ is nearly 0 ($d_j \approx 0$), the inverse explodes and the coefficient variance in that direction blows up. Equivalently, a small data perturbation is amplified: $\Delta\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top \Delta \mathbf{y}$. In practice: CV averages look similar while fold-wise coefficients, selected variables and prediction rankings change a lot. Check the stability of λ and of the selected variables, not only performance.

2 Regularization: Buying Variance With Bias

$$\mathbb{E}[(Y - \hat{f}(X))^2] = \mathrm{Bias}^2 + \mathrm{Var} + \sigma^2.$$

OLS aims at low bias; in high dimensions the variance term can be large. Regularization steps back from the training optimum and reduces the size of coefficients, the degrees of freedom, and sample sensitivity. Success is judged by CV and the final test, not by training loss. Low regularization tracks noise in the training data, adequate regularization is a candidate for minimal generalization risk, excessive regularization shrinks even the signal.

2.1 One template, three models

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \left[\frac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda P(\boldsymbol{\beta}) \right], \qquad P(\boldsymbol{\beta}) = \begin{cases} \|\boldsymbol{\beta}\|_2^2 & \text{Ridge} \\ \|\boldsymbol{\beta}\|_1 & \text{Lasso} \\ \alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2 & \text{Elastic Net} \end{cases}$$

The first term is data fit, the second is model complexity. $\lambda = 0$ gives OLS; larger λ means stronger shrinkage and simplification. The good λ is not the one with the smallest training loss but the one with the smallest loss on new data.

2.2 Standardization comes first

$$x_{ij}^{\mathrm{std}} = \frac{x_{ij} - \mu_j^{(\mathrm{train})}}{s_j^{(\mathrm{train})}}.$$

A penalty on $|\beta_j|$ depends on the unit of X_j (card spending in won versus number of clicks): rescaling X_j by c rescales its unpenalised coefficient by $1/c$ and so changes how hard it is penalized. Standardization makes the penalty unit-free. The intercept is usually not penalized (center y and X first). One-hot variables, sparse matrices and already meaningful units need separate judgement. Crucially, μ_j and s_j are estimated **on the training part only**.

3 Ridge

Motivation. With correlated or many predictors, least squares coefficients have huge variance. Ridge shrinks them toward zero, most strongly along poorly determined directions, and accepts a little bias for a large gain in stability.

3.1 Closed form and constrained form

$$\hat{\boldsymbol{\beta}}_\lambda^{\text{Ridge}} = \arg\min_{\boldsymbol{\beta}} \left[\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2\right] = (X^\top X + n\lambda I)^{-1} X^\top \mathbf{y}.$$

Proof [Supplement]. The gradient is $-\frac{2}{n}X^\top(\mathbf{y} - X\boldsymbol{\beta}) + 2\lambda\boldsymbol{\beta}$; setting it to zero gives $(X^\top X + n\lambda I)\boldsymbol{\beta} = X^\top \mathbf{y}$. The matrix is positive definite for every $\lambda > 0$, so the solution is unique even when $p \geq n$, and the objective is strictly convex. ■

Large coefficients cost more; most coefficients approach 0 but are not exactly 0. Ridge is natural when the predictive signal is spread over many variables. By Lagrangian duality the problem is equivalent to $\min \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2$ subject to $\|\boldsymbol{\beta}\|_2^2 \leq c$, for some $c = c(\lambda)$.

3.2 Ridge shrinks small singular directions most

Theorem: SVD form of Ridge

With $X = UDV^\top$ and $\tilde{\lambda} = n\lambda$ (the slides absorb n into λ),

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = V\,\text{diag}\Big(\frac{d_j}{d_j^2 + \tilde{\lambda}}\Big)U^\top \mathbf{y}, \qquad X\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \sum_j u_j \frac{d_j^2}{d_j^2 + \tilde{\lambda}} u_j^\top \mathbf{y}.$$

Proof [Supplement]. $X^\top X + \tilde{\lambda} I = V(D^2 + \tilde{\lambda} I)V^\top$ and $X^\top \mathbf{y} = VDU^\top \mathbf{y}$, so $(X^\top X + \tilde{\lambda} I)^{-1} X^\top \mathbf{y} = V(D^2 + \tilde{\lambda} I)^{-1} DU^\top \mathbf{y}$. ■

OLS multiplies direction j by $1/d_j$; Ridge by $d_j/(d_j^2 + \tilde{\lambda})$, which stays bounded by $1/(2\sqrt{\tilde{\lambda}})$. Directions with small d_j are those the data can hardly distinguish, and Ridge shrinks exactly those most strongly. The **effective degrees of freedom** $\text{df}(\lambda) = \sum_j d_j^2/(d_j^2 + \tilde{\lambda})$ decrease from p to 0 as λ increases. Numerical stability, coefficient stability and prediction on new data can improve simultaneously.

3.3 Ridge shares signal among correlated variables

Theorem: identical columns get identical Ridge coefficients

If $x_{(1)} = x_{(2)}$, then $\hat{\beta}_1^{\text{Ridge}} = \hat{\beta}_2^{\text{Ridge}}$ for every $\lambda > 0$.

Proof [Supplement]. The fit term depends on β_1, β_2 only through $\beta_1 + \beta_2$. Swapping $\beta_1 \leftrightarrow \beta_2$ leaves the objective unchanged, and the objective is strictly convex, so the unique minimizer must be symmetric. ■

OLS can produce opposite-sign coefficients such as $\hat{y} \approx x_1(12) + x_2(-11)$; the sum predicts similarly but each coefficient is unstable across resamples. Ridge tends towards $\hat{y}_{\text{Ridge}} \approx x_1(0.8) + x_2(0.7)$: it shrinks similar variables together so that no single one takes an excessive role (illustrative, not

literal numbers). If stable prediction rather than variable selection is the goal, Ridge is a strong starting point.

Data / goal	Why Ridge fits	What to check
many weak signals	keeps all coefficients a little	does it beat OLS?
strong multicollinearity	stabilizes unstable directions	standardization and coefficient paths
$p \geq n$ or sparse matrices	unique, computable solution	sparse implementation, memory
prediction over interpretation	targets generalization, not zeros	final metric and operating cost

4 Lasso

4.1 The L_1 penalty sets some coefficients exactly to zero

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1\Big], \qquad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

It pushes not only large but also small coefficients to 0, giving a **sparse** model: a short variable list together with predictions. What is *not* gained: one cannot conclude that unselected variables are truly irrelevant.

Theorem: soft-thresholding under an orthogonal design

If $X^\top X = nI$, let $z = X^\top \mathbf{y}/n$ (the OLS estimate). Then

$$\hat{\beta}_j^{\text{Lasso}} = S(z_j, \lambda/2) = \text{sign}(z_j)\big(|z_j| - \lambda/2\big)_+, \qquad \hat{\beta}_j^{\text{Ridge}} = \frac{z_j}{1+\lambda}.$$

Proof [Supplement]. $\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \frac{1}{n}\|\mathbf{y}\|^2 - 2\boldsymbol{\beta}^\top z + \|\boldsymbol{\beta}\|^2$, so the Lasso objective separates into $\sum_j\{(\beta_j - z_j)^2 + \lambda|\beta_j|\}$ up to a constant. For one coordinate the subgradient condition is $2(\beta_j - z_j) + \lambda s_j = 0$ with $s_j = \text{sign}(\beta_j)$ if $\beta_j \neq 0$ and $s_j \in [-1, 1]$ if $\beta_j = 0$. If $|z_j| \leq \lambda/2$, $\beta_j = 0$ satisfies it; otherwise $\beta_j = z_j - \frac{\lambda}{2}\text{sign}(z_j)$. For Ridge, $(\beta_j - z_j)^2 + \lambda\beta_j^2$ is minimized at $z_j/(1+\lambda)$. ■

Ridge shrinks proportionally and never reaches zero; Lasso subtracts a constant and sets small coefficients exactly to zero. This is the algebraic reason for sparsity; in general designs, coordinate descent applies exactly this soft-threshold to partial residuals one coordinate at a time.

4.2 Geometry: the ball and the diamond

In the constrained form the elliptical contours of the squared loss meet the constraint region. The Ridge ball $\|\boldsymbol{\beta}\|_2^2 \leq c$ has a smooth boundary, so the contact point is typically off the axes. The Lasso diamond $\|\boldsymbol{\beta}\|_1 \leq c$ has corners on the axes, so contact often happens at a corner, where some coefficients are 0. The coefficient paths show this: under Ridge many coefficients shrink together; under Lasso some disappear exactly.

4.3 Lasso is unstable among correlated variables

If several variables measure the same latent factor, Lasso may select one and set another to 0; with a slightly different sample a different name is selected. That a variable was not selected does not mean its predictive information is 0, and selection results must not be over-interpreted causally. Check selection frequencies across resamples or CV folds and whether variables are domain-equivalent; with many correlated signals keep Ridge or Elastic Net as candidates.

Remark: a limit on Lasso selection

When $p > n$, the Lasso selects at most n variables. For two identical columns the Lasso solution is not unique: any split (β_1, β_2) with the same sum and signs is optimal.

Situation	Why Lasso is attractive	Caveat
variables costly to measure or deploy	simpler prediction pipeline	compare selection frequency and loss in performance
very many variables, sparse signal suspected	shrinks unnecessary coefficients to 0	may keep only one of correlated signals
an explainable operating model is needed	easy to present a variable list	do not turn selection into causal claims
screening for later analysis	starting point of exploratory reduction	needs separate validation and domain review

5 Elastic Net

$$\hat{\boldsymbol{\beta}}^{\mathrm{EN}} = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\{\alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2\}\Big], \qquad \alpha \in [0,1].$$

$\alpha = 1$ is Lasso, $\alpha = 0$ is Ridge. The L_2 part makes the objective strictly convex, so the solution is unique, and gives a **grouping effect**: highly correlated variables receive similar coefficients and tend to enter or leave together. The L_1 part keeps sparsity. Under an orthogonal design the solution is $S(z_j, \lambda\alpha/2)/\{1 + \lambda(1-\alpha)\}$: soft-thresholding followed by proportional shrinkage. Both λ and α are tuned by CV.

6 Choosing λ Inside the Folds

Within each fold, standardization, transformations and fitting of each λ candidate must depend only on the training part; only then is the validation score fair. The CV curve gives two common rules:

- **minimum rule**: choose $\lambda_{\min}$ with the smallest mean CV loss, natural when predictive performance is all that matters;
- **1-SE rule**: $\lambda_{1\mathrm{SE}} = \max\{\lambda : \widehat{R}_{\mathrm{CV}}(\lambda) \le \widehat{R}_{\min} + \mathrm{SE}_{\min}\}$, the simplest (most regularized) model within one standard error of the minimum; useful when interpretability, computation and stability matter.

The final selection rule is fixed *before* the analysis. The lowest point of the CV curve is not the only answer: differences within the noise level of CV are not meaningful.

7 Regularized Logistic Regression

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\sum_{i=1}^{n} \ell\{y_i, \sigma(\beta_0 + x_i^\top \boldsymbol{\beta})\} + \lambda P(\boldsymbol{\beta})\Big], \qquad \ell(y,p) = -[y\log p + (1-y)\log(1-p)].$$

What changes: log loss replaces squared error. What stays: standardization, L_1/L_2 penalties, CV selection of λ, protection of the test set. The penalty also fixes the separation problem of Ch. 1: with $\lambda > 0$ the objective is coercive, so a finite minimizer exists.

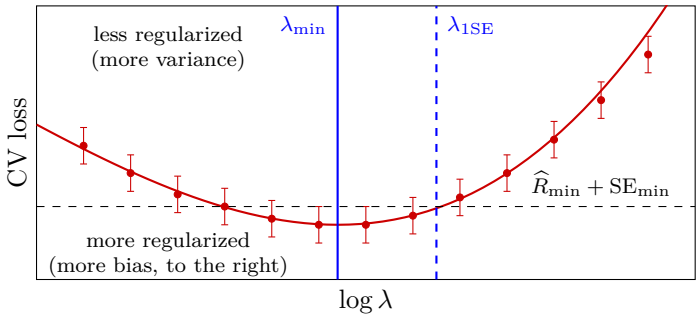


Figure 4: A typical CV curve with ±1 SE bars (schematic). The minimum rule takes $\lambda_{\min}$; the 1-SE rule moves right to the largest λ whose CV loss is still below the dashed line, giving a simpler, more stable model at almost the same estimated risk.

8 Regularization Versus Robust Regression

Regularization	Robust regression
reduces coefficient variance and overfitting caused by many variables, collinearity and large degrees of freedom	mitigates loss domination caused by a few large residuals, label errors and extreme cases
Ridge, Lasso, Elastic Net	Huber, LAD
shrinkage, selection, generalization risk	influence of outliers, conditional location

They address *different failure causes* and can be combined (e.g. Huber loss with an Elastic Net penalty).

9 Practical Selection and Reporting

Prediction-centered: prefer the lowest CV loss, but if differences are within the SE choose the simpler or faster candidate. **Operation-centered**: if the loss in performance is small prefer fewer variables, more stable selection and easier retraining. **Interpretation-centered**: look at signs, selection frequency and domain constraints together; one Lasso fit does not settle variable importance. **Final step**: fix the selection rule, refit on all training data, evaluate the protected test set once.

Example: reading a results table (illustrative)

Candidate	CV primary metric	Complexity	Interpretation
OLS	0.842 ± 0.031	180 variables	baseline; unstable under collinearity
Ridge	0.801 ± 0.019	180 shrunk coefficients	better performance and stability
Lasso	0.808 ± 0.026	24 selected	small loss in performance for simplicity
Elastic Net	0.799 ± 0.020	47 selected	compromise for correlated blocks

(Loss: smaller is better.) Do not copy the numbers; adopt the format that reads mean, variability, complexity and operating cost together.

Example: customer churn case

Situation. 8,000 customers, 600 derived variables; strong correlations among usage, payment and inquiry variables; churn rate 8%. **Candidates.** Standardized logistic baseline, Ridge, Lasso, Elastic Net. Primary metrics PR-AUC and log loss; operating metric recall@budget. **Split.** Protect the test period by time order; stratified CV within the training period. **Report.** CV mean and variability, chosen λ, number of selected variables, probability calibration and test performance in one table.

Summary

1. Many variables and collinearity make *coefficient variance and generalization risk*, rather than training fit, the main problem of OLS.
2. Ridge shrinks, Lasso selects, Elastic Net compromises; the right choice depends on the signal structure and the operating goal.
3. Standardization and λ selection are done inside CV folds; regularization does not automatically fix leakage or outliers.

10 Selected exercises

Motivation. The first exercise makes the bias–variance trade of Ridge exact, the second finds the smallest penalty at which the Lasso selects nothing (the start of every Lasso path), and the third proves the grouping effect of the Elastic Net stated above.

Exercise 3.1 (Some Ridge penalty always beats OLS)

Let X be fixed with $X^\top X = nI$, $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I$ and $\boldsymbol{\beta} \neq \mathbf{0}$. With the Ridge estimate $\hat{\boldsymbol{\beta}}_\lambda = z/(1+\lambda)$, $z = X^\top \mathbf{y}/n$, of the orthogonal-design theorem:
(a) show that $\mathbb{E}\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 = \dfrac{\lambda^2\|\boldsymbol{\beta}\|^2 + p\sigma^2/n}{(1+\lambda)^2}$;
(b) show that this risk decreases on $[0, \lambda^\star]$ and increases afterwards, where $\lambda^\star = p\sigma^2/(n\|\boldsymbol{\beta}\|^2)$, and that its minimum is strictly smaller than the OLS risk $p\sigma^2/n$.

Solution [Supplement].

(a)

Step 1. $z = X^\top(X\boldsymbol{\beta} + \boldsymbol{\varepsilon})/n = \boldsymbol{\beta} + X^\top\boldsymbol{\varepsilon}/n$, so $\mathbb{E}z = \boldsymbol{\beta}$ and $\mathrm{Var}(z) = \sigma^2 X^\top X/n^2 = (\sigma^2/n)I$.

Step 2. Hence $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta} = -\frac{\lambda}{1+\lambda}\boldsymbol{\beta}$ and $\mathrm{Var}(\hat{\boldsymbol{\beta}}_\lambda) = \frac{\sigma^2}{n(1+\lambda)^2} I$. The mean squared error is the squared bias plus the trace of the covariance:

$$\mathbb{E}\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 = \frac{\lambda^2\|\boldsymbol{\beta}\|^2}{(1+\lambda)^2} + \frac{p\sigma^2}{n(1+\lambda)^2}.$$

(b)

Step 3. Write $a = \|\boldsymbol{\beta}\|^2 > 0$, $c = p\sigma^2/n$ and $f(\lambda) = (\lambda^2 a + c)/(1+\lambda)^2$. Then

$$f'(\lambda) = \frac{2\lambda a(1+\lambda) - 2(\lambda^2 a + c)}{(1+\lambda)^3} = \frac{2(\lambda a - c)}{(1+\lambda)^3},$$

negative for $\lambda < c/a = \lambda^\star$ and positive for $\lambda > \lambda^\star$.

Step 4. Substituting $\lambda^\star = c/a$,

$$f(\lambda^\star) = \frac{c^2/a + c}{(1 + c/a)^2} = \frac{ac}{a + c} < c = f(0).$$

So the optimal penalty grows with the noise level and shrinks with the signal size, and the relative gain $c/(a + c)$ is largest when the signal-to-noise ratio a/c is small. ■

Exercise 3.2 (When the Lasso selects nothing)

For a general design X and $\lambda > 0$, consider the Lasso objective $Q(\boldsymbol{\beta}) = \frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1$.
(a) Show that $\mathbf{0}$ minimizes Q if and only if $\lambda \geq \lambda_{\max} := \frac{2}{n}\|X^\top \mathbf{y}\|_\infty$.
(b) Show that all minimizers of Q have the same fitted values $X\hat{\boldsymbol{\beta}}$, and deduce that for $\lambda \geq \lambda_{\max}$ the minimizer $\mathbf{0}$ is unique.

Solution [Supplement].

(a)

Step 1. Q is convex, so $\boldsymbol{\beta}$ is a minimizer iff $\mathbf{0} \in \partial Q(\boldsymbol{\beta}) = -\frac{2}{n}X^\top(\mathbf{y} - X\boldsymbol{\beta}) + \lambda\,\partial\|\boldsymbol{\beta}\|_1$.

Step 2. At $\boldsymbol{\beta} = \mathbf{0}$, $\partial\|\mathbf{0}\|_1 = [-1, 1]^p$, so the condition reads $\frac{2}{n}X^\top\mathbf{y} \in \lambda[-1, 1]^p$, i.e. $\frac{2}{n}|x_{(j)}^\top\mathbf{y}| \leq \lambda$ for every j, i.e. $\lambda \geq \lambda_{\max}$.

(b)

Step 3. Let $\hat{\boldsymbol{\beta}}_1, \hat{\boldsymbol{\beta}}_2$ be minimizers with $X\hat{\boldsymbol{\beta}}_1 \neq X\hat{\boldsymbol{\beta}}_2$, and $\bar{\boldsymbol{\beta}} = \frac{1}{2}(\hat{\boldsymbol{\beta}}_1 + \hat{\boldsymbol{\beta}}_2)$. The map $u \mapsto \|\mathbf{y} - u\|^2$ is strictly convex, so the fit term at $\bar{\boldsymbol{\beta}}$ is strictly below the average of the two fit terms, while $\|\bar{\boldsymbol{\beta}}\|_1 \leq \frac{1}{2}(\|\hat{\boldsymbol{\beta}}_1\|_1 + \|\hat{\boldsymbol{\beta}}_2\|_1)$. Then $Q(\bar{\boldsymbol{\beta}}) < \min Q$, a contradiction.

Step 4. If $\lambda \geq \lambda_{\max}$, $\mathbf{0}$ is a minimizer, so every minimizer has $X\hat{\boldsymbol{\beta}} = \mathbf{0}$ and $\min Q = \frac{1}{n}\|\mathbf{y}\|^2$. Then $\frac{1}{n}\|\mathbf{y}\|^2 + \lambda\|\hat{\boldsymbol{\beta}}\|_1 = \frac{1}{n}\|\mathbf{y}\|^2$, so $\hat{\boldsymbol{\beta}} = \mathbf{0}$.

Software therefore starts the λ grid at $\lambda_{\max}$. Under an orthogonal design, $\lambda \geq \lambda_{\max}$ means $|z_j| \leq \lambda/2$ for all j, which agrees with the soft-thresholding formula. ■

Exercise 3.3 (Grouping effect of the Elastic Net)

Let $\mathbf{y}$ be centered and the columns of X standardized ($\mathbf{1}^\top x_{(j)} = 0$, $\|x_{(j)}\|^2 = n$), with sample correlation $\rho_{ij} = x_{(i)}^\top x_{(j)}/n$. Let $0 \leq \alpha < 1$, $\lambda > 0$, and $\hat{\boldsymbol{\beta}}$ the Elastic Net estimate. Show that if $\hat{\beta}_i\hat{\beta}_j > 0$, then

$$|\hat{\beta}_i - \hat{\beta}_j| \leq \frac{\|\mathbf{y}\|}{\sqrt{n}} \cdot \frac{\sqrt{2(1 - \rho_{ij})}}{\lambda(1 - \alpha)}.$$

Solution [Supplement].

Step 1. Write $\hat{r} = \mathbf{y} - X\hat{\boldsymbol{\beta}}$. At a coordinate with $\hat{\beta}_k \neq 0$ the objective is differentiable in β_k, and the derivative vanishes at the minimizer:

$$-\frac{2}{n}x_{(k)}^\top\hat{r} + \lambda\alpha\,\mathrm{sign}(\hat{\beta}_k) + 2\lambda(1 - \alpha)\hat{\beta}_k = 0.$$

Step 2. For $k = i, j$ the signs agree, so subtracting the two equations gives

$$2\lambda(1 - \alpha)(\hat{\beta}_i - \hat{\beta}_j) = \frac{2}{n}(x_{(i)} - x_{(j)})^\top\hat{r}.$$

Step 3. By Cauchy–Schwarz, $|\hat{\beta}_i - \hat{\beta}_j| \leq \|x_{(i)} - x_{(j)}\|\,\|\hat{r}\|/\{n\lambda(1-\alpha)\}$, and $\|x_{(i)} - x_{(j)}\|^2 = 2n - 2n\rho_{ij}$.

Step 4. Comparing the objective at $\hat{\boldsymbol{\beta}}$ and at $\mathbf{0}$, $\frac{1}{n}\|\hat{r}\|^2 \le \frac{1}{n}\|\hat{r}\|^2 + \lambda P(\hat{\boldsymbol{\beta}}) \le \frac{1}{n}\|\mathbf{y}\|^2$, so $\|\hat{r}\| \le \|\mathbf{y}\|$. Combining,

$$|\hat{\beta}_i - \hat{\beta}_j| \le \frac{\sqrt{2n(1-\rho_{ij})}\,\|\mathbf{y}\|}{n\lambda(1-\alpha)} = \frac{\|\mathbf{y}\|}{\sqrt{n}} \cdot \frac{\sqrt{2(1-\rho_{ij})}}{\lambda(1-\alpha)}.$$

As $\rho_{ij} \to 1$ the two coefficients are forced together. For the Lasso ($\alpha = 1$) the bound is void, in line with the non-uniqueness for identical columns noted above. ■

Chapter 4. Decision Trees and Random Forests

Predictive Modeling, Lecture 3 (CART, impurity, pruning, bagging, OOB, random forest, importance)

1 After the Linear Baseline: Where Does the Relation Change?

The language of linear models expresses effects as a global linear combination; to add interactions or nonlinearity a person must design the terms. The language of trees splits the input space by if–then conditions and uses a simple constant prediction in each region. Typical questions that trees answer naturally: in which customer group does the price effect change? Does risk jump beyond a threshold? Do different combinations of variables need entirely different rules? **A tree is a flexible baseline that discovers interactions and thresholds; a random forest is the ensemble that stabilizes this flexibility.**

2 The Tree as a Piecewise-Constant Function

Motivation. A regression tree splits the input space into rectangles and predicts a constant on each. This view explains both its strengths (interactions and nonlinearity for free) and its weakness (high variance from greedy, discontinuous splits).

Definition: regression tree

$$\hat{f}(x) = \sum_{m=1}^{M} c_m \, \mathbb{I}(x \in R_m),$$

where $R_1, \ldots, R_M$ are the leaf regions (a partition of $\mathcal{X}$), c_m is the prediction in region m, and the boundaries are axis-aligned splits by a threshold on one variable.

Within a leaf the model is very simple, but with enough leaves the whole model can express nonlinearity and interactions. The price: if one split changes, every rule below it can change, so a single tree has high variance. The same tree has two views: in feature space each rectangle is a leaf; in the condition structure each root-to-leaf path is an if–then rule.

2.1 Nested conditions create interactions automatically

Two splits already give

$$\hat{f}(x) = \begin{cases} c_1, & x_1 < s_1, \\ c_2, & x_1 \geq s_1 \ \wedge \ x_2 < s_2, \\ c_3, & x_1 \geq s_1 \ \wedge \ x_2 \geq s_2. \end{cases}$$

Here the effect of x_2 exists only when $x_1 \geq s_1$: an interaction. A linear model needs a term such as $x_1 x_2$ inserted by hand, whereas the tree's split path produces it. Example: a high-price product's discount effect differs between new and existing customers; one rule path captures both conditions.

2.2 Leaf predictions

Theorem: optimal leaf constants

For a fixed partition, (i) under squared loss the optimal c_m is the leaf mean $\hat{c}_m = \frac{1}{N_m}\sum_{i:x_i\in R_m} y_i$;
(ii) under absolute loss it is the leaf median;
(iii) for classification, the leaf class proportions $\hat{p}_{mk} = \frac{1}{N_m}\sum_{i:x_i\in R_m}\mathbb{I}(y_i = k)$ minimize the log loss, and $\hat{y}(x) = \arg\max_k \hat{p}_{mk}$ minimizes misclassification.

Proof [Supplement]. The loss separates over leaves, and within a leaf the problem is the constant-prediction problem of Ch. 1 applied to the empirical distribution of the leaf (mean, median, class frequencies). ■

Intuitively, the tree divides customers into small groups by conditions and gives everyone in a leaf the same demand, risk or price prediction. At leaf boundaries the prediction jumps like a staircase; if a smooth relation is required, keep other models as candidates. The majority class is the default label but $\hat{p}_{mk}$ can be used as a probability estimate — before converting it into actions check calibration and thresholds separately; proportions in small leaves are very unstable.

3 Growing the Tree: CART

3.1 One variable, one threshold per node

$$R_1(j,s) = \{x : x_j < s\}, \qquad R_2(j,s) = \{x : x_j \geq s\}.$$

All candidate variables j and cut points s are examined; the best binary split of the current node is chosen; the procedure repeats in the child nodes. The tree does not search for the global optimum at once; it is **greedy**, choosing the best split at the current step. One axis-aligned split is simple, but stacking several builds rectangular regions and complex interactions.

3.2 Regression split criterion

$$\Delta_{\text{RSS}}(j,s) = \sum_{i\in R}(y_i - \bar{y}_R)^2 - \sum_{r\in\{1,2\}}\sum_{i\in R_r(j,s)}(y_i - \bar{y}_{R_r})^2.$$

Choose (j,s) with the largest Δ_{RSS}, i.e. the split that makes the targets inside the children most homogeneous.

Derivation: efficient search and the ANOVA form of the gain

For node R with $N = N_1 + N_2$, the within/between decomposition gives

$$\Delta_{\text{RSS}} = N_1(\bar{y}_{R_1} - \bar{y}_R)^2 + N_2(\bar{y}_{R_2} - \bar{y}_R)^2 = \frac{N_1 N_2}{N}(\bar{y}_{R_1} - \bar{y}_{R_2})^2 \geq 0.$$

For a continuous x_j, cut points between consecutive sorted values are the only candidates; after sorting, running sums $\sum y$ and counts update in $O(1)$ per candidate, so all cut points of a variable cost $O(N \log N)$. A categorical variable with K levels has $2^{K-1} - 1$ binary partitions; for regression (and binary classification) it suffices to sort the levels by their mean target and treat them as ordered, leaving $K - 1$ candidates.

A good split now does not guarantee global optimality once later splits are considered.

3.3 Classification split criterion

$$G(R) = 1 - \sum_{k=1}^{K} p_{Rk}^2 \quad \text{(Gini)}, \qquad H(R) = -\sum_{k=1}^{K} p_{Rk} \log p_{Rk} \quad \text{(entropy)}.$$

A leaf with one class has impurity 0. A good split strongly reduces the weighted average impurity $\frac{N_1}{N} I(R_1) + \frac{N_2}{N} I(R_2)$. Gini and entropy usually give similar trees in practice.

Theorem: interpretation and monotonicity of impurity

(i) $G(R)$ is the probability that two labels drawn independently from the leaf distribution differ, equivalently the misclassification rate of a rule that labels at random according to p_R. (ii) Gini and entropy are strictly concave functions of p; hence any split satisfies $\frac{N_1}{N} I(R_1) + \frac{N_2}{N} I(R_2) \leq I(R)$, with equality iff both children have the parent's class proportions.

Proof [Supplement]. (i) $P(\text{two draws differ}) = 1 - \sum_k p_k^2$.

(ii) The parent's proportions are the weighted average $p_R = \frac{N_1}{N} p_{R_1} + \frac{N_2}{N} p_{R_2}$; apply Jensen's inequality to the concave impurity. ■

The misclassification rate $1 - \max_k p_k$ is concave but piecewise linear, so many splits give zero improvement; this is why smoother Gini/entropy are used for growing. Distinguish the split criterion (an internal training objective) from the final evaluation by the problem's metric (AUC, log loss, cost).

3.4 Many candidate splits cause selection bias

Continuous variables offer many cut points; categorical variables offer many partitions, and the more levels a variable has the more chances to split. Statistically, the split that reduced training loss most need not be the best for generalization, especially when many candidates are searched in small nodes. Remedies: a minimum leaf size, pruning, external validation; check high-cardinality identifier variables for leakage first.

3.5 Greedy CART: fast but not globally optimal

Greedy growth chooses the split with the largest immediate gain and never revisits earlier choices. Strengths: handles continuous and categorical variables, nonlinearity and interactions naturally. Cost: one early split can lock in the later structure. Do not treat a tree's structure as "the discovered truth"; look at validation performance and stability together.

3.6 Axis-aligned splits approximate oblique boundaries by staircases

Well suited: threshold rules such as "fewer than 5 uses in the last 30 days and at least one payment failure". Hard: diagonal boundaries like $x_1 + x_2 > c$ or smooth functions need many rectangles. Growing deeper to gain expressiveness makes leaves smaller and variance larger, so a tree is not always best for nonlinear relations. Compare linear baselines, linear models with transformations and tree ensembles under the same validation protocol.

4 Controlling Complexity

A deeper tree always lowers training error but test error can rise again: depth increases expressiveness but also the risk that small leaves and complex boundaries memorize accidental patterns.

Approach	Example	Pros and cons
pre-pruning	maximum depth, minimum leaf size, minimum split gain	fast, but stopping too early can miss good structure
post-pruning	grow large, then remove branches	sees a wider set of candidates, but requires a separate complexity choice
validation-based choice	choose depth / leaf size / pruning strength by CV	targets generalization; must be done inside the pipeline

Not "never grow deep trees", but *choose depth by generalization performance.*

4.1 Minimum leaf size is a statistical regularizer

$$\hat{p}_m = \frac{n_{m,+}}{N_m}, \qquad \mathrm{SE}(\hat{p}_m) \approx \sqrt{\frac{\hat{p}_m(1-\hat{p}_m)}{N_m}}.$$

Too small a leaf: local means and class proportions are driven by a few observations, so training looks good while new cases wobble (e.g. $N_m = 5$, $\hat{p} = 0.4$ gives SE ≈ 0.22). Too large a leaf: different groups are mixed, local thresholds and interactions are missed, bias increases. The minimum leaf size is not merely a speed option; it controls the stability of local means and probabilities.

4.2 Cost-complexity pruning

$$R_\alpha(T) = R(T) + \alpha|T|,$$

where $R(T)$ is the training loss or impurity-based risk of tree T, $|T|$ the number of leaves and α the price of keeping one more leaf. A split that lowers training loss only a little is pruned once α exceeds its benefit. α is chosen by CV like any other hyperparameter.

Derivation: weakest-link pruning

For an internal node t with subtree T_t, collapsing T_t to a leaf changes R_α by $\{R(t) - R(T_t)\} - \alpha(|T_t| - 1)$. Collapsing pays off as soon as

$$\alpha \geq g(t) = \frac{R(t) - R(T_t)}{|T_t| - 1}.$$

Repeatedly collapsing the node with the smallest $g(t)$ yields a nested sequence $T_0 \supset T_1 \supset \cdots \supset \{\text{root}\}$ with breakpoints $\alpha_1 < \alpha_2 < \cdots$; for every α the optimal subtree is in this sequence (Breiman et al.). CV then only needs to choose among finitely many trees.

5 From One Tree to Many: Bagging

5.1 A single tree has high variance

Resampling experiment: changing a few training observations can change the root split, leaf boundaries, selected variables and predictions; observations near a boundary can flip an early split, and deep trees are more sensitive to small-sample accidents. With MSE = Bias^2 + Var + σ^2, the goal of an ensemble is to average several slightly different trees and cancel the accidental wobble of each. **Trees are ensembled not because they are weak, but because they are strong yet unstable.**

5.2 Bagging

Algorithm: bagging (bootstrap aggregating)

For $b = 1, \dots, B$: draw a bootstrap sample of size n with replacement; grow a deep tree $\hat{f}_b$. Aggregate by averaging (regression) or by averaging probabilities / majority vote (classification).

For a single tree one limits depth or prunes to lower variance, since that one tree is the final predictor. For bagging and random forests, individual trees are grown fairly deep for low bias and averaging reduces variance; overly strong pre-stopping would also cut the expressiveness of the ensemble. Not unlimited, though: with small samples, rare classes, probability quality or compute cost, minimum leaf size, maximum depth and weights still matter. Readability of an individual tree and predictive stability of the ensemble are different goals.

5.3 A bootstrap sample contains about 63.2% of the distinct observations

Theorem: inclusion probability

The probability that a given observation never appears in a bootstrap sample of size n is $(1 - 1/n)^n \to e^{-1} \approx 0.368$. Hence on average about $1 - 0.368 = 0.632$ of the distinct observations are used to grow one tree.

Proof [Supplement]. Each of the n independent draws misses the observation with probability $1 - 1/n$. Moreover $\log(1 - 1/n)^n = n \log(1 - 1/n) = -1 - \frac{1}{2n} + O(n^{-2}) \to -1$. ■

Some observations are drawn several times, others are left out; the left-out (out-of-bag, OOB) observations allow an internal evaluation.

5.4 Averages are sensitive to correlation between trees

Theorem: variance of an average of correlated predictors

If each $\hat{f}_b(x)$ has variance σ^2 and every pair has correlation $\rho \geq 0$, then

$$\operatorname{Var}\Big(\frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)\Big) = \sigma^2\Big[\rho + \frac{1-\rho}{B}\Big].$$

Proof [Supplement]. $\operatorname{Var}(\sum_b \hat{f}_b) = B\sigma^2 + B(B-1)\rho\sigma^2$. Divide by B^2: $\frac{\sigma^2}{B} + \frac{B-1}{B}\rho\sigma^2 = \sigma^2\{\rho + (1-\rho)/B\}$. ■

Increasing B shrinks the second term, but if ρ is large the average still wobbles: the variance floor is $\rho\sigma^2$. Averaging does not change the bias: $\mathbb{E}[\frac{1}{B}\sum_b \hat{f}_b] = \mathbb{E}\hat{f}_1$. Random forests add feature randomness precisely to lower ρ.

Task	Output of one tree	Ensemble output
regression	$\hat{f}_b(x)$	$\frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)$
binary classification	$\hat{p}_b(x)$ or class	average the probabilities then threshold, or majority vote
multiclass	class probability vector	argmax of the averaged class probabilities
ranking / policy	risk score, probability	combine with top-k, a threshold, a cost rule

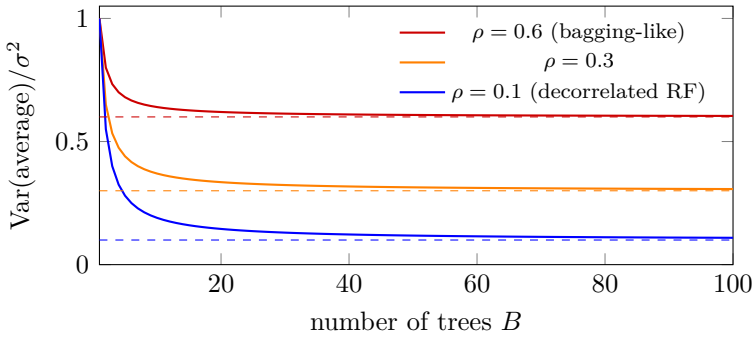


Figure 5: $\rho + (1-\rho)/B$: adding trees quickly removes the second term, but the variance of the ensemble levels off at the floor $\rho\sigma^2$ (dashed). Lowering the correlation between trees lowers the floor itself.

6 Random Forests

6.1 Bagging alone can let every tree use the same strong variable

Correlated trees: if every tree starts its root split with the same strong variable, the prediction boundaries look alike even though the bootstrap samples differ. Problem: even with large B, a high ρ limits the variance reduction of the average.

6.2 Randomness over both observations and features

Algorithm: random forest

For $b = 1, \ldots, B$: draw a bootstrap sample; grow a tree in which, **at each node**, only m_{try} randomly chosen variables are allowed as split candidates; grow deep (minimum leaf size small). Aggregate as in bagging.

The bootstrap makes the trees' data different; the feature subsampling makes their first splits and structure different. Trees explore different explanatory paths, their correlation falls, and the averaging effect grows.

6.3 m_{try}

$m_{\text{try}} \in \{1, \ldots, p\}$; $m_{\text{try}} = p$ is bagging. Small m_{try}: more diverse trees, lower correlation, but individual trees may get weaker. Large: each tree stronger but trees more similar. Conventional starting values are $\sqrt{p}$ for classification and about $p/3$ for regression, but they depend on data and implementation; choose by OOB or CV and compute cost.

6.4 OOB scores are convenient but do not always replace CV

For observation i, average only the trees for which i was out of bag: $\hat{f}^{\text{OOB}}(x_i) = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} \hat{f}_b(x_i)$, with $|\mathcal{B}_i| \approx 0.368B$; the OOB error is the average loss of these predictions.

Method	Strength	Caveat
OOB	fast internal estimate without a separate validation set	does not respect time/group structure; not independent validation
CV	compares preprocessing and candidates under the same rule	expensive; the pipeline must be repeated inside folds
final test	reports generalization of a fixed procedure	used once; not material for re-tuning

OOB is an excellent internal diagnostic for random forests, but does not remove the time/-group/leakage validation principles of Ch. 1.

7 Variable Importance Is Not Causal Importance

Impurity-decrease importance: sum over the forest the impurity reductions obtained by splits on each variable; fast and usually provided by default. **Permutation importance**: shuffle one column and see how much validation/OOB performance drops:

$$I_j = \text{Score}(D_{\text{eval}}) - \text{Score}(D_{\text{eval}}^{\pi(j)}),$$

where $D_{\text{eval}}^{\pi(j)}$ is the evaluation data with only column j permuted; for losses define the sign so that a "loss increase" is reported. Compute it on OOB or validation sets, not the final test, and record variability over repeated permutations.

High importance signals that the model's predictions *relied on* the variable. Correlated variables may share or substitute each other's importance: when one is shuffled, its substitute still carries the information, so its importance can look low. High-cardinality categorical variables are favored by impurity importance (more chances to split). Forbidden interpretation: high importance does not mean that changing the variable changes the outcome.

8 Comparison and Summary

Model	Randomness	Main strength	Main limitation
single tree	none or limited	rules are directly readable	high variance, unstable boundaries
bagging	bootstrap of rows	reduces variance of deep trees	gain limited if trees are correlated
random forest	rows + feature subsets	lower correlation, more stable prediction	hard to explain by one rule
boosting (next)	sequential residual / weight updates	reduces bias, strong prediction	more sensitive to tuning, overfitting, time structure

The key difference among tree ensembles is not "how many trees" but **how dissimilar the trees' errors are made**.

Summary

1. Trees split the input space by conditions and capture nonlinearity and interactions.
2. A deep single tree is expressive but has high variance and overfits.
3. Bagging averages bootstrap trees to lower variance; random forests add feature subsampling to lower correlation between trees.

4. OOB, importance and partial dependence are useful diagnostics, but do not replace independent validation or causal interpretation.

5. Trees must be compared with the linear baseline under the same split, CV and final-test rules.

9 Selected exercises

Motivation. These exercises prove the claim that sorting categorical levels by their mean suffices, show why misclassification error is a poor splitting criterion, and explain why averaging (bagging) helps under squared loss but can hurt a majority vote.

Exercise 4.1 (Ordering the levels of a categorical variable)

A node R contains N observations of a categorical feature with levels $1, \dots, K$, level k having $N_k \geq 1$ observations with mean target $\bar{y}_k$. A binary split sends a nonempty proper subset L of the levels left and the rest right. Show that some split maximizing Δ_{RSS} has the form $L = \{k : \bar{y}_k \leq \tau\}$ for a threshold τ, so only the $K-1$ splits along the sorted level means need to be examined. Show also that for $y_i \in \{0,1\}$ the Gini decrease equals $\frac{2}{N}\Delta_{\text{RSS}}$, so the same holds for binary classification with Gini.

Solution [Supplement].

Step 1. Let $W(L)$ be the within-children sum of squares, so $\Delta_{\text{RSS}}(L) = \text{SS}_{\text{tot}} - W(L)$, and maximizing Δ_{RSS} means minimizing W. Because means minimize sums of squares,

$$W(L) = \min_{c_1,c_2} C(L, c_1, c_2), \qquad C(L, c_1, c_2) = \sum_{i:\,\text{lev}(i)\in L} (y_i - c_1)^2 + \sum_{i:\,\text{lev}(i)\notin L} (y_i - c_2)^2.$$

Step 2. Within level k, $\sum_{i:\,\text{lev}(i)=k}(y_i - c)^2 = Q_k + N_k(\bar{y}_k - c)^2$, where Q_k does not depend on c.

Step 3. Let $L^\star$ be optimal with child means c_1, c_2. Replacing $L^\star$ by its complement gives the same split, so assume $c_1 \leq c_2$. If $c_1 = c_2$, the ANOVA form gives $\Delta_{\text{RSS}}(L^\star) = 0$; since $\Delta_{\text{RSS}} \geq 0$ for every split, all splits are optimal and we are done. So let $c_1 < c_2$, $\tau = \frac{1}{2}(c_1 + c_2)$ and $L' = \{k : \bar{y}_k \leq \tau\}$. By Step 2, level k costs less on the c_1 side iff $(\bar{y}_k - c_1)^2 \leq (\bar{y}_k - c_2)^2$ iff $\bar{y}_k \leq \tau$. Placing every level on its cheaper side,

$$W(L') \leq C(L', c_1, c_2) \leq C(L^\star, c_1, c_2) = W(L^\star).$$

Step 4. L' is a proper split: if it contained no level or all levels, $C(L', c_1, c_2)$ would be the sum of squares of all y_i about a single constant, which is at least SS_{tot}, giving $\Delta_{\text{RSS}}(L^\star) \leq 0$, whereas the ANOVA form gives $\Delta_{\text{RSS}}(L^\star) = \frac{N_1 N_2}{N}(c_1 - c_2)^2 > 0$. Hence L' is optimal and consists of the levels with the smallest means.

Step 5. For $y_i \in \{0,1\}$ with node proportion p, $G = 1 - p^2 - (1-p)^2 = 2p(1-p)$ and $\sum_i (y_i - \bar{y})^2 = Np(1-p)$, so $N\,G(R) = 2\,\text{SS}(R)$ for every node. Therefore

$$G(R) - \frac{N_1}{N}G(R_1) - \frac{N_2}{N}G(R_2) = \frac{2}{N}\{\text{SS}(R) - \text{SS}(R_1) - \text{SS}(R_2)\} = \frac{2}{N}\Delta_{\text{RSS}}.$$

■

Exercise 4.2 (Misclassification error ignores useful splits)

Let $E(R) = 1 - \max_k p_{Rk}$ be the misclassification impurity.
(a) Show that if a class $k^\star$ is a majority class of the parent and of both children, then $N_1 E(R_1) + N_2 E(R_2) = N\,E(R)$: the split has zero gain.

(b) A node has 400 cases of each of two classes. Split A gives children $(300, 100)$ and $(100, 300)$; split B gives $(200, 400)$ and $(200, 0)$. Compare the weighted impurities of the two splits under E and under Gini.

Solution [Supplement].

(a) With class counts n_k in the parent and $n_{r,k}$ in child r, $N\,E(R) = N - n_{k^\star}$ and $N_r E(R_r) = N_r - n_{r,k^\star}$. Adding the children, $N_1 + N_2 - (n_{1,k^\star} + n_{2,k^\star}) = N - n_{k^\star}$.

(b)

Step 1. Under E: split A makes $100 + 100 = 200$ errors, split B makes $200 + 0 = 200$ errors, so both have weighted impurity $200/800 = 0.25$ (parent 0.5).

Step 2. Under Gini, split A: both children have proportions $(\frac{3}{4}, \frac{1}{4})$ and $G = 1 - \frac{9}{16} - \frac{1}{16} = 0.375$, so the weighted impurity is 0.375. Split B: the first child has $(\frac{1}{3}, \frac{2}{3})$, $G = 1 - \frac{1}{9} - \frac{4}{9} = \frac{4}{9}$, weight $\frac{600}{800}$; the second is pure. The weighted impurity is $\frac{3}{4} \cdot \frac{4}{9} = \frac{1}{3} < 0.375$.

Gini prefers B, which creates a pure leaf that needs no further splitting; misclassification error cannot see the difference. Entropy gives the same ranking (0.562 versus 0.477 nats). This is the practical meaning of strict concavity in the impurity theorem. ■

Exercise 4.3 (Aggregation helps under squared loss, not always under a vote)

Let $\varphi(x, \mathcal{D})$ be a predictor fitted on a random training sample $\mathcal{D}$, and $\varphi_A(x) = \mathbb{E}_{\mathcal{D}}\,\varphi(x, \mathcal{D})$ the ideal aggregated predictor (assume $\mathbb{E}_{\mathcal{D}}\varphi(x, \mathcal{D})^2 < \infty$).
(a) Show that for every (x, y), $\mathbb{E}_{\mathcal{D}}\{y - \varphi(x, \mathcal{D})\}^2 = \{y - \varphi_A(x)\}^2 + \mathrm{Var}_{\mathcal{D}}\varphi(x, \mathcal{D})$, and conclude that for a test pair (X, Y) independent of $\mathcal{D}$ the risk of φ_A is at most the expected risk of φ.
(b) For classification let $\varphi(x, \mathcal{D}) \in \{0, 1\}$ and $\varphi_A(x) = \mathbb{I}\{P_{\mathcal{D}}(\varphi(x, \mathcal{D}) = 1) > \frac{1}{2}\}$ (the majority vote over training samples). Give a point x where the vote has a larger error than φ.

Solution [Supplement].

(a)

Step 1. Write $y - \varphi = (y - \varphi_A) + (\varphi_A - \varphi)$. Squaring and taking $\mathbb{E}_{\mathcal{D}}$, the cross term is $2(y - \varphi_A)\mathbb{E}_{\mathcal{D}}(\varphi_A - \varphi) = 0$, which gives the identity.

Step 2. Since (X, Y) is independent of $\mathcal{D}$, apply the identity at (X, Y) and take expectations (Fubini):

$$\mathbb{E}_{X,Y,\mathcal{D}}\{Y - \varphi(X, \mathcal{D})\}^2 = \mathbb{E}_{X,Y}\{Y - \varphi_A(X)\}^2 + \mathbb{E}_X \mathrm{Var}_{\mathcal{D}}\varphi(X, \mathcal{D}) \;\geq\; \mathbb{E}_{X,Y}\{Y - \varphi_A(X)\}^2.$$

The gain is exactly the variance of the base learner over training samples, which is why bagging helps most for unstable learners such as deep trees.

(b) Take x with $P(Y = 1 \mid x) = 1$ and a base learner with $P_{\mathcal{D}}(\varphi(x, \mathcal{D}) = 1) = 0.4$. Its expected error at x is 0.6, while $\varphi_A(x) = 0$ errs with probability 1. A vote amplifies whichever answer the base learner gives more often, so it hurts at points where the base learner is usually wrong; under squared loss, by (a), no such point exists. ■

Chapter 5. Boosting Methods

Predictive Modeling, Lecture 4 (AdaBoost, gradient boosting, XGBoost, LightGBM, CatBoost)

1 From Averaging to Sequential Correction

Random forest (Ch. 4)	Gradient boosting (this chapter)
$\hat{f}_{\mathrm{RF}}(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x)$	$F_M(x) = F_0(x) + \sum_{m=1}^{M} \eta\, h_m(x)$
trees learned **independently** on different bootstrap samples	h_m: a shallow tree learning the **error (residual) direction** left by the previous model
averaging reduces the instability of individual trees	η: the **learning rate**, shrinking each correction to prevent overfitting
every tree looks at the same target Y	small corrections ηh_m accumulate **sequentially**, not independently

Random forests mainly reduce variance; boosting mainly reduces bias by adding many weak, shallow learners. This chapter covers *by which principle* boosting learns on tabular data and *under which conditions* to choose it.

2 AdaBoost

Motivation. Boosting builds a strong classifier from many weak ones by focusing each new learner on the examples the current ensemble gets wrong. AdaBoost does this by reweighting, and it turns out to minimize the exponential loss stagewise.

2.1 Algorithm

Algorithm: AdaBoost (labels $y_i \in \{-1, +1\}$)

Initialize $D_1(i) = 1/N$. For $b = 1, \ldots, B$:

1. **Train**: fit a classifier $h_b(x) \in \{-1, +1\}$ using weights D_b.
2. **Evaluate**: weighted error $\epsilon_b = \sum_{i=1}^{N} D_b(i)\, \mathbb{I}\{h_b(x_i) \neq y_i\}$.
3. **Weight**: $\alpha_b = \frac{1}{2} \log \dfrac{1 - \epsilon_b}{\epsilon_b}$.
4. **Update**: $D_{b+1}(i) = \dfrac{D_b(i) \exp\{-\alpha_b y_i h_b(x_i)\}}{Z_b}$, Z_b the normalizer.

Output: $H(x) = \operatorname{sign}\Big(\sum_{b=1}^{B} \alpha_b h_b(x)\Big)$.

Core mechanism: the more an observation was misclassified by previous models, the larger its weight D_{b+1} in the next round ($\exp(+\alpha_b)$ for errors, $\exp(-\alpha_b)$ for correct cases); the better a model (smaller ϵ_b), the larger its say α_b in the final vote. A learner with $\epsilon_b = \frac{1}{2}$ gets $\alpha_b = 0$; one with $\epsilon_b > \frac{1}{2}$ would get a negative weight (it is flipped).

2.2 AdaBoost is forward stagewise fitting of the exponential loss

Theorem: derivation of α_b and the weight update

Let $F_{b-1} = \sum_{s<b} \alpha_s h_s$ and consider the exponential loss $L(y, F) = e^{-yF}$. Choosing (α, h) to minimize $\sum_i \exp\{-y_i(F_{b-1}(x_i) + \alpha h(x_i))\}$ gives: h_b minimizes the weighted error with weights $D_b(i) \propto e^{-y_i F_{b-1}(x_i)}$, and $\alpha_b = \frac{1}{2} \log \frac{1-\epsilon_b}{\epsilon_b}$. The weights evolve exactly by the AdaBoost update.

Proof [Supplement]. With $w_i = e^{-y_i F_{b-1}(x_i)}$, normalized to $D_b(i)$, and $y_i h(x_i) = \pm 1$,

$$\sum_i D_b(i) e^{-\alpha y_i h(x_i)} = e^{-\alpha}(1-\epsilon) + e^{\alpha}\epsilon = e^{-\alpha} + (e^{\alpha} - e^{-\alpha})\epsilon.$$

For any $\alpha > 0$ this is increasing in the weighted error ϵ, so h_b is the weighted-error minimizer. Differentiating in α: $-e^{-\alpha}(1-\epsilon) + e^{\alpha}\epsilon = 0 \iff e^{2\alpha} = (1-\epsilon)/\epsilon$. Finally $w_i^{\text{new}} = w_i e^{-\alpha_b y_i h_b(x_i)}$, which after normalization is the update rule. ■

Theorem: training-error bound

$Z_b = 2\sqrt{\epsilon_b(1-\epsilon_b)}$ and

$$\frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\{H(x_i) \neq y_i\} \leq \prod_{b=1}^{B} Z_b \leq \exp\Big(-2\sum_{b=1}^{B}\gamma_b^2\Big), \qquad \gamma_b = \tfrac{1}{2} - \epsilon_b.$$

Proof [Supplement]. Plugging α_b into $Z_b = e^{-\alpha_b}(1-\epsilon_b) + e^{\alpha_b}\epsilon_b$ gives $2\sqrt{\epsilon_b(1-\epsilon_b)}$. Unrolling the update, $D_{B+1}(i) = \frac{e^{-y_i F_B(x_i)}}{N\prod_b Z_b}$, and these sum to 1, so $\frac{1}{N}\sum_i e^{-y_i F_B(x_i)} = \prod_b Z_b$. Since $\mathbb{I}\{H(x_i) \neq y_i\} \leq e^{-y_i F_B(x_i)}$, the first inequality follows. Finally $2\sqrt{\epsilon(1-\epsilon)} = \sqrt{1-4\gamma^2} \leq e^{-2\gamma^2}$. ■

So if every weak learner is slightly better than chance ($\gamma_b \geq \gamma > 0$) the training error falls exponentially fast. The population minimizer of the exponential loss is $f^\star(x) = \frac{1}{2}\log\frac{P(Y=1|x)}{P(Y=-1|x)}$ (minimize $pe^{-f} + (1-p)e^{f}$), so AdaBoost estimates half the log-odds.

Example: AdaBoost on a toy case (classic illustration)

Ten points in the plane with decision stumps. Round 1: a vertical stump misclassifies three + points, $\epsilon_1 = 0.30$, $\alpha_1 = \frac{1}{2}\log(0.7/0.3) = 0.42$; the three misclassified points are enlarged in D_2. Round 2: another vertical stump, $\epsilon_2 = 0.21$, $\alpha_2 \approx 0.65$; three − points it misclassifies grow in D_3. Round 3: a horizontal stump, $\epsilon_3 = 0.14$, $\alpha_3 \approx 0.92$. The final classifier

$$H_{\text{final}} = \text{sign}(0.42\, h_1 + 0.65\, h_2 + 0.92\, h_3)$$

is a staircase boundary that no single stump can represent. (With the formula, 0.21 and 0.14 give 0.66 and 0.91; the figures round the weighted errors.)

3 Gradient Boosting Machines

3.1 Algorithm

Algorithm: GBM

Initialize $F_0(x) = \arg\min_\gamma \sum_{i=1}^N L(y_i, \gamma)$. For $b = 1, \dots, B$:

1. **Gradient**: pseudo-residuals $r_{ib} = -\Big[\dfrac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\Big]_{F=F_{b-1}}$.
2. **Fit**: fit a regression tree $h_b(x)$ (leaves R_{jb}) to the targets r_{ib}; optionally set each leaf value by a line search $\gamma_{jb} = \arg\min_\gamma \sum_{x_i \in R_{jb}} L(y_i, F_{b-1}(x_i) + \gamma)$.
3. **Update**: $F_b(x) = F_{b-1}(x) + \eta\, h_b(x)$.

Output: $F_B(x) = F_0(x) + \sum_{b=1}^B \eta\, h_b(x)$.

Core mechanism: if the loss L is squared error, the gradient r_{ib} coincides exactly with the ordinary residual $y_i - F_{b-1}(x_i)$. Unlike AdaBoost, **any differentiable loss** can be used. It is the principle of **gradient descent in function space**: the vector $(F(x_1), \dots, F(x_N))$ is moved in the negative gradient direction, and the tree h_b is the best approximation of that direction that can also be evaluated at new x.

3.2 Loss functions and their pseudo-residuals

Loss	$\Psi(y, f)$	$-\partial\Psi/\partial f$	F_0
squared (L_2)	$\frac{1}{2}(y-f)^2$	$y - f$	mean
absolute (L_1)	$\lvert y-f\rvert$	$\operatorname{sign}(y-f)$	median
Huber	$\frac{1}{2}(y-f)^2$ if $\lvert y-f\rvert \le \delta$; $\delta(\lvert y-f\rvert - \delta/2)$ otherwise	$y-f$ or $\delta\operatorname{sign}(y-f)$	robust center
quantile	$(1-\alpha)\lvert y-f\rvert$ if $y-f \le 0$; $\alpha\lvert y-f\rvert$ if $y-f>0$	α or $-(1-\alpha)$	α-quantile
Bernoulli	$\log(1+e^{-2\tilde{y}f})$, $\tilde{y} \in \{-1, 1\}$	$\dfrac{2\tilde{y}}{1+e^{2\tilde{y}f}}$	$\frac{1}{2}\log\frac{\bar{p}}{1-\bar{p}}$
exponential (AdaBoost)	$e^{-\tilde{y}f}$	$\tilde{y}e^{-\tilde{y}f}$	$\frac{1}{2}\log\frac{\bar{p}}{1-\bar{p}}$

In the Bernoulli loss, f is half the log-odds ($p = 1/(1+e^{-2f})$); with f equal to the log-odds the same loss is the usual logistic log loss with pseudo-residual $y-p$, $y \in \{0, 1\}$. With absolute or Huber loss a few large residuals cannot dominate the trees (Ch. 2).

3.3 Shrinkage, subsampling and early stopping

Small learning rates ($\eta = 0.01$–0.1) need more trees but generalize better: each tree only partially corrects the residual, which acts like regularization along the path. **Stochastic gradient boosting** fits each tree on a random subsample of rows (and/or columns), which reduces correlation and variance. The number of trees B is chosen by **early stopping** on a validation set: stop when validation loss has not improved for a given number of rounds. Learning rate, number of trees, tree depth (interaction order), subsampling and early stopping control the *same* overfitting problem from different places.

3.4 Limitations of the standard GBM

Standard GBM predicts very well but has practical limits: with too many trees it can overfit; it contains no explicit regularization of tree complexity; sequential tree fitting can be computationally inefficient. Modern implementations address these.

4 XGBoost: Regularized Second-Order Boosting

Motivation. Gradient boosting fits each tree to the negative gradient. XGBoost uses a second-order Taylor expansion of the loss, which gives closed-form leaf values and a split gain that includes explicit regularization.

XGBoost improves both the *mathematical optimization* and the *system computation*:

- **Regularization**: GBM has no built-in complexity penalty; XGBoost adds L_1/L_2 penalties of the leaf weights and a per-leaf cost to the objective.
- **Second-order optimization**: GBM follows only the first derivative (gradient) and can converge slowly; XGBoost uses a second-order Taylor expansion, i.e. the curvature (Hessian), to reach the optimum faster and more precisely.

Theorem: optimal leaf weights and split gain

At round t the new tree $f_t(x) = w_{q(x)}$ (leaf index $q(x)$, T leaves) minimizes

$$\mathcal{L}^{(t)} = \sum_i L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \gamma T + \tfrac{1}{2}\lambda \sum_{j=1}^{T} w_j^2.$$

With $g_i = \partial_{\hat{y}} L(y_i, \hat{y}_i^{(t-1)})$, $h_i = \partial^2_{\hat{y}} L(y_i, \hat{y}_i^{(t-1)})$, $G_j = \sum_{i \in I_j} g_i$, $H_j = \sum_{i \in I_j} h_i$, the second-order approximation gives

$$w_j^\star = -\frac{G_j}{H_j + \lambda}, \qquad \tilde{\mathcal{L}}^\star = -\frac{1}{2} \sum_{j=1}^{T} \frac{G_j^2}{H_j + \lambda} + \gamma T,$$

and the gain of splitting a leaf into L and R is

$$\text{Gain} = \frac{1}{2}\Big[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda}\Big] - \gamma.$$

Proof [Supplement]. Taylor: $L(y_i, \hat{y} + f) \approx L(y_i, \hat{y}) + g_i f + \frac{1}{2} h_i f^2$. Dropping constants and grouping by leaves, $\tilde{\mathcal{L}} = \sum_j \{G_j w_j + \frac{1}{2}(H_j + \lambda) w_j^2\} + \gamma T$, a separate quadratic in each w_j; its minimizer is $-G_j/(H_j + \lambda)$ with value $-\frac{1}{2} G_j^2/(H_j + \lambda)$. The gain is the objective of one leaf minus that of two leaves, minus the extra leaf cost γ. ■

Example: what $w^\star$ is for familiar losses

Squared loss $\frac{1}{2}(y - \hat{y})^2$: $g_i = \hat{y}_i - y_i$, $h_i = 1$, so $w_j^\star = \sum_{i \in I_j} (y_i - \hat{y}_i)/(n_j + \lambda)$, a *shrunken* mean residual (Ridge on leaf values). Logistic loss on the log-odds scale: $g_i = p_i - y_i$, $h_i = p_i(1 - p_i)$, so $w_j^\star$ is one Newton step for the leaf. The constraint `min_child_weight` bounds H_j from below; γ prunes splits whose gain is too small.

On the system side: sparsity-aware split finding learns a default direction for missing values, approximate (quantile-sketch) split candidates, column blocks and cache-aware parallel computation.

5 LightGBM

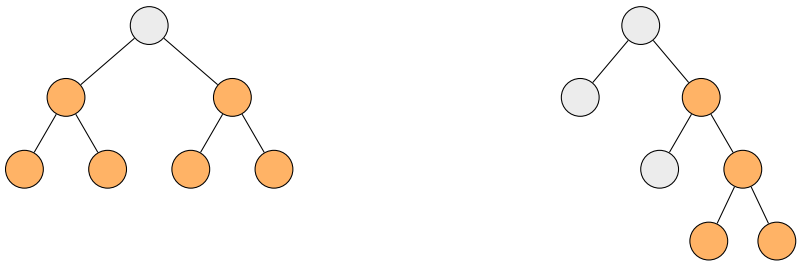
LightGBM is a highly efficient gradient-boosting implementation optimized for large and high-dimensional data:

- **Histogram-based learning**: continuous variables are bucketed into a small number of bins, drastically reducing computation and memory.
- **Leaf-wise tree growth**: the tree is not grown symmetrically; only the leaf whose split reduces loss most is expanded, deeply and selectively.
- **Efficient computation**: combining the two cuts training time and memory dramatically.

Aspect	Standard GBM	LightGBM
split search	pre-sorts every continuous value and scans all of them; bottleneck on large data	integer bins (histogram) cut computation and memory
tree growth	level-wise: grows all nodes of a level symmetrically, even unnecessary ones	leaf-wise: expands only the node that reduces the error most

Derivation: why histograms are cheap

With k bins (default 255), building a node's histogram of (G, H) sums costs $O(\#\text{rows in node})$ per feature, and finding the best split costs only $O(k)$ instead of $O(\#\text{rows})$. Moreover a child's histogram is the parent's minus the sibling's (*histogram subtraction*), so only the smaller child needs to be built. Coarse bins act as mild regularization.



level-wise (GBM, XGBoost default) leaf-wise (LightGBM)

Figure 6: Orange nodes are created by the most recent splits. Level-wise growth splits every node of a depth before going deeper; leaf-wise growth always splits the single leaf with the largest loss reduction, giving deep, unbalanced trees for the same number of leaves.

Remark: leaf-wise growth and overfitting

For the same number of leaves, leaf-wise growth attains lower training loss than level-wise growth, but it produces deep, unbalanced trees that overfit small data; control `num_leaves`, `max_depth`, `min_data_in_leaf`. LightGBM also uses gradient-based one-side sampling (GOSS: keep rows with large $|g_i|$, subsample the rest with reweighting) and exclusive feature bundling (EFB: merge sparse features that are rarely nonzero simultaneously).

6 CatBoost

Motivation. Target statistics for categorical features and ordinary gradient boosting both reuse a sample's own label, which leaks information and biases the fit. CatBoost's ordered scheme computes

everything for a sample from earlier samples only.

6.1 The curse of dimensionality of categorical variables

- GBM: hundreds of categories require one-hot encoding into hundreds of dummy columns; memory and trees become inefficient.
- CatBoost: however many categories, each is replaced *within a single column* by the proportion of target = 1 (target statistics), preventing the explosion of dimensions.

Example: one-hot (GBM) versus target statistics (CatBoost)

Category	Share of target = 1 in that category	CatBoost value (one column)
A	80 of 100 (80%)	$\to 0.80$
B	10 of 40 (25%)	$\to 0.25$
C	100 of 200 (50%)	$\to 0.50$
D	5 of 50 (10%)	$\to 0.10$

Instead of four new columns for A, B, C, D, one continuous column $0.80, 0.25, \ldots$ replaces them.

6.2 Naive target statistics leak the target

Theorem: target leakage of in-sample target encoding

If each row's encoding uses its own target, $\hat{x}_i = \frac{1}{n_{c_i}} \sum_{j:c_j=c_i} y_j$, then for a category observed once the encoded value equals y_i. In the extreme of all categories unique (e.g. an ID), the encoded feature equals the target on the training data and a single split fits it perfectly, while on new data the feature carries no information.

More generally, $\mathbb{E}[\hat{x}_i \mid y_i]$ depends on y_i through the term y_i/n_{c_i}, so the distribution of the encoded feature differs between training rows (which saw their own target) and test rows (*target shift*). CatBoost fixes this in two steps:

- **smoothing with a prior**: $\hat{x}_i = \dfrac{\sum_{j \in \mathcal{P}_i} \mathbb{I}(c_j = c_i)\, y_j + a\, \bar{p}}{\sum_{j \in \mathcal{P}_i} \mathbb{I}(c_j = c_i) + a}$, shrinking rare categories toward the global rate $\bar{p}$;
- **ordered target statistics**: draw a random permutation σ and let $\mathcal{P}_i = \{j : \sigma(j) < \sigma(i)\}$, i.e. each row is encoded using only rows that come *before* it, as in an online stream. A row's own target never enters its encoding, mimicking the test situation.

The same idea, **ordered boosting**, computes the residuals of a row with a model that has not seen that row, removing the "prediction shift" of standard GBM. CatBoost also uses symmetric (oblivious) trees, the same split at every node of a level, which are fast to evaluate and regularize the trees.

7 Summary and Practical Rules

Algorithm	Direction of development
XGBoost	explicit regularization controls overfitting; second-order information (Hessian) gives precise optimization
LightGBM	histograms and leaf-wise growth drastically reduce computation on large numerical/sparse data
CatBoost	ordered target statistics handle leakage and instability of categorical features

Rules for using boosting.

- Boosting adds shallow trees sequentially, updating the predictor in the direction that most reduces the current loss.
- Learning rate, number of iterations, tree complexity, subsampling and early stopping control the same overfitting problem from different places.
- Choose a model not by its name or one best score but by the performance and error structure repeated under a fixed validation procedure (time and group structure of Ch. 1 still apply; early stopping uses a validation fold, not the test set).

8 Selected exercises

Motivation. The exercises show why AdaBoost never reuses the learner it has just fitted, what boosting with a linear base learner does to each direction of the data, and how the XGBoost gain relates to the CART criterion and why its penalty λ already acts as a complexity cost.

Exercise 5.1 (The last weak learner becomes a coin flip)

In AdaBoost with $0 < \epsilon_b < 1$, show that the weighted error of h_b under the updated weights is exactly one half:

$$\sum_{i=1}^{N} D_{b+1}(i)\, \mathbb{I}\{h_b(x_i) \neq y_i\} = \tfrac{1}{2}.$$

Solution [Supplement].

Step 1. Misclassified cases have $y_i h_b(x_i) = -1$ and are multiplied by e^{α_b}; correct cases by $e^{-\alpha_b}$. With $e^{\alpha_b} = \sqrt{(1-\epsilon_b)/\epsilon_b}$, the total unnormalized weight of the errors is $\epsilon_b e^{\alpha_b} = \sqrt{\epsilon_b(1-\epsilon_b)}$ and that of the correct cases is $(1-\epsilon_b)e^{-\alpha_b} = \sqrt{\epsilon_b(1-\epsilon_b)}$.

Step 2. Hence $Z_b = 2\sqrt{\epsilon_b(1-\epsilon_b)}$, and the normalized weight of the errors is $\sqrt{\epsilon_b(1-\epsilon_b)}/Z_b = \frac{1}{2}$.

So under D_{b+1} the learner h_b is no better than chance, and the next round is forced to find a learner that is informative on a different weighting of the cases. ■

Exercise 5.2 (L_2 boosting with a linear smoother)

Use squared loss, $F_0 = \mathbf{0}$ (centered $\mathbf{y}$), and a base learner that maps a target vector r to the fitted vector Sr, where S is a symmetric $N \times N$ matrix with eigenvalues $\mu_1, \ldots, \mu_N \in [0, 1]$ (e.g. the OLS hat matrix, or a Ridge smoother). The fitted vector is updated by $F_b = F_{b-1} + \nu S(\mathbf{y} - F_{b-1})$ with $0 < \nu \leq 1$.
(a) Show that $F_B = \{I - (I - \nu S)^B\}\mathbf{y}$.

(b) With $S = \sum_k \mu_k u_k u_k^\top$ (orthonormal u_k), show that $F_B = \sum_k \{1 - (1 - \nu\mu_k)^B\}(u_k^\top \mathbf{y})\, u_k$, that each factor lies in $[0, 1]$ and increases to $\mathbb{I}\{\mu_k > 0\}$ as $B \to \infty$, and that the degrees of freedom $\mathrm{tr}\{I - (I - \nu S)^B\}$ increase in B.
(c) For $S = H$ a projection, show that $F_B = \{1 - (1 - \nu)^B\} H\mathbf{y}$.

Solution [Supplement].

(a) The pseudo-residuals are the residuals $R_{b-1} = \mathbf{y} - F_{b-1}$. Then $R_b = R_{b-1} - \nu S R_{b-1} = (I - \nu S) R_{b-1}$, so $R_B = (I - \nu S)^B \mathbf{y}$ and $F_B = \mathbf{y} - R_B$.

(b)

Step 1. $I - \nu S = \sum_k (1 - \nu\mu_k) u_k u_k^\top$, so $(I - \nu S)^B = \sum_k (1 - \nu\mu_k)^B u_k u_k^\top$ and, since $I = \sum_k u_k u_k^\top$, the formula for F_B follows from (a).

Step 2. $0 \le 1 - \nu\mu_k \le 1$, with $1 - \nu\mu_k < 1$ iff $\mu_k > 0$. Hence $(1 - \nu\mu_k)^B \in [0, 1]$ is non-increasing in B and tends to 0 if $\mu_k > 0$; it equals 1 for all B if $\mu_k = 0$.

Step 3. $\mathrm{tr}\{I - (I - \nu S)^B\} = \sum_k \{1 - (1 - \nu\mu_k)^B\}$ is a sum of non-decreasing terms.

(c) A projection has eigenvalues 0 and 1, so the factors are $1 - (1 - \nu)^B$ on the range of H and 0 on its complement, i.e. $F_B = \{1 - (1 - \nu)^B\} H\mathbf{y}$.

The number of rounds therefore plays the role of $1/\lambda$ in Ridge: few rounds or a small ν shrink every direction, weakly supported directions (small μ_k) most. In (c) boosting merely shrinks the OLS fit, so the gains of real boosting come from base learners (trees) that change with the residuals. ■

Exercise 5.3 (The XGBoost gain and the CART criterion)

(a) For $L = \frac{1}{2}(y - \hat{y})^2$ with current residuals $r_i = y_i - \hat{y}_i^{(t-1)}$, show that $w_j^\star = n_j \bar{r}_j / (n_j + \lambda)$ and that for $\lambda = \gamma = 0$ the gain equals $\frac{1}{2}\Delta_{\mathrm{RSS}}$ of Ch. 4 computed on the residuals.
(b) Show that for $\lambda = \gamma = 0$ and $h_i > 0$ the gain is never negative, but that for $\lambda > 0$ splitting a leaf into two halves with equal (G, H) has negative gain.

Solution [Supplement].

(a)

Step 1. $g_i = \hat{y}_i - y_i = -r_i$ and $h_i = 1$, so $G_j = -n_j \bar{r}_j$ and $H_j = n_j$, which gives $w_j^\star = n_j \bar{r}_j / (n_j + \lambda)$.

Step 2. With $\lambda = 0$, $G_j^2 / H_j = n_j \bar{r}_j^2$, and $n\bar{r} = n_L \bar{r}_L + n_R \bar{r}_R$ for the parent. Hence

$$\text{Gain} = \tfrac{1}{2}\big(n_L \bar{r}_L^2 + n_R \bar{r}_R^2 - n\bar{r}^2\big) = \tfrac{1}{2}\{n_L(\bar{r}_L - \bar{r})^2 + n_R(\bar{r}_R - \bar{r})^2\},$$

which is half the ANOVA form of Δ_{RSS} for the targets r_i.

(b)

Step 3. By Cauchy–Schwarz,

$$(G_L + G_R)^2 = \Big(\sqrt{H_L}\,\frac{G_L}{\sqrt{H_L}} + \sqrt{H_R}\,\frac{G_R}{\sqrt{H_R}}\Big)^2 \le (H_L + H_R)\Big(\frac{G_L^2}{H_L} + \frac{G_R^2}{H_R}\Big),$$

which is the claim Gain ≥ 0.

Step 4. With $G_L = G_R = g \neq 0$, $H_L = H_R = h$ and $\gamma = 0$,

$$\text{Gain} = \frac{1}{2}\Big[\frac{2g^2}{h + \lambda} - \frac{4g^2}{2h + \lambda}\Big] = -\frac{\lambda g^2}{(h + \lambda)(2h + \lambda)} < 0.$$

The two halves carry no information that the parent lacks (for $\lambda = 0$ their leaf values equal the parent's), but each has only half the curvature, so its weight is shrunk more strongly towards 0.

Thus even with $\gamma = 0$ the penalty λ discourages splits into small leaves whose gradients do not differ enough. ■

Chapter 6. Neural Networks and In-Context Learning for Tabular Data

Predictive Modeling, Lecture 5 (MLP, embeddings, backpropagation, Adam, regularization, TabPFN, TabICL)

1 When Tree Boosting Is Not Enough

Limitation of tree boosting		What neural networks offer
finds splits on given features but does not learn the representation of features	$\longrightarrow$	learns representations needed for prediction through **embeddings** and **hidden layers**
smooth relations may need many splits and trees	$\longrightarrow$	combines several features continuously to represent complex relations
text, image, time series must first be summarized into tabular features	$\longrightarrow$	different input types can be learned in one model
usually re-learns a model from scratch for each new data set	$\longrightarrow$	pre-training and in-context learning reuse information from other problems

$$\underbrace{\text{fixed features} \longrightarrow \text{split learning}}_{\text{tree boosting}} \qquad \underbrace{\text{representation learning} + \text{prediction learning}}_{\text{neural network}}$$

2 The MLP as a Composite Nonlinear Function

2.1 One hidden layer

$$z^{(1)} = W^{(1)}x + b^{(1)} \quad \text{weighted sums of the raw features,}$$
$$a^{(1)} = \mathrm{ReLU}(z^{(1)}) \quad \text{hidden representation after activation,}$$
$$\hat{y} = g\big(W^{(2)}a^{(1)} + b^{(2)}\big) \quad \text{final prediction from the hidden representation.}$$

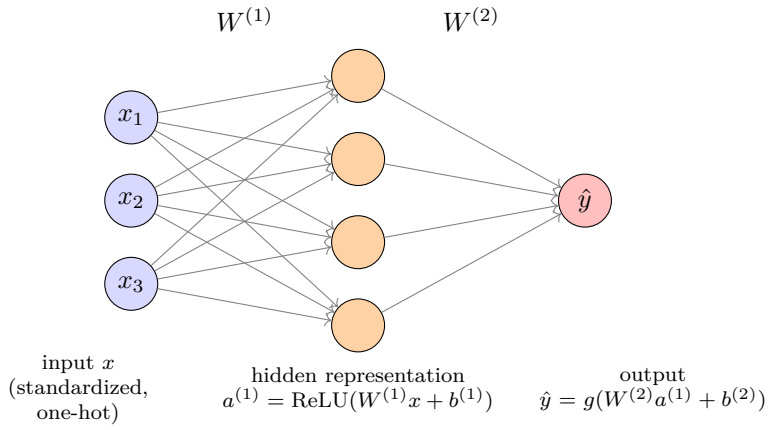


Figure 7: A one-hidden-layer MLP. Each hidden unit forms a weighted sum of all inputs and applies ReLU; the output layer combines the hidden units and applies g (identity, sigmoid or softmax). Backpropagation sends the error signal $\hat{y} - y$ back along the same edges.

Hidden layer: builds linear combinations of the features in many directions; the activation makes piecewise and interaction structure representable; more layers and width enlarge the function space

but also the number of parameters. A one-hidden-layer ReLU network is a continuous piecewise-linear function of x, whose pieces are cut by the hyperplanes $\{w_k^{(1)\top} x + b_k^{(1)} = 0\}$; with enough units it can approximate any continuous function on a compact set (universal approximation).

Output layer: the same hidden representation uses a different loss and output layer depending on the target:

Target	Output g	Loss	Gradient w.r.t. the output score
regression	identity $g(u) = u$	$\frac{1}{2}(y - \hat{y})^2$	$\hat{y} - y$
binary	sigmoid	log loss	$\hat{p} - y$
multiclass	softmax $p_k = e^{u_k} / \sum_l e^{u_l}$	cross-entropy $-\sum_k y_k \log p_k$	$p - y$

Proof of the softmax gradient [Supplement]. $\log p_k = u_k - \log \sum_l e^{u_l}$, so $\partial \log p_k / \partial u_m = \mathbb{I}(k = m) - p_m$. Then $\partial(-\sum_k y_k \log p_k)/\partial u_m = -y_m + p_m \sum_k y_k = p_m - y_m$ since $\sum_k y_k = 1$. ■

These "canonical" pairs give the same simple error signal $\hat{y} - y$ at the output, which is then propagated backwards.

3 Neural-Network Inputs Are Numbers

Suppose the raw record of one customer is:

Raw data	Conversion to numbers	Input
age 52	standardize with mean and SD (training data)	0.8
occupation: teacher	one-hot vector	$(0, 1, 0)$
income: missing	median imputation and missing indicator	$(40, 1)$

So the network actually receives $(0.8,\ 0,\ 1,\ 0,\ 40,\ 1)$. Unlike trees, MLPs are sensitive to scale: unscaled inputs (here 40) dominate the initial weighted sums and slow gradient descent, so continuous features are usually standardized (with training statistics only).

4 Categorical Embeddings Are Learned From Prediction Error

With K occupations, keep a learnable matrix $E \in \mathbb{R}^{K \times d}$ with $d \ll K$. Each occupation becomes a row index (teacher = 1, professor = 2, developer = 3, ...); a teacher input fetches row 1:

$$e_{\text{teacher}} = E[1, :] = (0.7, -0.2, 0.4), \qquad o_{\text{teacher}}^{\top} E = E[1, :],$$

i.e. the lookup is the same computation as a one-hot vector times a weight matrix.

Algorithm: how an embedding is learned

1. Initialize the entries of E with small random numbers.
2. Use the row of the observed category as network input.
3. Compute the error between truth and prediction.
4. Update, by backpropagation, only the row that was used:

$$E[1, :] \leftarrow E[1, :] - \eta \frac{\partial \ell}{\partial E[1, :]}.$$

Theorem: the gradient touches only the used row

If $e = o_k^\top E$ enters the network, then $\dfrac{\partial \ell}{\partial E} = o_k \Big(\dfrac{\partial \ell}{\partial e}\Big)^\top$, a $K \times d$ matrix whose only nonzero row is row k.

Proof [Supplement]. $e_m = \sum_l o_{k,l} E_{lm} = E_{km}$, so $\partial e_m / \partial E_{lm'} = \mathbb{I}(l = k)\mathbb{I}(m = m')$; apply the chain rule. ■

As teacher cases keep arriving, $E[1, :]$ keeps moving in a direction useful for prediction; categories with similar effects end up with nearby rows. The embedding matrix is not given in advance: it is learned together with the other MLP weights from the current training data. Rare categories receive few updates and remain noisy, which is one reason small tabular data favor trees.

5 Training an MLP: Forward, Backward, Update

Algorithm: minibatch training

Initialize the weights θ. For each epoch and each minibatch B_t:

1. **Forward**: compute predictions and the average loss $\widehat{R}_{B_t}(\theta) = \frac{1}{|B_t|} \sum_{i \in B_t} L\{y_i, f_\theta(x_i)\}$.
2. **Backward**: compute the gradient $g_t = \nabla_\theta \widehat{R}_{B_t}(\theta)$.
3. **Update**: move the weights in the opposite direction, $\theta \leftarrow \theta - \eta g_t$.

At the end of each epoch record the validation loss; at the end, restore the weights with the lowest validation loss.

The forward pass computes $\hat{y}_i = f_\theta(x_i)$ with the current weights and compares it with the truth, e.g. $L_i = \frac{1}{2}(y_i - \hat{y}_i)^2$ in regression. The backward pass uses the chain rule to compute the effect of each weight on the loss, $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w}$; the sign of the gradient gives the direction of correction and its size the strength of the influence.

Derivation: backpropagation for one hidden layer

For $z^{(1)} = W^{(1)} x + b^{(1)}$, $a^{(1)} = \mathrm{ReLU}(z^{(1)})$, $\hat{y} = W^{(2)} a^{(1)} + b^{(2)}$, $L = \frac{1}{2}\|y - \hat{y}\|^2$:

$$\delta^{(2)} = \hat{y} - y, \quad \frac{\partial L}{\partial W^{(2)}} = \delta^{(2)} a^{(1)\top}, \quad \frac{\partial L}{\partial b^{(2)}} = \delta^{(2)},$$

$$\delta^{(1)} = \big(W^{(2)\top} \delta^{(2)}\big) \odot \mathbb{I}\{z^{(1)} > 0\}, \quad \frac{\partial L}{\partial W^{(1)}} = \delta^{(1)} x^\top, \quad \frac{\partial L}{\partial b^{(1)}} = \delta^{(1)}.$$

Each layer reuses the error signal δ of the layer above, so all gradients cost about as much as one forward pass. Hidden units with $z \le 0$ pass no gradient ("dead" ReLU units).

5.1 Minibatches and Adam

A minibatch gradient is an unbiased estimate of the full-data gradient, $\mathbb{E}[g_t] = \nabla \widehat{R}(\theta)$ when batches are drawn uniformly; its noise helps escape poor regions and costs $O(|B_t|)$ per step, well suited to GPU matrix operations. Adam adapts the step size per parameter:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2,$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \ \hat{v}_t = \frac{v_t}{1-\beta_2^t}, \quad \theta \leftarrow \theta - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon},$$

(typically $\beta_1 = 0.9$, $\beta_2 = 0.999$). The bias corrections compensate for $m_0 = v_0 = 0$: if g_t has a constant mean μ, then $\mathbb{E}m_t = (1-\beta_1^t)\mu$. The ratio $\hat{m}/\sqrt{\hat{v}}$ is roughly scale-free, so parameters with consistently small gradients still move. Minibatch, Adam: *optimization* devices.

6 The Core Difficulty of Tabular MLPs: Overfitting

An MLP has many parameters and is flexible, so it can learn even the noise of the training data:

training loss keeps decreasing + validation loss rises again $\implies$ overfitting.

Device	Role	Signal to check
small model	limit depth and width, fewer parameters	does even the training loss stay high (underfitting)?
weight decay	penalize large weights	does the train–validation gap shrink?
dropout	randomly switch off some hidden units during training	does validation improve, or does it overfit?
early stopping	restore the weights of the epoch with minimal validation loss	from when does validation loss rise again?

Theorem: weight decay and dropout as regularizers

(i) Adding $\frac{\lambda}{2}\|\theta\|^2$ to the loss changes the gradient step to $\theta \leftarrow (1-\eta\lambda)\theta - \eta g_t$: every step multiplicatively shrinks the weights (Ridge, Ch. 3).
(ii) Inverted dropout replaces a unit a by $\tilde{a} = a\,m/(1-q)$ with $m \sim \text{Bernoulli}(1-q)$, so $\mathbb{E}\tilde{a} = a$. For a linear model $\hat{y} = \beta^\top \tilde{x}$ with dropout on the inputs,

$$\mathbb{E}_m\left[(y - \beta^\top \tilde{x})^2\right] = (y - \beta^\top x)^2 + \frac{q}{1-q}\sum_j \beta_j^2 x_j^2,$$

i.e. dropout behaves like a data-dependent Ridge penalty.

Proof [Supplement]. (i) $\nabla(\widehat{R} + \frac{\lambda}{2}\|\theta\|^2) = g_t + \lambda\theta$.
(ii) $\mathbb{E}[\beta^\top \tilde{x}] = \beta^\top x$ and the m_j are independent with $\text{Var}\{m_j/(1-q)\} = q/(1-q)$, so $\mathbb{E}(y - \beta^\top \tilde{x})^2 = (y - \beta^\top x)^2 + \text{Var}(\beta^\top \tilde{x})$, and $\text{Var}(\beta^\top \tilde{x}) = \sum_j \beta_j^2 x_j^2\, q/(1-q)$. ■

Weight decay, dropout, early stopping: *generalization* devices. Early stopping on a validation fold is also roughly equivalent to L_2 regularization for gradient descent on quadratic losses (fewer steps $\approx$ larger penalty).

7 Trees and MLPs Have Different Inductive Biases

	Tree boosting	MLP
representation	piecewise functions built from stacked feature thresholds	continuous hidden representation from weighted sums and activations
preprocessing	relatively insensitive to the scale of numerical features	sensitive to scaling, missing values, categorical encoding
data size	strong baseline on small/medium tabular data	benefits from sufficient data
computation	many efficient CPU implementations	suited to minibatches and GPU matrix operations

Why MLPs wobble on small tabular data: many weights and embeddings relative to observations; split rules that a tree learns directly must be built from combinations of many neurons; performance changes a lot with scaling and categorical encoding. Tabular features are often heterogeneous, irregular (thresholds, rotation-variant meaning of columns) and contain uninformative columns — properties that favor trees.

8 Another Route: In-Context Learning

Motivation. Instead of training a network on each dataset, a model pretrained on many synthetic tables can predict on a new table directly from the examples given as input.

8.1 Refitting versus conditioning

MLP and GBM: fit anew	TabPFN, TabICL: use as context
$\mathcal{D}_{\text{train}} \xrightarrow{\text{fit}} \theta_{\mathcal{D}} \xrightarrow{x_q} \hat{y}_q$	$\Big(\underbrace{\{(x_i, y_i)\}_{i \in C}}_{\text{labeled cases}}, \underbrace{x_q}_{\text{new case}}\Big) \xrightarrow{\text{fixed pre-trained model}} \hat{y}_q$
a new model is learned for each new data set	uses a model pre-trained on many tabular tasks
weights or trees are updated from the training data	predicts usually without re-training weights
estimates can be unstable with small data	

Both use the new data. MLP and GBM change the model with the new data; in-context learning keeps the model fixed and feeds the new data as input.

8.2 TabPFN pre-trains the learning algorithm itself

Algorithm: TabPFN

Pre-training. (1) Generate many synthetic tables from diverse generating structures (prior over structural causal models, MLPs, trees, noise, . . .).
(2) Split each table into rows with visible labels and rows with hidden labels.
(3) Train to predict the hidden labels from the visible rows.
(4) Repeat over many tabular tasks to learn Θ.

Prediction on a new data set. $(X_{\text{train}}, y_{\text{train}}, x_q) \xrightarrow{\text{fixed } \Phi_\Theta} \hat{p}(y_q)$: the pre-trained weights stay unchanged; a Transformer reads the structure of the whole new table; the inference procedure learned in pre-training computes the distribution of y_q.

TabPFN does not simply compare training rows with the new row; it runs, in one forward pass, the "procedure for inferring a prediction rule from a table" learned during pre-training.

Theorem: what prior-fitted training approximates

Let tasks $\phi \sim \pi$, and given ϕ draw a data set $\mathcal{D}$ and a query (x_q, y_q). If q_Θ ranges over all conditional distributions, the minimizer of the pre-training loss

$$\mathbb{E}_{\phi, \mathcal{D}, x_q, y_q}\big[-\log q_\Theta(y_q \mid x_q, \mathcal{D})\big]$$

is the **posterior predictive distribution** $p(y_q \mid x_q, \mathcal{D}) = \int p(y_q \mid x_q, \phi)\, p(\phi \mid \mathcal{D})\, d\phi$.

Proof [Supplement]. Conditioning on $(\mathcal{D}, x_q)$, the expected loss equals $\mathbb{E}_{\mathcal{D}, x_q}[\mathrm{KL}(p(\cdot \mid x_q, \mathcal{D}) \,\|\, q_\Theta(\cdot \mid x_q, \mathcal{D}))] + \mathbb{E}[H(p(\cdot \mid x_q, \mathcal{D}))]$. The entropy term does not depend on Θ and the

KL term is ≥ 0 with equality iff $q_\Theta = p(\cdot \mid x_q, \mathcal{D})$. The posterior predictive form follows from $p(y_q \mid x_q, \mathcal{D}) = \int p(y_q \mid x_q, \phi) p(\phi \mid x_q, \mathcal{D}) d\phi$ and $p(\phi \mid x_q, \mathcal{D}) = p(\phi \mid \mathcal{D})$. ■

So TabPFN approximates Bayesian prediction under its synthetic prior. It works well when the new table "looks like" a draw from that prior, and its quality depends on the prior's coverage.

8.3 TabICL summarizes rows first, then learns in context

Let the input table be $X = (x_{ij}) \in \mathbb{R}^{n \times m}$ with n rows and m features.

1. **Column-wise interpretation**: the meaning of each cell is understood within its column, $(x_{1j}, \ldots, x_{nj}) \longrightarrow (e_{1j}, \ldots, e_{nj})$. The same number 50 is represented differently in the age column and in the income column.
2. **Row-wise summary**: all cell representations of a row are merged into one vector, $(e_{i1}, \ldots, e_{im}) \longrightarrow h_i \in \mathbb{R}^d$, so every row becomes a fixed-length vector regardless of m.
3. **In-context learning**: the prediction rule of the new task is inferred from the row representations, $(\{(h_i, y_i)\}_{i=1}^n, h_q) \xrightarrow{\text{pre-trained TabICL}} \hat{y}_q$, applying the inference method learned in pre-training to the rows of the current table.

Compressing rows first makes the in-context step scale to more rows than attending over all cells.

Remark: cautions for in-context models

The context must contain only training rows of the current split (otherwise the leakage of Ch. 1 enters through the context). Performance is limited by the context length (number of rows and features the model accepts), by mismatch between the pre-training prior and the real data, and by compute and memory at prediction time, which grows with context size.

9 Summary

Data condition	Candidates to compare first	What to check
small/medium mixed-type data	regularized regression, CatBoost, LightGBM	is performance stable across splits?
enough data or useful categorical embeddings	tree boosting and a small MLP	were preprocessing and overfitting controlled?
pre-training and ICL usable	add TabPFN/TabICL to the existing baselines	do problem type and data size fit, within the compute budget?

MLP. Converts tabular input, through preprocessing and embeddings, into a continuous hidden representation. Minibatch and Adam are optimization devices; weight decay, dropout and early stopping are generalization devices.

TabPFN and TabICL. Pre-trained models that put labeled cases into the context and predict new query rows. They can be useful candidates for small/medium tabular tasks but are sensitive to context length, the pre-training prior and leakage.

10 Selected exercises

Motivation. The exercises explain why weights must be initialized at random, prove the representation claim for one hidden ReLU layer in one dimension, and make precise the statement that early stopping acts like an L_2 penalty.

Exercise 6.1 (Identical hidden units stay identical)

Consider the one-hidden-layer network of the backpropagation box with scalar output $\hat{y} = w^{(2)\top} a^{(1)} + b^{(2)}$ and any differentiable loss $L(y, \hat{y})$, trained by (minibatch) gradient descent; for ReLU use the convention $\mathbb{I}\{z > 0\}$ for its derivative. Suppose that at $t = 0$ hidden units j and k have the same incoming weights and bias, $W^{(1)}_{j\cdot} = W^{(1)}_{k\cdot}$, $b^{(1)}_j = b^{(1)}_k$, and the same outgoing weight, $w^{(2)}_j = w^{(2)}_k$. Show that these equalities hold at every step t.

Solution [Supplement].

Step 1. If the equalities hold at step t, then for every input x, $z_j = z_k$ and $a_j = a_k$.

Step 2. With $\delta^{(2)} = \partial L/\partial \hat{y}$ (a scalar), $\partial L/\partial w^{(2)}_j = \delta^{(2)} a_j = \partial L/\partial w^{(2)}_k$. Further $\delta^{(1)}_j = w^{(2)}_j \delta^{(2)} \mathbb{I}\{z_j > 0\} = \delta^{(1)}_k$, so $\partial L/\partial W^{(1)}_{j\cdot} = \delta^{(1)}_j x^\top = \partial L/\partial W^{(1)}_{k\cdot}$ and $\partial L/\partial b^{(1)}_j = \partial L/\partial b^{(1)}_k$. Averaging over a minibatch preserves these equalities.

Step 3. The update $\theta \leftarrow \theta - \eta g_t$ subtracts equal amounts from equal parameters, so the equalities hold at step $t + 1$; induction on t finishes the proof. The same holds for Adam, whose update of a coordinate depends only on that coordinate's gradient history.

If all hidden units start equal, the network stays equivalent to one with a single hidden unit, whatever its nominal width. Random initialization breaks this symmetry. ■

Exercise 6.2 (One hidden ReLU layer and piecewise-linear functions)

(a) Let $f : [a, b] \to \mathbb{R}$ be continuous and linear on each $[t_{k-1}, t_k]$, $a = t_0 < t_1 < \cdots < t_K = b$, with slope m_k there. Show that for $x \in [a, b]$

$$f(x) = f(a) + m_1 \,\mathrm{ReLU}(x - a) + \sum_{k=1}^{K-1} (m_{k+1} - m_k) \,\mathrm{ReLU}(x - t_k),$$

so a network with K hidden ReLU units represents f exactly on $[a, b]$.
(b) Deduce that every continuous f on $[a, b]$ is a uniform limit of such networks.

Solution [Supplement].

(a)

Step 1. Call the right side g. On $[t_{k-1}, t_k]$, each term $\mathrm{ReLU}(x - t_l)$ is 0 if $l \geq k$ and $x - t_l$ if $l \leq k - 1$, and $\mathrm{ReLU}(x - a) = x - a$; so g is continuous and linear on each piece.

Step 2. The slope of g on (t_{k-1}, t_k) is $m_1 + \sum_{l=1}^{k-1} (m_{l+1} - m_l) = m_k$, and $g(a) = f(a)$ because every ReLU term vanishes at a.

Step 3. $f - g$ is continuous, piecewise linear with slope 0 on every piece, and 0 at a; hence $f = g$ on $[a, b]$. The hidden units have input weight 1 and biases $-a, -t_1, \ldots, -t_{K-1}$; the output weights are $m_1, m_2 - m_1, \ldots, m_K - m_{K-1}$ and the output bias is $f(a)$.

(b)

Step 4. Let $\varepsilon > 0$. f is uniformly continuous, so there is $h > 0$ with $|f(x) - f(x')| < \varepsilon$ whenever $|x - x'| \leq h$. Take equally spaced knots with mesh at most h and let ℓ interpolate f linearly between them.

Step 5. For $x \in [t_{k-1}, t_k]$, $\ell(x)$ is a convex combination of $f(t_{k-1})$ and $f(t_k)$, both within ε of $f(x)$; hence $|\ell(x) - f(x)| < \varepsilon$. By (a), ℓ is a one-hidden-layer ReLU network. ■

Exercise 6.3 (Early stopping and Ridge)

Let $L(\theta) = \frac{1}{2}\|\mathbf{y} - X\theta\|^2$ with thin SVD $X = UDV^\top$, $d_1 \geq \cdots \geq d_r > 0$, and run gradient descent $\theta_{t+1} = \theta_t - \eta\nabla L(\theta_t)$ from $\theta_0 = \mathbf{0}$ with $0 < \eta < 1/d_1^2$.
(a) Show that $\theta_t = V\mathrm{diag}(\phi_t(d_j)/d_j)U^\top\mathbf{y}$ with $\phi_t(d) = 1 - (1 - \eta d^2)^t$.
(b) The Ridge solution of $\frac{1}{2}\|\mathbf{y}-X\theta\|^2+\frac{\lambda}{2}\|\theta\|^2$ is $V\mathrm{diag}(\psi_\lambda(d_j)/d_j)U^\top\mathbf{y}$ with $\psi_\lambda(d) = d^2/(d^2+\lambda)$ (SVD form of Ridge). Show that for $\lambda = 1/(\eta t)$

$$\psi_\lambda(d) \leq \phi_t(d) \leq 2\,\psi_\lambda(d) \qquad \text{for all } d \in (0, d_1].$$

Solution [Supplement].

(a)

Step 1. $\nabla L(\theta) = -X^\top(\mathbf{y} - X\theta)$, so $\theta_{t+1} = (I - \eta X^\top X)\theta_t + \eta X^\top\mathbf{y}$. Both $X^\top\mathbf{y} = VDU^\top\mathbf{y}$ and $X^\top X = VD^2V^\top$ act within the column space of V, and $\theta_0 = \mathbf{0}$, so $\theta_t = Vc_t$ for all t.

Step 2. In these coordinates, $c_{t+1,j} = (1 - \eta d_j^2)c_{t,j} + \eta d_j\, u_j^\top\mathbf{y}$. The fixed point is $c_j^\star = u_j^\top\mathbf{y}/d_j$, and $c_{t,j} - c_j^\star = (1 - \eta d_j^2)^t(0 - c_j^\star)$, so $c_{t,j} = \phi_t(d_j)\, u_j^\top\mathbf{y}/d_j$.

(b)

Step 3. Put $u = \eta d^2 \in (0, 1)$. Then $\psi_\lambda(d) = \eta t d^2/(\eta t d^2 + 1) = ut/(1 + ut)$ and $\phi_t(d) = 1 - (1 - u)^t$.

Step 4. (Lower bound.) $(1 - u)^t \leq e^{-ut} \leq 1/(1 + ut)$, so $\phi_t \geq 1 - 1/(1 + ut) = \psi_\lambda$.

Step 5. (Upper bound.) Bernoulli's inequality $(1 - u)^t \geq 1 - ut$ gives $\phi_t \leq \min(1, ut)$. If $ut \leq 1$, then $\psi_\lambda = ut/(1 + ut) \geq ut/2$; if $ut > 1$, then $\psi_\lambda > \frac{1}{2}$. In both cases $\min(1, ut) \leq 2\psi_\lambda$.

So stopping after t steps keeps directions with $d_j^2 \gg 1/(\eta t)$ almost fully and damps directions with $d_j^2 \ll 1/(\eta t)$ by about $\eta t d_j^2$, as Ridge with $\lambda = 1/(\eta t)$ does: more steps correspond to a smaller penalty. ■

Chapter 7. AutoML and AutoGluon

Predictive Modeling, Lecture 6 (model selection, OOF predictions, fold bagging, stacking, weighted ensembles)

1 When There Are Many Candidates, Model Selection Becomes a Problem

Candidates built in Lectures 1–5: regularized linear models; random forests and boosting; MLPs and tabular pre-trained models. No model is always best on every tabular data set. The decision an ensemble needs: *how should several candidates be combined?*

Stage	What AutoGluon does
fit candidates	trains several candidates: linear models, tree boosting, neural networks, ...
internal validation	non-bagged models on a holdout set, bagged models with OOF predictions
ensemble	combines candidates by fold bagging, stacking, weighted ensembles
leaderboard	compares internal validation score, fit time, prediction time, stack level
final prediction	predicts new cases with the selected single model or ensemble

AutoGluon is not a new algorithm but an AutoML system that **manages the fitting, validation and combination of many algorithms**. The validation principles of Ch. 1 remain the user's responsibility: AutoGluon's internal splits are random/stratified unless told otherwise, so time or group structure and a protected test set must be imposed from outside.

2 Stacking: Candidate Predictions Become New Inputs

For a case x_i, several candidate models each predict:

$$z_i = (h_{\text{linear}}(x_i),\ h_{\text{boost}}(x_i),\ h_{\text{MLP}}(x_i)), \qquad \text{e.g. } z_i = (0.30,\ 0.75,\ 0.60)$$

as binary-class probabilities. A higher-level model uses these predictions as new features:

$$\hat{y}_i = g(x_i, z_i),$$

where z_i are the candidates' predictions and g is a higher-level model that uses the original features and the candidates' predictions. Stacking does not pick the single best model; it **turns the predictions of several models into new features and learns again**, so it can exploit that candidates err on different cases (e.g. trust the MLP where the boosting model is uncertain).

2.1 Why the level-1 inputs must be out-of-fold

Theorem: in-sample stacking rewards overfitting

If g is trained on in-sample predictions $z_i = h(x_i)$ of models that were fitted on the same rows, a model that memorizes the training data (e.g. 1-nearest neighbor, whose in-sample prediction is exactly y_i) looks perfect, and g puts all its weight on it, although its out-of-sample error can be the worst of all candidates.

The training inputs of g must therefore have the same distribution as the inputs it will see at prediction time: predictions on cases *that the candidate did not see*. That is what OOF predictions provide.

3 K-fold Bagging and OOF Predictions

Definition: fold bagging

Split the training data into folds $F_1, \ldots, F_K$ and fit the same model K times, the kth time without F_k, giving $h^{(-1)}, \ldots, h^{(-K)}$ (same algorithm and settings, different training rows).

- **Training data: OOF predictions.** $i \in F_k \Longrightarrow \hat{y}_i^{\text{OOF}} = h^{(-k)}(x_i)$: each case is predicted by the model that did not use it.
- **New cases: averaged prediction.** $x_{\text{new}} \mapsto \frac{1}{K}\sum_{k=1}^{K} h^{(-k)}(x_{\text{new}})$.

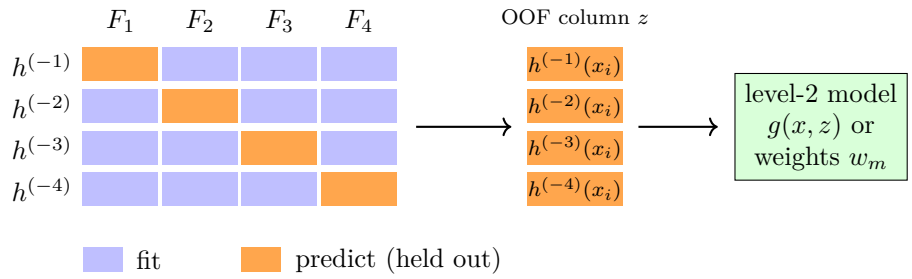


Figure 8: Fold bagging with $K = 4$. Model $h^{(-k)}$ is fitted without fold F_k and predicts only F_k; stacking the four held-out blocks gives an out-of-fold column z for every training row. The level-2 learner (stacker or weighted ensemble) is trained on these honest predictions; for a new case the four fold models' predictions are averaged.

Fold bagging creates OOF predictions and, for new cases, averages the predictions of the K fold models. The OOF vector gives (i) an honest internal validation score of the bagged model and (ii) leakage-free inputs for stacking. Averaging K models trained on overlapping $(K-1)/K$ subsets also lowers variance (Ch. 4). AutoGluon can repeat the fold split with different seeds (`num_bag_sets`) and average again.

Remark: a subtle mismatch

OOF predictions come from single fold models, while test predictions are averages of K fold models, which are slightly smoother. With K moderately large (5–10) the difference is small in practice.

4 Weighted Ensembles

Motivation. Models with similar accuracy often make different errors. Averaging them with weights chosen on held-out predictions reduces variance, and stacking learns the weights.

Example: three candidates

	linear	boosting	MLP
prediction	0.30	0.75	0.60
weight	0.10	0.60	0.30

$$\hat{y} = 0.10(0.30) + 0.60(0.75) + 0.30(0.60) = 0.66.$$

A weighted ensemble $\hat{y} = \sum_{m=1}^{M} w_m h_m(x)$ has one fixed weight per candidate, applied to every case. The weights are chosen to improve the internal validation (OOF) performance.

Algorithm: greedy ensemble selection (Caruana et al.)

Start from the empty ensemble. Repeat R times: add (with replacement) the candidate whose inclusion gives the best OOF score of the *averaged* ensemble. The weight of model m is (number of times selected)$/R$.

The weights are nonnegative and sum to 1; a model can be selected repeatedly, and useless models get weight 0. Selecting with replacement keeps the procedure from overfitting the validation set as easily as an unconstrained regression of y on z.

Theorem: ambiguity decomposition

For weights $w_m \geq 0$, $\sum_m w_m = 1$, and $\hat{y} = \sum_m w_m h_m$, under squared loss

$$(y - \hat{y})^2 = \sum_m w_m (y - h_m)^2 - \sum_m w_m (h_m - \hat{y})^2.$$

Hence the ensemble's error is never larger than the weighted average of the members' errors, and the improvement equals the weighted spread ("ambiguity") of the members' predictions.

Proof [Supplement]. $\sum_m w_m (y - h_m)^2 = \sum_m w_m \{(y - \hat{y}) + (\hat{y} - h_m)\}^2 = (y - \hat{y})^2 + 2(y - \hat{y}) \sum_m w_m (\hat{y} - h_m) + \sum_m w_m (\hat{y} - h_m)^2$, and the middle sum is $\hat{y} - \hat{y} = 0$. ■

Combining pays off when candidates are individually good *and* disagree; combining near-identical models gains nothing. The same convexity argument holds for any convex loss (e.g. log loss) via Jensen's inequality, without the exact decomposition.

5 How the Three Mechanisms Fit Together in AutoGluon

	Fold bagging	Stacking	Weighted ensemble
learning	same model on K folds	higher model g on candidates' OOF predictions z and original features x	repeatedly selects candidates that improve internal validation
new case	average of fold predictions	feeds candidates' predictions and original features to $g(x, z)$	sums candidates' predictions with fixed weights
prediction	$\frac{1}{K} \sum_{k=1}^{K} h^{(-k)}(x)$	$g(x, z)$	$\sum_{m=1}^{M} w_m h_m(x)$
core role	less prediction variance, OOF generation	learns new prediction rules from candidates' error patterns	combines the selected candidates into one final prediction

Fold bagging creates the OOF predictions needed by stacking; stacking and the weighted ensemble use them to combine the candidates' predictions. Stacking can be repeated in layers (`num_stack_levels`): the level-2 models are themselves fold-bagged, produce their own OOF predictions, and a final weighted ensemble combines the top layer. Because the original features are passed to each level ($g(x, z)$), a higher layer can still use them directly.

6 Setting the Search Space

```
predictor = TabularPredictor(label=label, eval_metric=metric)
predictor.fit(train_data, presets=..., time_limit=..., hyperparameters=...)
```

Setting	Role
`eval_metric`	metric used to rank candidates and to select the ensemble
`presets`	bundles the candidate portfolio, ensembling, refitting and cost settings (e.g. quality versus speed)
`time_limit`	approximate time budget for `fit()`
`hyperparameters`	the candidate portfolio: which model families and which settings of each
`num_bag_folds`	number of folds used to create OOF predictions
`num_stack_levels`	number of extra stacking layers that use lower models' predictions as inputs

The metric must match the decision of Ch. 1 (e.g. log loss for probability quality, PR-AUC for rare positives); otherwise the ensemble optimizes the wrong target. The time limit is part of the experiment: report it together with the leaderboard.

Summary

- No single tabular model is always best; combining candidates is a decision in its own right.
- Stacking needs out-of-fold predictions; in-sample predictions reward memorization.
- Fold bagging creates OOF predictions and averages fold models for new cases; stacking learns $g(x, z)$; the weighted ensemble combines selected candidates with fixed convex weights.
- A convex combination's loss is at most the average member loss; gains come from diverse, individually strong candidates.
- AutoGluon manages fitting, validation and combination; the validation design and the final test remain the analyst's responsibility.

7 Selected exercises

Motivation. The exercises compute the best blend of two models, prove that the internal score of the selected candidate is optimistic, and prove the Jensen argument for log loss mentioned after the ambiguity decomposition.

Exercise 7.1 (The best blend of two models)

Let $e_m = Y - h_m(X)$, $m = 1, 2$, with $s_m^2 = \mathbb{E}e_m^2$, $c = \mathbb{E}(e_1 e_2)$ and $\Delta = s_1^2 + s_2^2 - 2c = \mathbb{E}(e_1 - e_2)^2 > 0$. For $w \in \mathbb{R}$ let $h_w = wh_1 + (1 - w)h_2$.
(a) Show that $\mathbb{E}\{Y - h_w(X)\}^2$ is minimized at $w^\star = (s_2^2 - c)/\Delta$, with minimum $M = (s_1^2 s_2^2 - c^2)/\Delta$.
(b) Show that $M \le \min(s_1^2, s_2^2)$, with equality iff $c = \min(s_1^2, s_2^2)$.
(c) Show that $w^\star \in [0, 1]$ iff $c \le \min(s_1^2, s_2^2)$.

Solution [Supplement].

(a) $Y - h_w = we_1 + (1 - w)e_2$, so the risk is

$$q(w) = w^2 s_1^2 + 2w(1 - w)c + (1 - w)^2 s_2^2 = \Delta w^2 - 2(s_2^2 - c)w + s_2^2,$$

a quadratic with leading coefficient $\Delta > 0$. Its minimizer is $w^\star = (s_2^2 - c)/\Delta$, and

$$q(w^\star) = s_2^2 - \frac{(s_2^2 - c)^2}{\Delta} = \frac{s_2^2(s_1^2 + s_2^2 - 2c) - (s_2^2 - c)^2}{\Delta} = \frac{s_1^2 s_2^2 - c^2}{\Delta}.$$

(b) Suppose $s_1^2 \le s_2^2$ (the other case is symmetric). Then

$$s_1^2 - M = \frac{s_1^2(s_1^2 + s_2^2 - 2c) - s_1^2 s_2^2 + c^2}{\Delta} = \frac{(s_1^2 - c)^2}{\Delta} \ge 0,$$

with equality iff $c = s_1^2$.

(c) $w^\star \ge 0$ iff $c \le s_2^2$, and $w^\star \le 1$ iff $s_2^2 - c \le \Delta$ iff $c \le s_1^2$.

Blending helps unless the better model's error is already orthogonal to the difference of the errors ($c = s_1^2$). When the errors are strongly positively correlated ($c > \min s_m^2$), the optimal weight leaves $[0, 1]$ and extrapolates; convex weights, as in greedy ensemble selection, give this up in exchange for stability. ■

Exercise 7.2 (The winner's internal score is optimistic)

Candidates $k = 1, \dots, K$ have true risks R_k and internal validation estimates $\widehat{R}_k$ with $\mathbb{E}\widehat{R}_k = R_k$. Let $\hat{k} = \arg\min_k \widehat{R}_k$.
(a) Show that $\mathbb{E}\widehat{R}_{\hat{k}} \le \min_k R_k \le \mathbb{E}R_{\hat{k}}$.
(b) If $R_1 = \cdots = R_K = R$ and $\widehat{R}_k = R + \tau Z_k$ with $Z_k \overset{\text{iid}}{\sim} N(0,1)$, show that the optimism is $R - \mathbb{E}\widehat{R}_{\hat{k}} = \tau\, \mathbb{E}\max_k Z_k$, and that it equals $\tau/\sqrt{\pi}$ for $K = 2$ and does not decrease in K.

Solution [Supplement].

(a) $\widehat{R}_{\hat{k}} = \min_k \widehat{R}_k \le \widehat{R}_j$ for every j, so $\mathbb{E}\widehat{R}_{\hat{k}} \le \mathbb{E}\widehat{R}_j = R_j$; minimize over j. The second inequality holds because $R_{\hat{k}} \ge \min_k R_k$ pointwise.

(b)

Step 1. $\min_k(R + \tau Z_k) = R - \tau \max_k(-Z_k)$, and $(-Z_k)_k$ has the same law as $(Z_k)_k$, so $\mathbb{E}\widehat{R}_{\hat{k}} = R - \tau\mathbb{E}\max_k Z_k$.

Step 2. For $K = 2$, $\max(Z_1, Z_2) = \frac{1}{2}(Z_1 + Z_2) + \frac{1}{2}|Z_1 - Z_2|$ and $Z_1 - Z_2 \sim N(0, 2)$, so

$$\mathbb{E}\max(Z_1, Z_2) = \tfrac{1}{2}\mathbb{E}|Z_1 - Z_2| = \tfrac{1}{2}\sqrt{2}\sqrt{2/\pi} = 1/\sqrt{\pi} \approx 0.56.$$

Step 3. Adding a candidate can only increase the maximum, so $\mathbb{E}\max_k Z_k$ is non-decreasing in K.

Even when no candidate is better than another, the best internal score looks better than the truth by about half a standard error for two candidates, and more for many. This is why the final number must come from a test set not used for selection. ■

Exercise 7.3 (Convex combinations under log loss)

Let $\ell(y, p) = -y \log p - (1-y)\log(1-p)$ for $y \in \{0, 1\}$, $p \in (0, 1)$, members $p_1, \dots, p_M \in (0, 1)$, weights $w_m \ge 0$ with $\sum_m w_m = 1$, and $\bar{p} = \sum_m w_m p_m$.
(a) Show that $\ell(y, \bar{p}) \le \sum_m w_m \ell(y, p_m)$, with equality iff all p_m with $w_m > 0$ are equal.
(b) Two cases, both with $y = 1$; member 1 predicts $(0.9, 0.1)$ and member 2 predicts $(0.1, 0.9)$. Compare the average log losses of the members and of their equal-weight average.

Solution [Supplement].

(a)

Step 1. $p \mapsto \ell(1, p) = -\log p$ and $p \mapsto \ell(0, p) = -\log(1 - p)$ have second derivatives $1/p^2 > 0$ and $1/(1-p)^2 > 0$ on $(0, 1)$, so both are strictly convex.

Step 2. Jensen's inequality for a strictly convex function gives $\ell(y, \bar{p}) \le \sum_m w_m \ell(y, p_m)$, with equality iff the p_m with $w_m > 0$ coincide.

Applying (a) case by case with the same weights shows that, on any data set, the ensemble's average log loss is at most the weighted average of the members' average log losses.

(b) Each member has average loss $\frac{1}{2}(-\log 0.9 - \log 0.1) \approx \frac{1}{2}(0.105 + 2.303) = 1.204$. The average predicts 0.5 on both cases, with loss $\log 2 \approx 0.693$. The ensemble beats *both* members because their confident errors fall on different cases. ■

Chapter 8. Conditional Independence and Gaussian Graphical Models

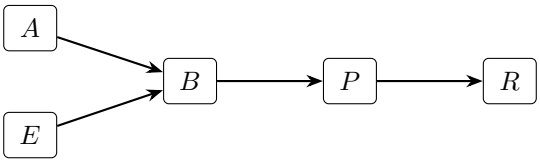
Graphical models for discovering relations, Lecture 1 (partial correlation, precision matrix, GGM, graphical lasso, neighborhood selection)

Roadmap of the causal-discovery part

Lecture	Condition on the data	Relation to be discovered
1 (Ch. 8)	Gaussian observational data	undirected graph of conditional relations
2 (Ch. 9)	linear SEM with given directions	conditional independences a graph implies; limits of direction
3 (Ch. 10)	all main variables observed	the equivalence class of DAGs that data can distinguish
4 (Ch. 11)	hidden common causes allowed	latent confounding, relations and directions compatible with it
5 (Ch. 12)	extra assumption on the error distribution	more specific directions inside the equivalence class
6 (Ch. 13)	further identifiable models	directions for which causal discovery is possible

A running toy health example

The same five-variable health data are reused through all six lectures. The data-generating structure is



A	age	E	amount of exercise
B	body-mass index (BMI)	P	blood pressure
R	disease risk		

i.e. $E \to B \to P \to R$ and $A \to B$. Lecture 1 does not use directions and looks for conditional relations given the remaining variables; Lectures 2–3 study how directions are represented and how far they can be discovered; Lecture 4 adds a hidden lifestyle variable U, and Lecture 6 adds measurement error.

1 Correlation Mixes Direct and Indirect Relations

Let three standardized Gaussian variables be generated by

$$X_1 \sim N(0,1), \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.8X_2 + \epsilon_3, \qquad \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0, 0.36),$$

with $X_1, \epsilon_2, \epsilon_3$ mutually independent, so all three variances equal $0.64 + 0.36 = 1$.

Pair	Correlation	Why the relation arises
X_1, X_2	0.8	X_1 and X_2 are directly connected
X_2, X_3	0.8	X_2 and X_3 are directly connected
X_1, X_3	0.64	the relation is transmitted through X_2

Indeed $\mathrm{Cov}(X_1, X_3) = \mathrm{Cov}(X_1, 0.64X_1 + 0.8\epsilon_2 + \epsilon_3) = 0.64$. The right question is not "is there a correlation?" but "**does a relation remain after accounting for the other variables?**"

2 Conditional Independence

Definition: conditional independence

$X \perp\!\!\!\perp Y \mid Z$ if $p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$ for all z, equivalently $p(x \mid y, z) = p(x \mid z)$: among cases with the same Z, knowing Y gives no further information about X.

Marginal relation: $X_1 \not\perp\!\!\!\perp X_3$ — knowing X_1 changes what we know about X_3 (here Corr $= 0.64$). Conditional relation: $X_1 \perp\!\!\!\perp X_3 \mid X_2$ — comparing only cases with the same X_2, X_1 gives no additional information about X_3:

$$p(x_1, x_3 \mid x_2) = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Conditional independence does not fix causal direction, but it **narrows the candidate edges** of conditional dependence among variables.

3 Partial Correlation

Motivation. Two variables can be correlated only because both depend on a third. Partial correlation removes the linear effect of the other variables and asks whether a direct association remains; zero partial correlations define the edges missing from a Gaussian graphical model.

3.1 Definition by residuals

Adjusting X_1 and X_3 for X_2:

$$X_1 = \alpha_1 + \beta_1 X_2 + r_1, \qquad X_3 = \alpha_3 + \beta_3 X_2 + r_3.$$

Quantity	Meaning
r_1	the part of X_1 not explained by X_2
r_3	the part of X_3 not explained by X_2
$\mathrm{Corr}(r_1, r_3)$	the partial correlation of X_1, X_3 given X_2

$$\rho_{13\cdot 2} = \mathrm{Corr}(r_1, r_3),$$

with population (linear-projection) residuals. For a set S, $\rho_{jk\cdot S}$ is defined in the same way after regressing both variables on X_S.

Theorem: first-order partial correlation

$$\rho_{13\cdot 2} = \frac{\rho_{13} - \rho_{12}\rho_{23}}{\sqrt{(1-\rho_{12}^2)(1-\rho_{23}^2)}}.$$

Proof [Supplement]. Standardize all variables. The projection coefficients are $\beta_1 = \rho_{12}$, $\beta_3 = \rho_{23}$, so $r_1 = X_1 - \rho_{12}X_2$ and $r_3 = X_3 - \rho_{23}X_2$. Then $\mathrm{Cov}(r_1, r_3) = \rho_{13} - \rho_{12}\rho_{23} - \rho_{23}\rho_{12} + \rho_{12}\rho_{23} = \rho_{13} - \rho_{12}\rho_{23}$, $\mathrm{Var}(r_1) = 1 - \rho_{12}^2$ and $\mathrm{Var}(r_3) = 1 - \rho_{23}^2$. ■

Example: a correlation of 0.64 can disappear

With $\rho_{12} = 0.8$, $\rho_{23} = 0.8$, $\rho_{13} = 0.64$,

$$\rho_{13\cdot 2} = \frac{0.64 - (0.8)(0.8)}{\sqrt{(1-0.8^2)(1-0.8^2)}} = 0.$$

Under joint Gaussianity we therefore conclude $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

4 The Precision Matrix

4.1 Zeros of the precision matrix are conditional independences

Let $X = (X_1, \ldots, X_p)^\top \sim N_p(\mu, \Sigma)$ and call $\Omega = \Sigma^{-1}$ the **precision matrix**.

Precision matrix	Interpretation
$\Omega_{jk} = 0$	$X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$
$\Omega_{jk} \neq 0$	a relation between X_j, X_k remains given all the other variables

Theorem: precision matrix and full-order partial correlation

For $X \sim N_p(\mu, \Sigma)$ with $\Sigma \succ 0$ and $\Omega = \Sigma^{-1}$,

$$\rho_{jk\cdot -\{j,k\}} = -\frac{\Omega_{jk}}{\sqrt{\Omega_{jj}\Omega_{kk}}}, \qquad \text{so} \qquad \Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}.$$

Proof [Supplement]. Let $A = \{j, k\}$ and B the rest. For a Gaussian vector, $X_A \mid X_B$ is Gaussian with covariance equal to the Schur complement $\Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$. By the block-inversion formula this Schur complement equals $(\Omega_{AA})^{-1}$, where Ω_{AA} is the 2×2 block of Ω. Hence

$$\mathrm{Cov}(X_A \mid X_B) = \begin{pmatrix} \Omega_{jj} & \Omega_{jk} \\ \Omega_{jk} & \Omega_{kk} \end{pmatrix}^{-1} = \frac{1}{\Omega_{jj}\Omega_{kk} - \Omega_{jk}^2} \begin{pmatrix} \Omega_{kk} & -\Omega_{jk} \\ -\Omega_{jk} & \Omega_{jj} \end{pmatrix},$$

whose correlation is $-\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$. The conditional covariance does not depend on x_B, so this is the partial correlation. For a bivariate Gaussian, zero correlation is equivalent to independence. ■

One matrix inversion gives, for every pair simultaneously, the relation after accounting for all other variables.

4.2 Turning the precision matrix into a graph: the GGM

Definition: Gaussian graphical model

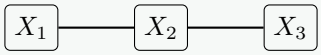
The GGM of $N_p(\mu, \Sigma)$ is the undirected graph on $\{1, \ldots, p\}$ with

$$(j, k) \text{ is an edge} \iff \Omega_{jk} \neq 0.$$

Example: the three-variable chain

$$R = \begin{pmatrix} 1 & 0.8 & 0.64 \\ 0.8 & 1 & 0.8 \\ 0.64 & 0.8 & 1 \end{pmatrix}, \qquad \Omega = R^{-1} \approx \begin{pmatrix} 2.78 & -2.22 & 0 \\ -2.22 & 4.56 & -2.22 \\ 0 & -2.22 & 2.78 \end{pmatrix}.$$

Exactly: $\Omega_{11} = \Omega_{33} = 1/0.36$, $\Omega_{22} = 1.64/0.36$, $\Omega_{12} = \Omega_{23} = -0.8/0.36$, $\Omega_{13} = 0$ (check $R\Omega = I$). So the GGM is



The conditional relations $X_1 - X_2$ and $X_2 - X_3$ remain; $\Omega_{13} = 0$ removes the edge $X_1 - X_3$. The partial correlation of X_1, X_2 given X_3 is $0.8/\sqrt{1.64} \approx 0.62$, smaller than the marginal 0.8 (the same value follows from the first-order formula).

Remark: from pairwise to global statements

The definition uses the *pairwise* Markov property. For a positive density (e.g. a nondegenerate Gaussian) it is equivalent to the *global* Markov property: if a set S separates node sets A and B in the graph (every path from A to B passes through S), then $X_A \perp\!\!\!\perp X_B \mid X_S$ (Hammersley–Clifford). In the chain, $\{X_2\}$ separates X_1 from X_3.

5 Estimating the Graph

5.1 Inverting the sample covariance produces spurious edges

With the sample covariance S, the simplest estimate is $\widehat{\Omega} = S^{-1}$.

Data condition	Problem
finite sample	because of sampling error, entries that are truly 0 are estimated as small nonzero values
many variables	S^{-1} is unstable and most entries are nonzero
$p \geq n$	S is singular and the inverse may not exist

To discover relations we need a **sparse** precision-matrix estimate.

Method	Core idea	Suitable when
partial-correlation tests	compute partial correlations from S^{-1} and test $H_0 : \rho_{jk\cdot -\{j,k\}} = 0$	$p < n$ and sample large enough
graphical lasso	ℓ_1 penalty on the whole precision matrix, some entries estimated as 0	a sparse positive-definite precision matrix is needed
neighborhood selection	Lasso regression of each variable on the rest; choose neighbors variable by variable	many variables, node-wise relations
CLIME	directly estimate a sparse Ω satisfying $S\Omega \approx I$	estimating the precision matrix in high dimension

The edge definition of the GGM is the same; the estimation method depends on the data conditions.

5.2 Testing a zero partial correlation

For Gaussian data, $H_0 : X_j \perp\!\!\!\perp X_k \mid X_S \iff H_0 : \rho_{jk\cdot S} = 0$. Let $\hat{\rho} = \hat{\rho}_{jk\cdot S}$ be the sample partial correlation and $q = |S|$.

Test	Statistic	Reference distribution
partial-correlation t-test	$T = \hat{\rho}\sqrt{\dfrac{n-q-2}{1-\hat{\rho}^2}}$	t_{n-q-2}
Fisher-z test	$Z = \sqrt{n-q-3}\,\operatorname{arctanh}(\hat{\rho})$	approximately $N(0,1)$
conditional distance correlation	$\mathrm{dCor}(X_j, X_k \mid X_{-\{j,k\}})$	p-value by resampling

Derivation: where the degrees of freedom come from

Regressing on q variables plus an intercept uses $q+1$ degrees of freedom, so the residual correlation behaves like an ordinary sample correlation from $n-q$ observations. For an ordinary correlation from m pairs, $T = r\sqrt{(m-2)/(1-r^2)} \sim t_{m-2}$ under $\rho = 0$, and $\operatorname{arctanh}(r) \approx N(\operatorname{arctanh}\rho, 1/(m-3))$ by the delta method with variance stabilization ($\mathrm{Var}(r) \approx (1-\rho^2)^2/m$ and $\frac{d}{d\rho}\operatorname{arctanh}\rho = 1/(1-\rho^2)$). Substituting $m = n-q$ gives the table.

Failing to reject H_0 does *not* prove conditional independence.

5.3 Graphical lasso

Derivation: Gaussian log-likelihood of Ω

For centered data, $\log p(x_i) = \frac{1}{2}\log\det\Omega - \frac{1}{2}x_i^\top\Omega x_i + c$, and $\sum_i x_i^\top\Omega x_i = \operatorname{tr}(\Omega\sum_i x_i x_i^\top) = n\operatorname{tr}(S\Omega)$. So the log-likelihood is $\frac{n}{2}\{\log\det\Omega - \operatorname{tr}(S\Omega)\} + c$, maximized at $\Omega = S^{-1}$ when $S \succ 0$.

The graphical lasso combines Gaussian fit with sparsity:

$$\widehat{\Omega}_\lambda = \arg\min_{\Omega\succ 0}\Big[\operatorname{tr}(S\Omega) - \log\det(\Omega) + \lambda\sum_{j\neq k}|\Omega_{jk}|\Big].$$

Part of the objective	Role
$\operatorname{tr}(S\Omega) - \log\det(\Omega)$	matches the covariance structure of the observed data
$\lambda\sum_{j\neq k}\lvert\Omega_{jk}\rvert$	sets small off-diagonal entries exactly to 0

The objective is convex in Ω; its optimality condition is $\Omega^{-1} - S - \lambda\Gamma = 0$ with $\Gamma_{jk} \in \partial|\Omega_{jk}|$ (and $\Gamma_{jj} = 0$). Consequently $|(\widehat{\Omega}^{-1})_{jk} - S_{jk}| \le \lambda$ for all $j \neq k$, with equality on the estimated edges. The solution is positive definite even when $p \ge n$. λ is chosen by CV log-likelihood, EBIC or stability selection.

5.4 Neighborhood selection

Theorem: conditional distribution of one node

For centered $X \sim N_p(0,\Sigma)$ and $\Omega = \Sigma^{-1}$,

$$X_j \mid X_{-j} \sim N\Big(-\frac{\Omega_{j,-j}}{\Omega_{jj}}X_{-j},\ \frac{1}{\Omega_{jj}}\Big),$$

so the population coefficients of the regression of X_j on the other variables are $\beta_k^{(j)} = -\Omega_{jk}/\Omega_{jj}$, and $\beta_k^{(j)} = 0 \iff \Omega_{jk} = 0$.

Proof [Supplement]. Apply the Schur-complement argument with $A = \{j\}$: $\mathrm{Var}(X_j \mid X_{-j}) = (\Omega_{jj})^{-1}$, and the conditional mean $\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}X_{-j}$ equals $-\Omega_{jj}^{-1}\Omega_{j,-j}X_{-j}$ by the block-inverse identity $\Omega_{j,-j} = -\Omega_{jj}\Sigma_{j,-j}\Sigma_{-j,-j}^{-1}$. ■

In practice the population coefficients are estimated sparsely by the Lasso:

$$\hat{\beta}^{(j)} = \arg\min_{\beta} \Big\{ \frac{1}{2n}\|X_j - X_{-j}\beta\|_2^2 + \lambda\|\beta\|_1 \Big\}.$$

If $\hat{\beta}_k^{(j)} \neq 0$, $j - k$ is an edge candidate. In samples the two regressions (j on k and k on j) may disagree, so they are combined by the **AND** rule (both nonzero) or the **OR** rule (at least one nonzero). Each regression costs one Lasso, which parallelises over nodes and scales to large p; the price is that the result is a graph, not a positive-definite $\widehat{\Omega}$.

5.5 CLIME

$$\widehat{\Omega} = \arg\min_{\Omega} \|\Omega\|_1 \quad \text{subject to} \quad \|S\Omega - I\|_\infty \leq \lambda,$$

a linear program that splits into one column at a time; the estimate is symmetrized afterwards. It estimates the sparsest matrix that is approximately an inverse of S.

6 GGM Edges Are Conditional Associations, Not Causal Directions

Interpretable from a GGM	Not interpretable from a GGM alone
Does a relation remain after accounting for the other observed variables?	Which variable is the cause and which the effect?
What are the sign and size of the partial correlation?	Is an edge a direct causal effect?
Which variables are conditionally connected to many others?	Is there a hidden common cause?

For example, $X_j - X_k$ means that the two variables remain related after accounting for the remaining observed variables. It means neither $X_j \to X_k$ nor $X_j \leftarrow X_k$. Worse, conditioning on a common *effect* can create an edge between two independent causes (Ch. 9). To infer directions we need DAGs and additional structure-learning assumptions.

Summary

- Correlation mixes direct and indirect relations; conditional independence asks what remains given the other variables.
- Partial correlations are correlations of residuals; for Gaussians, $\Omega_{jk} = 0 \iff X_j \perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ and $\rho_{jk\cdot\mathrm{rest}} = -\Omega_{jk}/\sqrt{\Omega_{jj}\Omega_{kk}}$.
- The GGM puts an edge where $\Omega_{jk} \neq 0$; it is estimated by partial-correlation tests, graphical lasso, neighborhood selection or CLIME.
- GGM edges are undirected conditional associations; directions require the DAG machinery of the next chapters.

7 Selected exercises

Motivation. The proofs above rest on the block-inverse identity and on Gaussian conditioning, which are proved here; the remaining exercises find the penalty at which the graphical lasso returns no edge and connect partial correlations with the two node-wise regressions.

Exercise 8.1 (Block inverse and Gaussian conditioning)

Let $\Sigma \succ 0$ be partitioned into blocks A, B, $\Omega = \Sigma^{-1}$, and $M = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$.
(a) Show that $M \succ 0$, $\Omega_{AA} = M^{-1}$ and $\Omega_{AB} = -\Omega_{AA}\Sigma_{AB}\Sigma_{BB}^{-1}$.
(b) For $X \sim N_p(\mu, \Sigma)$ show that $X_A \mid X_B = x_B \sim N\big(\mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(x_B - \mu_B),\, M\big)$.

Solution [Supplement].

(a)

Step 1. For $v \neq 0$ let $u = (v^\top, -(\Sigma_{BB}^{-1}\Sigma_{BA}v)^\top)^\top \neq 0$. Expanding, $u^\top \Sigma u = v^\top M v$, which is positive since $\Sigma \succ 0$. So $M \succ 0$.

Step 2. Write $K = \Sigma_{AB}\Sigma_{BB}^{-1}$ and consider

$$\Omega' = \begin{pmatrix} M^{-1} & -M^{-1}K \\ -K^\top M^{-1} & \Sigma_{BB}^{-1} + K^\top M^{-1} K \end{pmatrix}.$$

Using $\Sigma_{AA} - K\Sigma_{BA} = M$ and $\Sigma_{BB}K^\top = \Sigma_{BA}$, the four blocks of $\Sigma\Omega'$ are

$$\begin{aligned} \Sigma_{AA}M^{-1} - \Sigma_{AB}K^\top M^{-1} &= MM^{-1} = I, \\ -\Sigma_{AA}M^{-1}K + \Sigma_{AB}\Sigma_{BB}^{-1} + \Sigma_{AB}K^\top M^{-1}K &= -MM^{-1}K + K = 0, \\ \Sigma_{BA}M^{-1} - \Sigma_{BB}K^\top M^{-1} &= 0, \\ -\Sigma_{BA}M^{-1}K + I + \Sigma_{BB}K^\top M^{-1}K &= I. \end{aligned}$$

Hence $\Omega = \Omega'$, which gives $\Omega_{AA} = M^{-1}$ and $\Omega_{AB} = -M^{-1}K = -\Omega_{AA}\Sigma_{AB}\Sigma_{BB}^{-1}$.

(b)

Step 3. Let $R = X_A - \mu_A - K(X_B - \mu_B)$. Then $\mathrm{Cov}(R, X_B) = \Sigma_{AB} - K\Sigma_{BB} = 0$ and

$$\mathrm{Var}(R) = \Sigma_{AA} - K\Sigma_{BA} - \Sigma_{AB}K^\top + K\Sigma_{BB}K^\top = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA} = M.$$

Step 4. (R, X_B) is a linear transform of X, hence jointly Gaussian; uncorrelated jointly Gaussian blocks are independent. So $R \sim N(0, M)$ independently of X_B, and given $X_B = x_B$, $X_A = \mu_A + K(x_B - \mu_B) + R$ has the stated law. The conditional covariance $M = (\Omega_{AA})^{-1}$ does not depend on x_B. ∎

Exercise 8.2 (When the graphical lasso returns no edge)

Let $S_{jj} > 0$ for all j and $\lambda \geq \max_{j \neq k} |S_{jk}|$, $\lambda > 0$. Show that $\widehat{\Omega}_\lambda = \mathrm{diag}(1/S_{11}, \ldots, 1/S_{pp})$, and that this minimizer is unique.

Solution [Supplement].

Step 1. (Strict convexity.) For $\Omega \succ 0$ and a symmetric direction $\Delta \neq 0$,

$$\frac{d^2}{dt^2}\{-\log\det(\Omega + t\Delta)\}\Big|_{t=0} = \mathrm{tr}(\Omega^{-1}\Delta\Omega^{-1}\Delta) = \|\Omega^{-1/2}\Delta\Omega^{-1/2}\|_F^2 > 0.$$

So $-\log\det$ is strictly convex on the positive definite cone, and the objective $f(\Omega) = \operatorname{tr}(S\Omega) - \log\det\Omega + \lambda\sum_{j\neq k}|\Omega_{jk}|$ is strictly convex. It has at most one minimizer, and any point at which 0 is a subgradient is that minimizer.

Step 2. (Optimality condition.) The gradient of $\operatorname{tr}(S\Omega) - \log\det\Omega$ is $S - \Omega^{-1}$, so $0 \in \partial f(\Omega)$ means $\Omega^{-1} - S - \lambda\Gamma = 0$ with $\Gamma_{jj} = 0$, $\Gamma_{jk} = \operatorname{sign}(\Omega_{jk})$ if $\Omega_{jk} \neq 0$ and $\Gamma_{jk} \in [-1, 1]$ otherwise.

Step 3. (Check at the diagonal candidate.) For $\Omega_0 = \operatorname{diag}(1/S_{jj})$, $\Omega_0^{-1} = \operatorname{diag}(S_{jj})$. Diagonal entries: $S_{jj} - S_{jj} = 0$. Off-diagonal entries: $\Omega_{0,jk} = 0$, and $\Gamma_{jk} = -S_{jk}/\lambda$ lies in $[-1, 1]$ by assumption and makes $0 - S_{jk} - \lambda\Gamma_{jk} = 0$.

So $\lambda_{\max} = \max_{j\neq k}|S_{jk}|$ is the start of the graphical-lasso path, the analogue of the Lasso's $\lambda_{\max}$; for standardized data it is the largest absolute sample correlation. ■

Exercise 8.3 (Partial correlation from the two node-wise regressions)

With $\beta_k^{(j)}$ the population coefficient of X_k in the regression of X_j on X_{-j}, show that for $j \neq k$

$$\rho^2_{jk\cdot-\{j,k\}} = \beta_k^{(j)}\beta_j^{(k)}, \qquad \beta_k^{(j)} = \rho_{jk\cdot-\{j,k\}}\sqrt{\frac{\operatorname{Var}(X_j \mid X_{-j})}{\operatorname{Var}(X_k \mid X_{-k})}},$$

so $\beta_k^{(j)}$, $\beta_j^{(k)}$ and $\rho_{jk\cdot-\{j,k\}}$ have the same sign and vanish together. Conclude that the AND and OR rules agree in the population.

Solution [Supplement].

Step 1. By the node-wise theorem, $\beta_k^{(j)} = -\Omega_{jk}/\Omega_{jj}$ and $\beta_j^{(k)} = -\Omega_{kj}/\Omega_{kk}$, with $\Omega_{jj}, \Omega_{kk} > 0$ as diagonal entries of a positive definite matrix.

Step 2. The product is $\Omega_{jk}^2/(\Omega_{jj}\Omega_{kk}) = \rho^2_{jk\cdot-\{j,k\}}$ by the precision-matrix theorem.

Step 3. $-\Omega_{jk} = \rho_{jk\cdot-\{j,k\}}\sqrt{\Omega_{jj}\Omega_{kk}}$, so $\beta_k^{(j)} = \rho_{jk\cdot-\{j,k\}}\sqrt{\Omega_{kk}/\Omega_{jj}}$, and $1/\Omega_{jj} = \operatorname{Var}(X_j \mid X_{-j})$, $1/\Omega_{kk} = \operatorname{Var}(X_k \mid X_{-k})$.

In samples the two Lasso regressions estimate these quantities with different errors and may disagree; the AND/OR choice matters only through estimation error. ■

Chapter 9. Gaussian Linear SEMs and Directed Relations

Graphical models, Lecture 2 (DAGs, chain/fork/collider, d-separation, faithfulness, Markov equivalence, CPDAG)

1 From Undirected to Directed Graphs

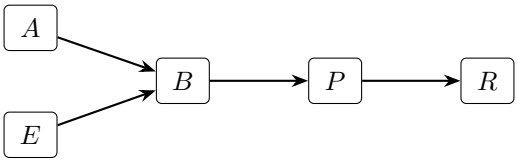
Lecture 1: conditional-dependence graph	Lecture 2: directed graph
$X_j - X_k \iff X_j \not\perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ finds relations that remain after accounting for the others edges but no directions	$X_k \to X_j$ expresses each arrow by a generating equation per variable reads the conditional independences the graph creates; checks which directions data cannot distinguish

The key question: **how far can the direction of relations be inferred from Gaussian observational data?**

2 A Gaussian Linear SEM Is a Bundle of Regressions on a DAG

2.1 The health example as equations

Consider the generating relations $A \to B \leftarrow E$ and $B \to P \to R$:



$$A = \epsilon_A, \quad E = \epsilon_E, \quad B = \beta_{AB}A + \beta_{EB}E + \epsilon_B, \quad P = \beta_{BP}B + \epsilon_P, \quad R = \beta_{PR}P + \epsilon_R,$$
$$\epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma_j^2).$$

Each variable is a linear combination of its parents plus its own error; with independent Gaussian errors the whole vector is multivariate Gaussian.

Definition: DAG and Gaussian linear SEM

A directed acyclic graph (DAG) G has directed edges and no directed cycle. With $\text{pa}(j)$ the parents of node j,

$$X_j = \sum_{k \in \text{pa}(j)} \beta_{kj} X_k + \epsilon_j, \qquad \epsilon_1, \ldots, \epsilon_p \text{ independent}, \ \epsilon_j \sim N(0, \sigma_j^2).$$

Variable	Parents	Local regression
A, E	none	$A = \epsilon_A$, $E = \epsilon_E$
B	A, E	$B = \beta_{AB}A + \beta_{EB}E + \epsilon_B$
P	B	$P = \beta_{BP}B + \epsilon_P$
R	P	$R = \beta_{PR}P + \epsilon_R$

For instance exercise E does not enter the equation of blood pressure P directly, but is related to P through the path $E \to B \to P$.

2.2 Matrix form and the implied covariance

Let B be the $p \times p$ matrix with $B_{jk} = \beta_{kj}$ (row j holds the coefficients of the equation of X_j) and $D = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_p^2)$. Then $X = BX + \epsilon$, and because G is acyclic, B can be permuted to be strictly lower triangular, so $I - B$ is invertible:

$$X = (I - B)^{-1}\epsilon, \qquad \Sigma = (I - B)^{-1} D (I - B)^{-\top}, \qquad \Omega = \Sigma^{-1} = (I - B)^{\top} D^{-1} (I - B).$$

The density factorizes along the DAG (**Markov factorization**):

$$p(x_1, \ldots, x_p) = \prod_{j=1}^{p} p(x_j \mid x_{\mathrm{pa}(j)}).$$

Theorem: the GGM of an SEM is its moral graph

For $j \neq k$,

$$\Omega_{jk} = -\frac{B_{jk}}{\sigma_j^2} - \frac{B_{kj}}{\sigma_k^2} + \sum_{l \neq j,k} \frac{B_{lj} B_{lk}}{\sigma_l^2}.$$

Hence $\Omega_{jk} \neq 0$ only if j, k are adjacent in G or have a common child; generically (no cancellation) exactly then. The GGM of Ch. 8 is the **moral graph**: the skeleton plus edges "marrying" the parents of each child.

Proof [Supplement]. $\Omega_{jk} = \sum_l (I - B)_{lj}(I - B)_{lk}/\sigma_l^2$. The term $l = j$ gives $(1)(-B_{jk})/\sigma_j^2$, the term $l = k$ gives $(-B_{kj})(1)/\sigma_k^2$, and $l \neq j, k$ gives $B_{lj}B_{lk}/\sigma_l^2$, nonzero only if l is a child of both. ■

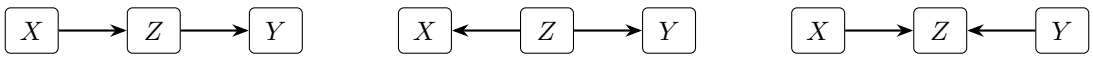
In the health example A and E are independent, yet the GGM contains $A - E$, because both are parents of B: conditioning on B (among all other variables) connects them. This is the first warning that undirected conditional relations are not causal edges.

3 Chains, Forks and Colliders

Motivation. Every path in a DAG is built from three local patterns. Chains and forks transmit association unless the middle variable is conditioned on; colliders block it unless conditioned on. These rules are the basis of d-separation and of all constraint-based discovery.

Structure	Graph	Marginally	Conditioning on Z
chain	$X \to Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
fork	$X \leftarrow Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
collider	$X \to Z \leftarrow Y$	$X \perp\!\!\!\perp Y$	X and Y related

These are the typical patterns observed when the Markov condition and faithfulness hold (§4). Conditioning **closes** chains and forks and **opens** colliders.



chain: open given Z: blocked fork: open given Z: blocked collider: blocked given Z: open

Figure 9: The three elementary paths and what conditioning on the middle node does to the flow of association between X and Y. Only the collider reverses the pattern, which is why v-structures are detectable from data. Conditioning on a *descendant* of a collider also opens it.

Theorem: the three patterns in a Gaussian SEM

Let all errors be independent with variances 1 and coefficients $a, b \neq 0$.

1. Chain $Z = aX + \epsilon_Z$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.
2. Fork $X = aZ + \epsilon_X$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.
3. Collider $Z = aX + bY + \epsilon_Z$ with $X \perp\!\!\!\perp Y$: $\mathrm{Cov}(X, Y) = 0$ but $\rho_{XY \cdot Z} = -\dfrac{ab}{\sqrt{(1 + b^2)(1 + a^2)}} \neq 0$.

Proof [Supplement]. (1) $Y = abX + b\epsilon_Z + \epsilon_Y$ so $\mathrm{Cov}(X, Y) = ab$; the residual of Y on Z is ϵ_Y, independent of (X, Z), so the partial correlation is 0.

(2) $\mathrm{Cov}(X, Y) = ab\mathrm{Var}Z$; residuals of X, Y on Z are ϵ_X, ϵ_Y, independent.

(3) $\mathrm{Cov}(X, Y) = 0$. With $\mathrm{Var}Z = a^2 + b^2 + 1$, $\mathrm{Cov}(X, Z) = a$, $\mathrm{Cov}(Y, Z) = b$, the residual covariance is $0 - ab/(a^2 + b^2 + 1)$ and the residual variances are $1 - a^2/(a^2 + b^2 + 1) = (1 + b^2)/(a^2 + b^2 + 1)$ and $(1 + a^2)/(a^2 + b^2 + 1)$; divide. ■

In a collider, telling us that the effect Z has a given value makes information about one cause change what we know about the other ("explaining away"): if BMI is high and age is low, exercise is probably low. **The change of conditional independence is the key information that distinguishes a collider from the other two structures.**

4 d-Separation, the Markov Condition and Faithfulness

Definition: d-separation

A path between X and Y is **blocked** by a set S if it contains (i) a non-collider (chain or fork node) that is in S, or (ii) a collider that is not in S and has no descendant in S. X and Y are d-separated by S, written $X \perp_G Y \mid S$, if every path between them is blocked.

Definition: causal Markov condition and faithfulness

Markov: $X \perp_G Y \mid S \Rightarrow X \perp\!\!\!\perp Y \mid S$ in the distribution (equivalent to the Markov factorization). **Faithfulness**: the converse, $X \perp\!\!\!\perp Y \mid S \Rightarrow X \perp_G Y \mid S$: every conditional independence in the distribution is implied by the graph.

Under both, the conditional independences of the data are *exactly* the d-separations of G, which is what structure learning (Ch. 10) exploits.

Example: how faithfulness fails — cancellation

Let $X \sim N(0, 1)$, $Z = aX + \epsilon_Z$, $Y = cX + bZ + \epsilon_Y$. Then $\mathrm{Cov}(X, Y) = c + ab$. If $c = -ab$, the direct and indirect effects cancel and $X \perp\!\!\!\perp Y$ although $X \to Y$ is an edge. The set of such coefficient values has Lebesgue measure zero, which is the usual justification of faithfulness; near-cancellation, however, makes tests unreliable in finite samples.

Example: d-separation in the health example

$A \perp E$ (the only path $A \to B \leftarrow E$ has an unconditioned collider), but $A \not\perp E \mid P$ (P is a descendant of the collider B). $E \perp R \mid B$ and $E \perp R \mid P$ (the chain is blocked). $A \perp P \mid B$.

5 Markov Equivalence

5.1 Directions with the same conditional independences cannot be distinguished

The three DAGs

$$X_1 \to X_2 \to X_3, \qquad X_1 \leftarrow X_2 \to X_3, \qquad X_1 \leftarrow X_2 \leftarrow X_3$$

all imply the same conditional independence $X_1 \perp\!\!\!\perp X_3 \mid X_2$ and nothing else. Hence from observational data alone the three directions cannot be told apart. DAGs that imply the same conditional independences form a **Markov equivalence class**.

Theorem: the three DAGs define the same Gaussian model

A Gaussian distribution on (X_1, X_2, X_3) factorizes according to one of the three DAGs iff $X_1 \perp\!\!\!\perp X_3 \mid X_2$; so all three describe exactly the same set of covariance matrices.

Proof [Supplement]. If $X_1 \perp\!\!\!\perp X_3 \mid X_2$ then $p = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_2)$ (first DAG). By Bayes' rule $p(x_1)p(x_2 \mid x_1) = p(x_2)p(x_1 \mid x_2)$, giving $p(x_2)p(x_1 \mid x_2)p(x_3 \mid x_2)$ (fork), and similarly $p(x_2 \mid x_3)p(x_3)$ replaces $p(x_2)p(x_3 \mid x_2)$ (third DAG). Conversely each factorization implies the conditional independence. Each conditional of a Gaussian is a Gaussian linear regression, so every factorization is a Gaussian linear SEM. ■

The collider $X_1 \to X_2 \leftarrow X_3$ has a different pattern, $X_1 \perp\!\!\!\perp X_3$ but $X_1 \not\perp\!\!\!\perp X_3 \mid X_2$, so it *can* be distinguished.

5.2 Skeleton and unshielded colliders

Definition: skeleton and v-structure

The **skeleton** is the adjacency structure left after deleting arrowheads: $X \to Z \to Y \mapsto X - Z - Y$. An **unshielded collider** (v-structure) is $X \to Z \leftarrow Y$ with X and Y non-adjacent.

Theorem: Verma–Pearl characterization

Two DAGs are Markov equivalent iff they have the same skeleton and the same unshielded colliders.

Idea [Supplement]. Adjacency is detectable: X, Y are adjacent iff no set d-separates them (an edge is never blocked; for non-adjacent nodes, the parents of the later one in a topological order d-separate them). Given the skeleton, an unshielded triple $X - Z - Y$ is a collider iff Z is *not* in any set separating X and Y, which again is a CI statement. Conversely, if skeletons and v-structures agree, one shows that every path is blocked by the same sets in both graphs. ■

Without further distributional assumptions, observational data usually identify not one DAG but this equivalence class.

5.3 CPDAG

Definition: CPDAG

The completed partially directed acyclic graph of an equivalence class has a directed edge $X \to Y$ if every DAG in the class contains $X \to Y$, and an undirected edge $X - Y$ if the direction varies within the class.

For instance the CPDAG $X - Z - Y$ represents $X \to Z \to Y$, $X \leftarrow Z \to Y$ and $X \leftarrow Z \leftarrow Y$ together; the collider $X \to Z \leftarrow Y$ is its own class and its CPDAG has both arrows. For the health example, the CPDAG is $A \to B \leftarrow E$, $B \to P \to R$: the v-structure at B fixes $A \to B$ and $E \to B$, and then $B \to P$ is forced (otherwise $A \to B \leftarrow P$ would be a new v-structure), and likewise $P \to R$. **An undirected edge does not mean "no relation"; it means the direction could not be distinguished.**

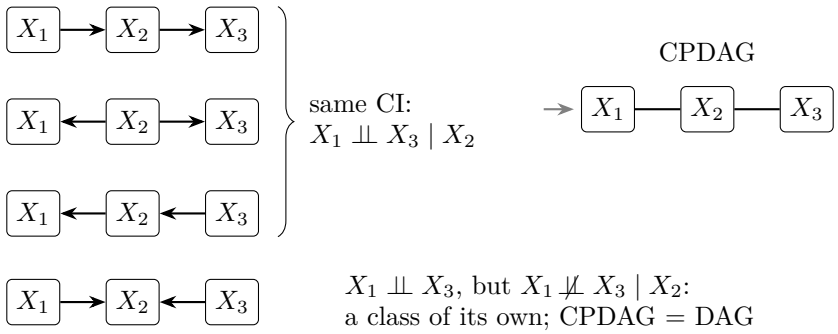


Figure 10: Markov equivalence. The chain, fork and reversed chain share the skeleton and have no v-structure, so they imply the same conditional independences; their CPDAG keeps both edges undirected. The collider has a v-structure and is identified.

Summary

- A Gaussian linear SEM writes each variable as a regression on its parents in a DAG; $\Sigma = (I - B)^{-1} D (I - B)^{-\top}$ and the GGM is the moral graph.
- Chains and forks are closed by conditioning on the middle node; colliders are opened. d-separation formalizes this.
- Under the Markov condition and faithfulness, conditional independences equal d-separations.
- Markov-equivalent DAGs share skeleton and v-structures; Gaussian observational data identify only this class, summarized by the CPDAG.

The key question of the next chapters: in Gaussian observational data, **how far can the direction of relations be inferred**, and with which algorithms?

6 Selected exercises

Motivation. The first exercise shows that each structural equation is an ordinary regression on the parents and yields the local Markov property; the second checks the rule that conditioning on a descendant of a collider opens it; the third counts the Markov equivalence class of a path.

Exercise 9.1 (Structural errors and non-descendants)

In a Gaussian linear SEM on a DAG G, let $\mathrm{nd}(j)$ be the set of non-descendants of j other than j.
(a) Show that ϵ_j is independent of $X_{\mathrm{nd}(j)}$.
(b) Deduce that the population least-squares regression of X_j on $X_{\mathrm{pa}(j)}$ has coefficients β_{kj} and residual variance σ_j^2, and that $X_j \perp\!\!\!\perp X_{\mathrm{nd}(j)\setminus\mathrm{pa}(j)} \mid X_{\mathrm{pa}(j)}$ (local Markov property).

Solution [Supplement].

(a)

Step 1. Order the variables topologically; then B is strictly lower triangular, $B^p = 0$, and $(I-B)^{-1} = \sum_{m=0}^{p-1} B^m$. The entry $(B^m)_{kl}$ is a sum over directed paths $l \to \cdots \to k$ of length m of products of edge coefficients, so $\{(I-B)^{-1}\}_{kl} \neq 0$ only if $l = k$ or l is an ancestor of k.

Step 2. If $k \in \mathrm{nd}(j)$, then $j \neq k$ and j is not an ancestor of k, so $X_k = \sum_l \{(I-B)^{-1}\}_{kl}\epsilon_l$ involves only $(\epsilon_l)_{l\neq j}$. The errors are independent, so ϵ_j is independent of $(\epsilon_l)_{l\neq j}$ and hence of $X_{\mathrm{nd}(j)}$.

(b)

Step 3. A parent of j cannot be a descendant of j (that would close a directed cycle), so $\mathrm{pa}(j) \subseteq \mathrm{nd}(j)$. Write $X_P = X_{\mathrm{pa}(j)}$ and $\beta = (\beta_{kj})_{k\in\mathrm{pa}(j)}$. Since $\mathbb{E}(X_P\epsilon_j) = 0$, β solves the normal equations $\mathbb{E}\{X_P(X_j - X_P^\top\beta)\} = 0$. These have a unique solution because $\mathrm{Var}(X_P)$ is a principal submatrix of $\Sigma = (I-B)^{-1}D(I-B)^{-\top} \succ 0$. The residual is ϵ_j, with variance σ_j^2.

Step 4. Let $X_N = X_{\mathrm{nd}(j)\setminus\mathrm{pa}(j)}$. Given $(X_P, X_N) = (x_P, x_N)$, $X_j = x_P^\top\beta + \epsilon_j$ with ϵ_j independent of (X_P, X_N) by (a), so the conditional law $N(x_P^\top\beta, \sigma_j^2)$ does not depend on x_N. This is $X_j \perp\!\!\!\perp X_N \mid X_P$. ■

Exercise 9.2 (Conditioning on a descendant of a collider)

Let $X, Y, \epsilon_Z, \epsilon_W$ be independent $N(0,1)$, $Z = aX + bY + \epsilon_Z$ and $W = cZ + \epsilon_W$ with $a, b, c \neq 0$. Show that

$$\rho_{XY\cdot W} = -\frac{c^2ab}{\sqrt{\{c^2(1+b^2)+1\}\{c^2(1+a^2)+1\}}} \neq 0,$$

and that $|\rho_{XY\cdot W}|$ is strictly smaller than the collider value $|\rho_{XY\cdot Z}| = |ab|/\sqrt{(1+a^2)(1+b^2)}$.

Solution [Supplement].

Step 1. $\mathrm{Cov}(X, W) = c\,\mathrm{Cov}(X, Z) = ca$, $\mathrm{Cov}(Y, W) = cb$, and $v := \mathrm{Var}W = c^2(a^2+b^2+1)+1$.

Step 2. The residuals of X and Y on W are $X - \frac{ca}{v}W$ and $Y - \frac{cb}{v}W$. Their covariance is $\mathrm{Cov}(X,Y) - \frac{c^2ab}{v} = -\frac{c^2ab}{v}$, and their variances are $1 - \frac{c^2a^2}{v} = \frac{c^2(1+b^2)+1}{v}$ and $\frac{c^2(1+a^2)+1}{v}$. Dividing gives the formula, which is nonzero since $a, b, c \neq 0$.

Step 3. Dividing numerator and denominator by c^2,

$$|\rho_{XY\cdot W}| = \frac{|ab|}{\sqrt{(1+b^2+c^{-2})(1+a^2+c^{-2})}} < \frac{|ab|}{\sqrt{(1+a^2)(1+b^2)}}.$$

So W opens the path $X \to Z \leftarrow Y$, as d-separation predicts, but less than Z itself; the effect approaches the collider value as $c^2 \to \infty$, when W becomes an almost noiseless copy of Z. ■

Exercise 9.3 (The equivalence class of a path)

Consider DAGs whose skeleton is the path $X_1 - X_2 - \cdots - X_p$, $p \geq 2$. Show that exactly p of them have no v-structure, namely those with a single source X_s,

$$X_1 \leftarrow \cdots \leftarrow X_{s-1} \leftarrow X_s \rightarrow X_{s+1} \rightarrow \cdots \rightarrow X_p, \qquad s = 1, \ldots, p,$$

that these form one Markov equivalence class, and that its CPDAG is the undirected path.

Solution [Supplement].

Step 1. The skeleton has no cycle, so every orientation of its edges is acyclic. Encode the edge between X_i and X_{i+1} by $e_i = \mathsf{R}$ if $X_i \rightarrow X_{i+1}$ and $e_i = \mathsf{L}$ otherwise.

Step 2. Two edges share a node only if they are consecutive. The triple (X_{i-1}, X_i, X_{i+1}) is unshielded, since X_{i-1} and X_{i+1} are not adjacent, and it is a collider iff $e_{i-1} = \mathsf{R}$ and $e_i = \mathsf{L}$. So the DAG has no v-structure iff the word $e_1 \cdots e_{p-1}$ contains no factor RL.

Step 3. If $e_i = \mathsf{R}$ and $e_j = \mathsf{L}$ for some $i < j$, let i' be the largest index below j with $e_{i'} = \mathsf{R}$; then $e_{i'+1} = \mathsf{L}$ (either $i' + 1 = j$, or $e_{i'+1} \neq \mathsf{R}$ by maximality), so RL occurs. Hence a word without RL has all its L's before all its R's: it is $\mathsf{L}^{s-1}\mathsf{R}^{p-s}$ for some $s \in \{1, \ldots, p\}$, which is exactly the DAG with single source X_s. Conversely these p words contain no RL.

Step 4. By the Verma–Pearl theorem the p DAGs are Markov equivalent (same skeleton, no v-structures), and any DAG equivalent to them has the same skeleton and no v-structure, so it is one of them. The edge e_i is R for $s \leq i$ and L for $s > i$, so its direction varies within the class, and every edge of the CPDAG is undirected. ■

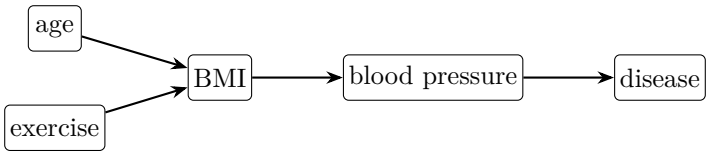
Chapter 10. Structure Learning: PC, GES and MMHC

Graphical models, Lecture 3 (constraint-based and score-based discovery of the Markov equivalence class)

1 What Observational Data Can Reveal

Questions of this lecture: which relations remain after accounting for the other variables? Which edges does conditional independence remove? How far can directions be fixed from observational data alone?

The toy example used throughout:



In a real analysis this graph is unknown and relations are searched for using observational data only. **The output of structure learning is the set of relations that the assumptions and the data allow; observational data do not fix every arrow.**

2 Assumptions Matter as Much as Data

Assumption	Practical meaning
acyclicity	relations at one time point are represented by a DAG without cycles
causal Markov condition	paths blocked in the graph appear as conditional independences in the data
faithfulness	no accidental cancellation of coefficients makes relations look independent
causal sufficiency	no unobserved common cause of two analyzed variables
same generating process	all samples come from the same system, without selection bias

PC and GES rely in particular on causal sufficiency. If hidden common causes are suspected, FCI is needed (Ch. 11).

3 The PC Algorithm

Motivation. Testing independence given every subset of variables is infeasible. The PC algorithm starts from the complete graph and removes edges using conditioning sets of increasing size taken only from current neighbours, then orients edges from the colliders it finds.

3.1 Skeleton phase

Algorithm: PC, skeleton

1. Connect all pairs of variables by undirected edges.
2. For $\ell = 0, 1, 2, \ldots$: for each adjacent pair (X_j, X_k) and each conditioning set $S \subseteq$

adj$(X_j) \setminus \{X_k\}$ of size ℓ, test

$$H_0 : X_j \perp\!\!\!\perp X_k \mid X_S.$$

3. If H_0 is not rejected for some S, remove the edge $X_j - X_k$,
4. and store that S as the separating set $\mathrm{Sep}(X_j, X_k)$.

Stop when no node has more than ℓ remaining neighbors.

The tests must suit the data: Gaussian continuous data use partial correlation with the Fisher-z test (Ch. 8); categorical data use the χ^2 or G^2 test; nonlinear relations need nonlinear CI tests (Ch. 11). The number of remaining edges depends on the significance level and sample size. The first output of PC is an undirected **skeleton**; an edge means a relation that remains after accounting for the other variables.

Example: skeleton of the toy example

Pair	Conditioning set	Decision	Action
age, exercise	$\varnothing$	independent	remove edge
age, blood pressure	BMI	conditionally independent	remove edge
exercise, blood pressure	BMI	conditionally independent	remove edge
BMI, disease	blood pressure	conditionally independent	remove edge

(Pairs such as age–disease and exercise–disease are removed similarly, e.g. given BMI.) The remaining skeleton is

$$\mathcal{E}_{\text{skel}} = \{\text{age–BMI},\ \text{exercise–BMI},\ \text{BMI–blood pressure},\ \text{blood pressure–disease}\}.$$

At this stage only the backbone of conditional relations is left; no direction has been decided yet.

3.2 Orientation phase

Colliders tell directions from observational data. The edge between age and exercise was removed with no conditioning set: $\mathrm{Sep}(\text{age}, \text{exercise}) = \varnothing$. Both variables are adjacent to BMI, they are not adjacent to each other, and BMI is not in their separating set. So PC orients

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}.$$

Before conditioning on BMI, age and exercise are independent; given BMI they can become dependent. If two non-adjacent variables point into the same middle variable, the middle variable is a collider.

Algorithm: PC, orientation

1. **v-structures**: for every unshielded triple $X - Z - Y$ (X, Y non-adjacent) with $Z \notin \mathrm{Sep}(X, Y)$, orient $X \to Z \leftarrow Y$.
2. **Meek rules**, applied until nothing changes:

   R1 (no new collider) $X \to Z - Y$, X, Y non-adjacent $\Rightarrow Z \to Y$;

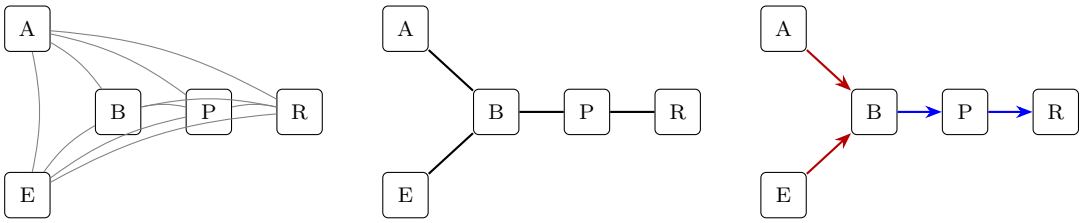
   R2 (no cycle) $X \to Z \to Y$ and $X - Y \Rightarrow X \to Y$;

   R3 $X - Z_1 \to Y$, $X - Z_2 \to Y$, $X - Y$, Z_1, Z_2 non-adjacent $\Rightarrow X \to Y$.

Preventing new colliders: if $A \to B - C$ and $A \not\sim C$, orienting $C \to B$ would create $A \to B \leftarrow C$, which the data did not support; hence $B \to C$. **Preventing cycles**: if $A \to B \to C$ and $A - C$, then $C \to A$ would create a directed cycle; hence $A \to C$. In the example the following directions are fixed in turn:

$$\text{age} \to \text{BMI} \leftarrow \text{exercise}, \qquad \text{BMI} \to \text{blood pressure} \to \text{disease}.$$

PC directions are interpreted only within the range forced by the conditional-independence pattern and the assumptions.



(1) complete graph (2) skeleton after CI tests (3) v-structure + Meek R1

Figure 11: PC on the toy example (A age, E exercise, B BMI, P blood pressure, R disease). Step 2 removes every edge whose endpoints become conditionally independent. Step 3 orients the v-structure $A \to B \leftarrow E$ (red, since $B \notin \text{Sep}(A, E) = \varnothing$) and then propagates $B \to P \to R$ (blue) to avoid creating new colliders.

3.3 Why PC is correct

Theorem: correctness of PC with a CI oracle

Assume the Markov condition, faithfulness and causal sufficiency, and that every CI query is answered correctly. Then PC returns the CPDAG of the true DAG.

Proof [Supplement]. *Adjacencies.* An edge of G is never removed: by faithfulness, adjacent variables are dependent given every set. If X, Y are non-adjacent, one of them, say Y, is not a descendant of the other; by the local Markov property $X \perp\!\!\!\perp Y \mid \text{pa}(X)$. Since true edges are never removed, $\text{pa}(X) \subseteq \text{adj}(X)$ at every stage, so this set (or another separating set) is found when $\ell = |\text{pa}(X)|$. *v-structures.* Take an unshielded triple $X - Z - Y$. If Z is a non-collider on $X \cdot Z \cdot Y$, every set separating X, Y must contain Z (else that path is open); if Z is a collider, no separating set contains Z (conditioning on Z opens the path). So the single stored set $\text{Sep}(X, Y)$ decides correctly. *Remaining edges.* The Meek rules are sound (each alternative orientation creates a new v-structure or a cycle) and complete for this situation (Meek, 1995), giving the CPDAG. ■

Remark: finite-sample issues

With real tests, an early type-II error removes an edge that later cannot be recovered and changes the separating sets used for orientation; errors propagate. The output also depends on the order in which pairs are tested; *PC-stable* removes this order dependence by updating adjacencies only after each level ℓ. Conflicting v-structures (both $X \to Z \leftarrow Y$ and $Z \to Y$ demanded) signal test errors or violated assumptions.

4 Score-Based Search

4.1 BIC for Gaussian linear SEMs

Score-based search compares candidate graphs by goodness of fit and complexity. For a candidate DAG G,

$$S_{\mathrm{BIC}}(G) = \ell(\hat{\theta}_G) - \frac{d_G}{2}\log n,$$

where $\ell(\hat{\theta}_G)$ measures how well the graph explains the data and d_G is the number of estimated parameters. Adding an edge can improve fit but increases the complexity penalty.

Theorem: BIC decomposes into local scores

For a Gaussian linear SEM,

$$S_{\mathrm{BIC}}(G) = \sum_{j=1}^{p} S_j\{X_j, \mathrm{pa}_G(j)\}, \qquad S_j = -\frac{n}{2}\log \hat{\sigma}^2_{j|\mathrm{pa}(j)} - \frac{|\mathrm{pa}(j)|+1}{2}\log n + c,$$

where $\hat{\sigma}^2_{j|\mathrm{pa}(j)}$ is the residual variance of the least-squares regression of X_j on its parents.

Proof [Supplement]. By the Markov factorization the log-likelihood is a sum of the log-likelihoods of the local regressions, whose parameters are separate. The maximized Gaussian log-likelihood of one regression is $-\frac{n}{2}\log(2\pi\hat{\sigma}^2) - \frac{n}{2}$, and that regression has $|\mathrm{pa}(j)|$ coefficients plus one variance (centered data). ■

Decomposability means that a local move (adding, deleting or reversing one edge) changes only one or two local scores, which makes search cheap. Markov-equivalent DAGs give the same maximum likelihood and the same number of parameters, so BIC is **score equivalent**: score-based methods also cannot distinguish directions within a Markov equivalence class from data alone.

4.2 GES and hill climbing

	GES	Hill climbing
search space	CPDAGs representing Markov equivalence classes	individual DAGs
moves	forward phase adds relations, backward phase removes them	add, delete or reverse an edge
typical output	a high-scoring CPDAG	one high-scoring DAG
caution	remaining undirected edges are undetermined	not every output direction is identified

Algorithm: Greedy Equivalence Search

Forward: start from the empty graph; repeatedly apply the single edge insertion (in equivalence-class space) that increases the score most, until no insertion helps. **Backward**: from there, repeatedly apply the best edge deletion until no deletion helps. Output the CPDAG.

Theorem: consistency of GES (Chickering, 2002)

If the distribution is faithful to a DAG and the score is consistent and decomposable (e.g. BIC), then GES returns the true CPDAG with probability tending to 1 as $n \to \infty$, despite

being greedy.

Hill climbing over DAGs can get stuck in local optima; tabu lists and random restarts help. Computing a DAG as output and identifying arrows from data are two different things.

5 Computational Burden and Hybrid Methods

	Why the cost grows	Remaining problem
PC	repeats CI tests over many conditioning sets	tests multiply when the graph is dense; early test errors propagate to later edges and directions
GES	compares scores of many edge additions/deletions at each step	with many variables score computations grow; greedy results depend on sample and search path

For a maximal degree k, the number of PC tests is polynomial in p (roughly of order p^{k+2}), but it is exponential in k. Ways to reduce the burden:

- **MMHC**: reduce candidate edges by conditional-independence tests, then run a score search only within the restricted space;
- **FLOP**: search efficiently over variable orders and parent sets to cut the cost of score computations.

5.1 MMHC

CI tests $\longrightarrow$ candidate skeleton $\longrightarrow$ score search $\longrightarrow$ DAG candidate

Stage 1 (candidate relations): for each variable, find the neighbors it may be directly connected to by CI tests (Max-Min Parents and Children); keep only those pairs as candidate edges. **Stage 2 (restricted score search)**: within the candidate edges, repeat additions, deletions and reversals, and select a DAG with high BIC or a similar score. An edge dropped in the first stage cannot be restored in the second, and not every direction of the output DAG is identified.

6 Comparison

	PC	GES	MMHC
approach	constraint (CI) based	score based	both, in sequence
search	removes edges when independent	changes the CPDAG to increase the score	restricts candidates by tests, then score search over DAGs
typical output	CPDAG	CPDAG	DAG candidate
key settings	independence test, significance level	score, complexity penalty	tests, significance level, score
sensitive to	test errors	score and search path	edges lost in stage 1

Summary

- Under acyclicity, Markov, faithfulness and causal sufficiency, observational data identify the Markov equivalence class.

- PC removes edges by CI tests, orients v-structures from the separating sets, then propagates orientations with the Meek rules; it is correct with a CI oracle.
- BIC decomposes into local regression scores and is score equivalent; GES searches equivalence classes and is consistent.
- MMHC combines both; its first-stage errors are irreversible.

7 Selected exercises

Motivation. The exercises check score equivalence and the consistency of BIC on a single edge, prove the soundness of the Meek rules used in the correctness proof of PC, and justify the polynomial bound on the number of tests.

Exercise 10.1 (BIC on one edge)

Two centered variables are observed n times, with sample variances s_X^2, s_Y^2 (divisor n) and sample correlation r, $|r| < 1$.
(a) Using the local scores of the BIC theorem, show that $S_{\mathrm{BIC}}(X \to Y) = S_{\mathrm{BIC}}(Y \to X)$ and that the edge beats the empty graph iff $-n \log(1 - r^2) > \log n$.
(b) For i.i.d. bivariate Gaussian data with correlation ρ, show that the probability that BIC chooses the correct adjacency tends to 1, both for $\rho = 0$ and for $\rho \neq 0$.

Solution [Supplement].

(a)

Step 1. The residual variance of Y on X is $s_Y^2 - s_{XY}^2/s_X^2 = s_Y^2(1 - r^2)$. Hence

$$S_{\mathrm{BIC}}(X \to Y) = -\frac{n}{2}\{\log s_X^2 + \log s_Y^2 + \log(1 - r^2)\} - \frac{3}{2}\log n + 2c,$$

which is symmetric in X and Y, so it equals $S_{\mathrm{BIC}}(Y \to X)$.

Step 2. The empty graph scores $-\frac{n}{2}(\log s_X^2 + \log s_Y^2) - \log n + 2c$. The difference is $-\frac{n}{2}\log(1 - r^2) - \frac{1}{2}\log n$, positive iff $-n \log(1 - r^2) > \log n$.

(b)

Step 3. ($\rho = 0$.) $\sqrt{n}\, r \xrightarrow{d} N(0, 1)$ (the t or Fisher-z results of Ch. 8 with $q = 0$), so nr^2 is bounded in probability and $r^2 \xrightarrow{p} 0$. For $0 \le u \le \frac{1}{2}$, $-\log(1 - u) = \sum_{k \ge 1} u^k/k \le u/(1 - u) \le 2u$. On the event $\{r^2 \le \frac{1}{2}\}$, whose probability tends to 1,

$$P\{-n \log(1 - r^2) > \log n\} \le P\{2nr^2 > \log n\} + P\{r^2 > \tfrac{1}{2}\} \to 0,$$

because $\log n \to \infty$ while nr^2 stays bounded in probability.

Step 4. ($\rho \neq 0$.) $r \xrightarrow{p} \rho$, so $-\log(1 - r^2) \xrightarrow{p} -\log(1 - \rho^2) > 0$, and $-n \log(1 - r^2)/\log n \xrightarrow{p} \infty$ because $n/\log n \to \infty$. Hence the edge is chosen with probability tending to 1.

The penalty $\frac{1}{2}\log n$ grows, so null partial correlations of size $O_p(n^{-1/2})$ are eventually rejected, yet slowly enough that fixed nonzero correlations are always kept. ■

Exercise 10.2 (Soundness of the Meek rules)

Let $\mathcal{C}$ be a Markov equivalence class and P a partially directed graph with the common skeleton such that every directed edge of P has that orientation in every DAG of $\mathcal{C}$, and every

v-structure of $\mathcal{C}$ appears in P with both edges directed. Show that if the premise of R1, R2 or R3 holds in P, then every DAG of $\mathcal{C}$ contains the edge that the rule orients.

Solution [Supplement]. Fix $G' \in \mathcal{C}$. The two variables in each conclusion are adjacent, so G' orients their edge one way or the other; we exclude the wrong way.

Step 1. (R1: $X \to Z - Y$, X, Y non-adjacent.) G' contains $X \to Z$. If it contained $Y \to Z$, then $X \to Z \leftarrow Y$ would be a v-structure of G', hence of $\mathcal{C}$, and would appear in P with $Y \to Z$ directed; but $Z - Y$ is undirected in P. So G' contains $Z \to Y$.

Step 2. (R2: $X \to Z \to Y$, $X - Y$.) If G' contained $Y \to X$, then $X \to Z \to Y \to X$ would be a directed cycle. So G' contains $X \to Y$.

Step 3. (R3: $X - Z_1 \to Y$, $X - Z_2 \to Y$, $X - Y$, Z_1, Z_2 non-adjacent.) Suppose G' contains $Y \to X$. For $i = 1, 2$, $X \to Z_i$ would create the cycle $X \to Z_i \to Y \to X$, so G' contains $Z_i \to X$. Then $Z_1 \to X \leftarrow Z_2$ with Z_1, Z_2 non-adjacent is a v-structure of $\mathcal{C}$ and must appear directed in P; but $X - Z_1$ is undirected in P. So G' contains $X \to Y$.

After the v-structure step of PC (with a CI oracle) the graph satisfies the assumptions on P, and each orientation keeps them, so all directions added by the Meek rules are correct. ■

Exercise 10.3 (How many tests PC needs)

Assume the conditions of the PC correctness theorem and let every node of the true DAG have at most k neighbours, $p \geq 3$. Show that the skeleton phase runs only the levels $\ell = 0, \dots, k$ and performs at most $(k+1)p^{k+2}$ CI tests.

Solution [Supplement].

Step 1. By the proof of the correctness theorem, a non-adjacent pair X, Y (with Y not a descendant of X) is separated by $\mathrm{pa}(X)$, so its edge is removed at the latest at level $\ell = |\mathrm{pa}(X)| \leq k$; true edges are never removed. So after level k the current graph is the true skeleton.

Step 2. At that point no node has more than k neighbours, so the stopping rule ends the skeleton phase; levels $\ell \geq k+1$ are never run.

Step 3. At level ℓ, an ordered adjacent pair (X_a, X_b) is tested with at most $\binom{|\mathrm{adj}(X_a)|-1}{\ell} \leq \binom{p-2}{\ell} \leq (p-2)^\ell$ sets, and there are at most $p(p-1)$ ordered pairs. Hence the total is at most

$$p(p-1)\sum_{\ell=0}^{k}(p-2)^\ell \leq p^2(k+1)(p-2)^k \leq (k+1)p^{k+2}.$$

The bound is polynomial in p for fixed sparsity, but the exponent grows with the maximal degree: dense graphs make PC expensive, and they also require large conditioning sets, for which the tests lose power. ■

Chapter 11. Relational Inference Matched to the Data Structure

Graphical models, Lecture 4 (hidden confounders, FCI and PAGs, nonlinear CI tests, PCMCI+, feedback)

1 PC Assumes All Common Causes Are Observed

Premises used by PC: relations among the observed variables are represented by one DAG; no common cause of the analyzed variables is missing; the Markov condition and faithfulness hold and conditional independences are tested properly.

Problem: an unobserved common cause. Suppose $E \leftarrow U \rightarrow P$ with U a hidden lifestyle factor. The observed data contain only exercise E and blood pressure P. E and P are related, but they are not directly connected. Because PC assumes all common causes are observed, it may represent this relation as a direct edge. **If hidden common causes are possible, an edge found by PC cannot be interpreted as a direct relation.**

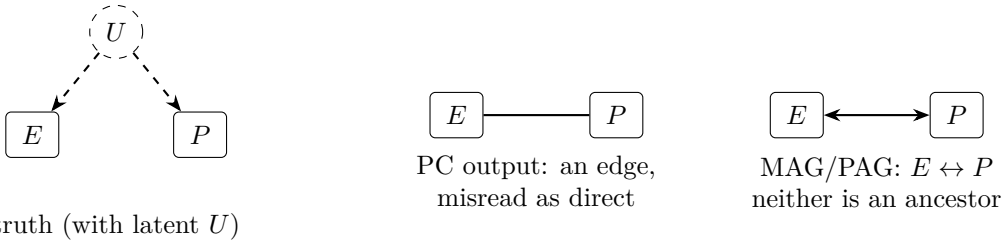


Figure 12: A hidden common cause. Marginalizing U leaves a dependence between E and P with no direct effect. A DAG over the observed variables must pick a direction; the ancestral-graph language of FCI has a bidirected edge for exactly this situation (identifiable when enough other variables are observed).

1.1 Observational data may not distinguish three explanations

Suppose only X and Y are observed and they are related.

Possible structure	Explanation	Observed conditional independences
$X \rightarrow Y$	X causes Y	none
$X \leftarrow Y$	Y causes X	none
$X \leftarrow U \rightarrow Y$	hidden common cause U	none

Additional observed variables and their conditional independences can exclude some explanations, but a single DAG is not always determined.

2 FCI Infers a PAG Instead of a DAG

2.1 Ancestral graphs and PAGs

When latent variables are marginalized out of a DAG, the conditional independences among the observed variables are represented by a **maximal ancestral graph** (MAG) with directed ($\rightarrow$) and bidirected ($\leftrightarrow$) edges: $X \rightarrow Y$ means X is an ancestor of Y; $X \leftrightarrow Y$ means neither is an ancestor of the other (a latent confounder). Markov-equivalent MAGs are summarized by a **partial ancestral graph** (PAG), in which an endpoint mark is shown only if it is common to all MAGs of the class; otherwise a circle $\circ$ is drawn.

	Object represented	Meaning of relations
PC's CPDAG	DAG equivalence class under the assumption of no missing common causes among analyzed variables	only directions common to all equivalent DAGs are fixed
FCI's PAG	equivalence class of observed-variable structures allowing hidden common causes	only endpoints common to all compatible structures are fixed

FCI (Fast Causal Inference) uses conditional independences but does *not* assume causal sufficiency. The circles and arrowheads of a PAG are not an incomplete result: they display the range that observational data can distinguish.

2.2 Reading PAG endpoints

PAG edge	How to read it
$X \to Y$	apart from selection bias, in every compatible structure X is an ancestor of Y and Y is not an ancestor of X
$X \leftrightarrow Y$	neither is an ancestor of the other; compatible with a structure with a hidden common cause
$X \circ\!\!\to Y$	it is certain that Y is not an ancestor of X; the mark at X is undetermined
$X \circ\!\!-\!\!\circ Y$	the two variables are adjacent but both endpoints are undetermined

An arrowhead means that variable is *not an ancestor* of the other. A circle may be a tail or an arrowhead depending on the compatible structure. $X \to Y$ in a PAG is an ancestral relation, not the single direct effect of a latent DAG.

2.3 FCI can still find the arrowheads of colliders

Suppose CI tests give the skeleton $X - Y - Z$ with $X \not\sim Z$, and $X \perp\!\!\!\perp Z$ unconditionally but $X \not\perp\!\!\!\perp Z \mid Y$. Without conditioning on Y, the path is blocked at the collider and X, Z are independent; conditioning on the collider opens the path and they become related. Hence FCI orients the arrowheads into Y:

$$X \circ\!\!\to Y \leftarrow\!\!\circ Z.$$

Whether X (or Z) is a cause of Y or shares a hidden cause with it remains open.

2.4 FCI checks wider separating sets, then orients conservatively

Algorithm: FCI

Edge removal.

1. Start from the complete graph on the observed variables.
2. If $X \perp\!\!\!\perp Y \mid S$ is found, remove $X - Y$ (as in PC's skeleton phase).
3. Widen the candidate separating sets beyond current neighbors to the variables on possible paths (Possible-D-Sep) and check again: because of hidden variables, a separating set may not lie among the current neighbors.

Endpoint determination.

1. Record the separating set used when an edge was removed.
2. Mark as colliders the middle variables not contained in the separating set.

3. Propagate only those endpoints that are forced, using the colliders and the PAG orientation rules (Zhang's rules R1–R10).

Undetermined endpoints remain circles $\circ$.

Remark: why neighbors are not enough

In a DAG, a non-adjacent pair is always separated by the parents of one of them, which are neighbors. In a MAG this fails: a separating set may have to contain a node W that reaches X only through a path whose inner nodes are all colliders (e.g. $W \leftrightarrow V \leftrightarrow X$ created by latent variables), and such a W need not be adjacent to X or Y. The set D-Sep(X, Y) collects such nodes; since it depends on the unknown MAG, FCI uses the computable superset Possible-D-Sep from the intermediate graph. PC, which only searches among neighbors, would wrongly keep the edge. The extra tests are the price of not assuming causal sufficiency.

Like PC, FCI starts from conditional independences, but checks the additional separation possibilities needed because of hidden variables. **RFCI** (really fast CI) skips most of the Possible-D-Sep search and runs targeted extra tests on triples, reducing computation at the cost of a slightly less informative (but still sound) output.

3 When Basic Assumptions Break, Methods Change

Basic assumption	Situation where it fails	Candidate methods
all common causes observed	unobserved common causes exist	FCI, RFCI
the chosen test captures the relation	partial correlation misses nonlinear relations	KCI, distance correlation
independent observations	autocorrelation and time lags	PCMCI+
no cycles in the graph	feedback among variables	CCD, cyclic LiNGAM

The choice of algorithm depends less on the data size than on **which basic assumptions can be maintained**.

3.1 Nonlinear relations: change the independence test

Partial correlation measures linear relations.

Example: dependence with zero correlation

Let $X \sim \text{Unif}(-1, 1)$ and $Y = X^2 + \varepsilon$ with $\varepsilon \perp\!\!\!\perp X$. Then $\text{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$ because the distribution of X is symmetric, although Y depends strongly on X. A Fisher-z test would declare them independent and PC would remove the edge.

CI test	When to use
Fisher-z	linear conditional relations in continuous Gaussian data
kernel CI test (KCI)	general nonlinear conditional relations (based on cross-covariance operators in reproducing-kernel spaces)
distance correlation	nonlinear relations without strong assumptions on the distribution; $\text{dCor}(X, Y) = 0$ iff $X \perp\!\!\!\perp Y$

The search procedures of PC and FCI stay the same; only the CI test is replaced. Nonparametric tests are more flexible but lose power with small samples or many conditioning variables, and are computationally heavier.

3.2 Time series: PCMCI+ for lagged relations

Ordinary PC treats rows as independent, but in a time series the past of a variable is related to its present. Relations of interest are

$$X^j_{t-\tau} \longrightarrow X^k_t, \qquad \tau = 0, 1, \ldots, \tau_{\max}.$$

Algorithm: PCMCI+ (outline)

1. Among the past values of all variables, first reduce the candidate parents that may be related to each present value (a PC-type condition-selection step on lagged links).
2. Re-test each candidate link conditioning on the other past parents of both endpoints (momentary conditional independence, MCI), which removes autocorrelation effects.
3. Distinguish time-lagged relations from same-time (contemporaneous) relations and represent them in a time-series graph; contemporaneous links are oriented with PC-type rules.

Example output: $X^1_{t-2} \to X^2_t$ and $X^2_{t-1} \to X^3_t$. Feeding time series into ordinary PC can create false edges through autocorrelation: two independent but strongly autocorrelated series often show large sample correlations. Check the maximal lag $\tau_{\max}$ and stationarity.

3.3 Feedback: the DAG assumption fails

Economic and biological systems can have feedback such as $X_1 \to X_2 \to X_3 \to X_1$.

Method	Core idea
CCD	uses conditional independences to find relations common to the equivalence class of cyclic graphs
cyclic LiNGAM	uses linearity and non-Gaussian errors to further identify the cyclic structure

A linear cyclic SEM $X = BX + \epsilon$ is read as an equilibrium of a dynamic system, requiring $I - B$ invertible (stability typically needs the spectral radius of B below 1). CCD allows cycles, but directions the observational data cannot distinguish remain undetermined. Before using a cyclic algorithm, decide whether the feedback is *simultaneous* or actually a *time-lagged* relation that PCMCI+ should handle.

4 Choosing a Method for the Data Structure

Feature of the data	Method	Main points to check
hidden common causes suspected	FCI or RFCI	read PAG circles and arrowheads separately
nonlinear relations expected	PC/FCI with nonparametric tests	test power and computing time
time lags and autocorrelation	PCMCI+	maximal lag and stationarity
simultaneous feedback possible	CCD or cyclic LiNGAM	distinguish cycles from time lags

Summary

- With hidden confounders, PC's edges are not direct relations; FCI outputs a PAG whose arrowheads mean "not an ancestor" and whose circles mean "undetermined".
- FCI widens the search for separating sets (Possible-D-Sep) because separating sets may lie outside the neighbors; RFCI trades some information for speed.
- Nonlinear relations need kernel or distance-based CI tests; time series need lag-aware methods (PCMCI+); feedback needs cyclic methods (CCD, cyclic LiNGAM).

5 Selected exercises

Motivation. The exercises show that a hidden common cause can reproduce a direct effect exactly and how one extra observed variable separates the two, quantify the spurious correlation between independent autocorrelated series, and justify the equilibrium reading of a cyclic SEM.

Exercise 11.1 (A hidden common cause mimics a direct effect)

Let $U \sim N(0,1)$ be latent, $X = aU + \epsilon_X$, $Y = bU + \epsilon_Y$, with $U, \epsilon_X, \epsilon_Y$ independent, $\epsilon_X \sim N(0, \sigma_X^2)$, $\epsilon_Y \sim N(0, \sigma_Y^2)$, $\sigma_Y^2 > 0$.
(a) Find c and $\tau^2 > 0$ such that the SEM $X' \sim N(0, a^2 + \sigma_X^2)$, $Y' = cX' + \eta$, $\eta \sim N(0, \tau^2)$ independent of X', has the same distribution as (X, Y).
(b) Add an observed $Z \sim N(0,1)$, independent of $(U, \epsilon_X, \epsilon_Y)$, with $X = dZ + aU + \epsilon_X$, $d \neq 0$. Show that $Z \perp\!\!\!\perp Y$, whereas in any Gaussian SEM $Z \to X \to Y$ with nonzero coefficients Z and Y are dependent.

Solution [Supplement].

(a)

Step 1. (X, Y) is bivariate Gaussian with mean 0, $\mathrm{Var}X = a^2 + \sigma_X^2$, $\mathrm{Var}Y = b^2 + \sigma_Y^2$ and $\mathrm{Cov}(X, Y) = ab$. A centered Gaussian law is determined by its covariance matrix.

Step 2. In the second SEM, $\mathrm{Cov}(X', Y') = c(a^2 + \sigma_X^2)$ and $\mathrm{Var}Y' = c^2(a^2 + \sigma_X^2) + \tau^2$. Matching gives

$$c = \frac{ab}{a^2 + \sigma_X^2}, \qquad \tau^2 = b^2 + \sigma_Y^2 - \frac{a^2 b^2}{a^2 + \sigma_X^2} = \sigma_Y^2 + \frac{b^2 \sigma_X^2}{a^2 + \sigma_X^2} > 0.$$

So no procedure using (X, Y) alone can tell $X \leftarrow U \to Y$ from $X \to Y$ (or, symmetrically, from $Y \to X$).

(b)

Step 3. $Y = bU + \epsilon_Y$ is a function of (U, ϵ_Y), which is independent of Z; hence $Z \perp\!\!\!\perp Y$.

Step 4. If $X = dZ + \epsilon'_X$ and $Y = cX + \eta$ with independent errors and $c, d \neq 0$, then $\mathrm{Cov}(Z, Y) = cd \neq 0$, so Z and Y are dependent.

The extra variable turns the question into a conditional-independence pattern: $Z \perp\!\!\!\perp Y$ together with $Z \not\perp\!\!\!\perp X$ and $X \not\perp\!\!\!\perp Y$ is exactly the evidence FCI uses to orient $Z \circ\!\!\to X \leftarrow\!\!\circ Y$; the arrowhead at X says that X is not an ancestor of Y, which rules out $X \to Y$. ■

Exercise 11.2 (Independent but autocorrelated series)

Let $X_t = \phi X_{t-1} + u_t$ and $Y_t = \phi Y_{t-1} + v_t$ be stationary AR(1) processes, $|\phi| < 1$, where (u_t) and (v_t) are i.i.d. $(0, 1)$ sequences independent of each other. Let $C_n = \frac{1}{n}\sum_{t=1}^{n} X_t Y_t$. Show that $\mathbb{E}C_n = 0$ and

$$n\mathrm{Var}(C_n) \longrightarrow \gamma(0)^2 \frac{1+\phi^2}{1-\phi^2}, \qquad \gamma(0) = \frac{1}{1-\phi^2},$$

whereas for $\phi = 0$ the limit is $\gamma(0)^2 = 1$.

Solution [Supplement].

Step 1. The stationary solution $X_t = \sum_{m\geq 0} \phi^m u_{t-m}$ has mean 0 and autocovariance $\gamma(h) = \phi^{|h|}/(1-\phi^2)$; the same holds for Y. The processes are independent, so $\mathbb{E}(X_t Y_t) = \mathbb{E}X_t\,\mathbb{E}Y_t = 0$.

Step 2. By independence of the two processes,

$$\mathrm{Cov}(X_s Y_s, X_t Y_t) = \mathbb{E}(X_s X_t)\,\mathbb{E}(Y_s Y_t) = \gamma(t-s)^2.$$

Step 3. Hence

$$n\mathrm{Var}(C_n) = \frac{1}{n}\sum_{s=1}^{n}\sum_{t=1}^{n} \gamma(t-s)^2 = \sum_{|h|<n} \Big(1 - \frac{|h|}{n}\Big)\gamma(h)^2.$$

Since $\sum_h \gamma(h)^2 < \infty$, dominated convergence gives the limit

$$\sum_{h\in\mathbb{Z}} \gamma(h)^2 = \gamma(0)^2\Big(1 + 2\sum_{h\geq 1}\phi^{2h}\Big) = \gamma(0)^2\Big(1 + \frac{2\phi^2}{1-\phi^2}\Big) = \gamma(0)^2\,\frac{1+\phi^2}{1-\phi^2}.$$

The sample variances converge to $\gamma(0)$, so the sample correlation $r_n \approx C_n/\gamma(0)$ has variance about $\frac{1}{n}\cdot\frac{1+\phi^2}{1-\phi^2}$ instead of $\frac{1}{n}$; for $\phi = 0.9$ this is 19 times larger. A Fisher-z test that treats the rows as independent therefore declares many spurious edges, which PCMCI+ avoids by conditioning on the past. ■

Exercise 11.3 (Equilibrium of a linear cyclic SEM)

Let B be a $p \times p$ matrix with spectral radius $\varrho(B) < 1$.
(a) Show that $I - B$ is invertible and $(I-B)^{-1} = \sum_{m\geq 0} B^m$.
(b) Show that for a fixed ϵ and any start $X^{(0)}$, the iteration $X^{(t+1)} = BX^{(t)} + \epsilon$ converges to $(I-B)^{-1}\epsilon$.
(c) For the feedback loop $X_1 = bX_2 + \epsilon_1$, $X_2 = aX_1 + \epsilon_2$, show that $\varrho(B) < 1$ iff $|ab| < 1$, and find the equilibrium.

Solution [Supplement].

(a)

Step 1. By Gelfand's formula $\varrho(B) = \lim_m \|B^m\|^{1/m}$ for any matrix norm, so for $\varrho(B) < q < 1$ there is m_0 with $\|B^m\| \leq q^m$ for $m \geq m_0$. Hence $\sum_m \|B^m\| < \infty$ and $\sum_m B^m$ converges.

Step 2. $(I-B)\sum_{m=0}^{M} B^m = I - B^{M+1} \to I$, so $(I-B)\sum_{m\geq 0} B^m = I$; the same holds with the factors in the other order.

(b) By induction, $X^{(t)} = B^t X^{(0)} + \sum_{m=0}^{t-1} B^m \epsilon$. The first term tends to 0 by Step 1 and the second to $(I-B)^{-1}\epsilon$.

(c) Here $B = \begin{pmatrix} 0 & b \\ a & 0 \end{pmatrix}$ has characteristic polynomial $\lambda^2 - ab$, so its eigenvalues have modulus $\sqrt{|ab|}$ and $\varrho(B) < 1$ iff $|ab| < 1$. Solving the two equations,

$$X_1 = \frac{\epsilon_1 + b\epsilon_2}{1-ab}, \qquad X_2 = \frac{a\epsilon_1 + \epsilon_2}{1-ab}.$$

The observed data are the fixed point of a dynamic system in which each shock is passed around the loop repeatedly ($\sum_m B^m$); if $|ab| \geq 1$ the shocks are amplified and no stable equilibrium exists. ■

Chapter 12. Identifiable Causal Discovery Models: LiNGAM

Graphical models, Lecture 5 (non-Gaussian errors, residual independence, ICA-LiNGAM, DirectLiNGAM, variants)

1 Directions That Conditional Independence Cannot Distinguish

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

All three structures satisfy $X \perp\!\!\!\perp Z \mid Y$. Using conditional independence alone we find the skeleton $X - Y - Z$ but cannot tell the three directions apart (Ch. 9). Structural equations can add information:

$$X = BX + \epsilon, \qquad \epsilon_j \text{ independent.}$$

The *distribution* of the errors, not only their covariance, can carry direction information.

2 The LiNGAM Model

Motivation. With Gaussian noise, $X \to Y$ and $Y \to X$ fit the data equally well. If the noise is non-Gaussian the symmetry breaks: only in the true direction is the residual independent of the regressor, and this makes the whole causal order identifiable.

Definition: LiNGAM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad X = BX + \epsilon.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering of variables, B is strictly lower triangular
independent errors	$\epsilon_1, \ldots, \epsilon_p$ are mutually independent: no hidden common cause among the analyzed variables
non-Gaussianity	the structural errors are not normally distributed, which leaves information that distinguishes directions

Theorem: identifiability of LiNGAM (Shimizu et al., 2006)

Under linearity, acyclicity, mutually independent errors and non-Gaussian errors (at most one Gaussian error is allowed), the coefficient matrix B — hence the DAG and the causal order — is uniquely determined by the distribution of X.

3 In the True Direction, Cause and Regression Residual Are Independent

3.1 Two variables

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X \perp\!\!\!\perp \epsilon_Y, \quad \epsilon_X, \epsilon_Y \text{ non-Gaussian.}$$

Regressing Y on X leaves the residual $r_Y = Y - bX = \epsilon_Y$, so $X \perp\!\!\!\perp r_Y$.

Reverse direction $Y \to X$: fit $X = cY + r_X$. Then Y and r_X generally both contain ϵ_X and ϵ_Y, so for non-Gaussian data typically $Y \not\perp\!\!\!\perp r_X$.

Theorem: Darmois–Skitovich

Let $\epsilon_1, \ldots, \epsilon_m$ be independent and $U = \sum_i \alpha_i \epsilon_i$, $V = \sum_i \beta_i \epsilon_i$. If $U \perp\!\!\!\perp V$, then every ϵ_i with $\alpha_i \beta_i \neq 0$ is Gaussian.

Theorem: the reverse regression leaves a dependent residual

In the model above with $b \neq 0$ and $\sigma_X^2 = \mathrm{Var}\epsilon_X$, $\sigma_Y^2 = \mathrm{Var}\epsilon_Y$, the least-squares coefficient is $c = \mathrm{Cov}(X, Y)/\mathrm{Var}Y = b\sigma_X^2/(b^2\sigma_X^2 + \sigma_Y^2)$ and

$$r_X = X - cY = \frac{\sigma_Y^2}{b^2\sigma_X^2 + \sigma_Y^2}\,\epsilon_X - c\,\epsilon_Y, \qquad Y = b\,\epsilon_X + \epsilon_Y.$$

r_X is uncorrelated with Y but, if either error is non-Gaussian, not independent of it.

Proof [Supplement]. $r_X = (1 - cb)\epsilon_X - c\epsilon_Y$ and $1 - cb = \sigma_Y^2/(b^2\sigma_X^2 + \sigma_Y^2)$. Both coefficients of r_X are nonzero, and both coefficients of Y are nonzero. If $r_X \perp\!\!\!\perp Y$, Darmois–Skitovich forces ϵ_X and ϵ_Y to be Gaussian. Uncorrelatedness is the normal equation of least squares. ■

If instead both errors are Gaussian, (r_X, Y) is jointly Gaussian and uncorrelated, hence independent: *both* directions produce independent residuals, which is exactly why Gaussian SEMs are not identifiable. **LiNGAM regresses in both directions and picks the direction in which explanatory variable and residual are more independent.**

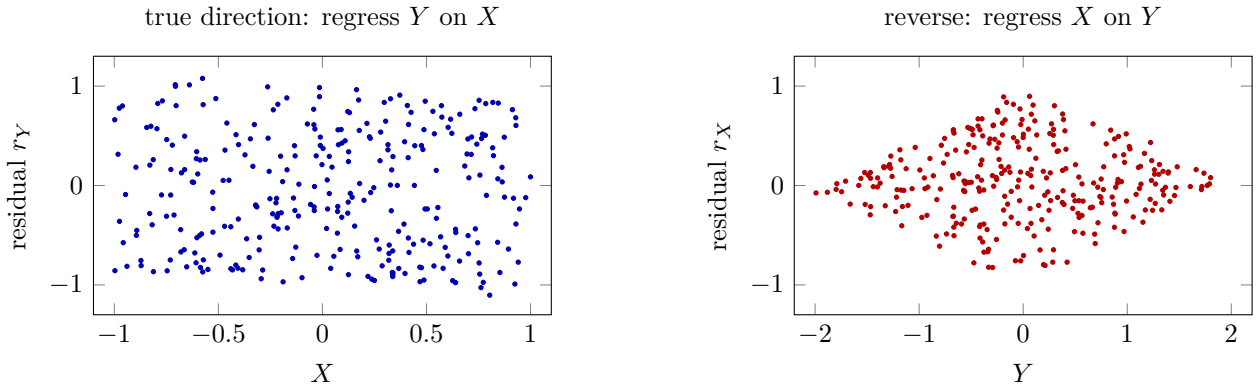


Figure 13: Simulation with $X = \epsilon_X$, $Y = X + \epsilon_Y$, $\epsilon_X, \epsilon_Y \overset{\text{iid}}{\sim} \mathrm{Unif}(-1, 1)$, $n = 300$. Both residuals are uncorrelated with their regressor (least squares guarantees it). In the true direction the cloud is a rectangle: $r_Y = \epsilon_Y$ is independent of X. In the reverse direction the cloud is a diamond whose vertical spread shrinks at the extremes of Y: uncorrelated but dependent (the correlation of the squares is 0.07 versus −0.42).

Example: a simulation picture

Take $\epsilon_X, \epsilon_Y \sim \text{Unif}(-1, 1)$ and $Y = X + \epsilon_Y$. The scatterplot of (X, r_Y) is a rectangle (independent), while (Y, r_X) forms a rotated square (diamond) whose vertical spread of r_X shrinks at the extremes of Y: uncorrelated but clearly dependent.

4 Estimation

4.1 ICA-LiNGAM

Since B is permutable to strictly lower triangular form, $X = (I - B)^{-1}\epsilon = A\epsilon$: X is a linear mixture of independent non-Gaussian sources, the setting of independent component analysis (ICA).

Algorithm: ICA-LiNGAM

1. Run ICA to estimate an unmixing matrix W with WX having independent components; W equals $I - B$ up to an unknown row permutation and row scaling.
2. Find the row permutation of W giving no zeros (in practice, no small entries) on the diagonal; because $I - B$ has a unit diagonal and is triangular after ordering, this permutation is unique.
3. Divide each row by its diagonal element to get W', and set $\hat{B} = I - W'$.
4. Find the causal order by permuting $\hat{B}$ to be as close as possible to strictly lower triangular; prune small coefficients (e.g. by bootstrap or adaptive Lasso).

ICA is an iterative non-convex optimization and can converge to local optima; the result can depend on initial values.

4.2 DirectLiNGAM finds exogenous variables one by one

Theorem: characterization of an exogenous variable

In a LiNGAM, let $r_i^{(j)} = X_i - \dfrac{\text{Cov}(X_i, X_j)}{\text{Var}(X_j)} X_j$ for $i \neq j$. Then X_j is exogenous (has no parents) iff $X_j \perp\!\!\!\perp r_i^{(j)}$ for all $i \neq j$.

Proof ($\Rightarrow$) and idea ($\Leftarrow$) [Supplement]. If X_j is exogenous, $X_j = \epsilon_j$. Every $X_i = a_{ij}\epsilon_j + \sum_{l \neq j} a_{il}\epsilon_l$ with $A = (I - B)^{-1}$, and $\text{Cov}(X_i, X_j)/\text{Var}(X_j) = a_{ij}$, so $r_i^{(j)} = \sum_{l \neq j} a_{il}\epsilon_l$, independent of ϵ_j. Conversely, if X_j has a parent, then X_j contains the error of some ancestor, and some residual $r_i^{(j)}$ shares a non-Gaussian error with nonzero coefficients with X_j; Darmois–Skitovich then gives dependence (Shimizu et al., 2011). ■

Moreover, after regressing out an exogenous variable, the residuals $(r_i^{(j)})_{i \neq j}$ again follow a LiNGAM with the remaining causal order, so the procedure can be repeated.

Algorithm: DirectLiNGAM

For each candidate X_j, regress every other X_i on X_j to obtain residuals $r_i^{(j)}$, and add up the

dependence between the candidate and all residuals:

$$T(j) = \sum_{i \neq j} \mathrm{Dep}(X_j, r_i^{(j)}).$$

1. Evaluate every variable as an exogenous candidate.
2. Choose the variable with the smallest $T(j)$ as the first in the order.
3. Replace the remaining variables by the residuals with the chosen variable's effect removed.
4. Repeat the same process on the remaining variables.

Dep is a measure of statistical dependence, e.g. a kernel-based independence measure or a mutual-information estimate. Unlike ICA, DirectLiNGAM has no iterative non-convex step and, given a consistent dependence measure, converges to the correct order in a fixed number of steps $(p-1)$.

4.3 Repeating residual independence yields the causal order

Example: three variables

Let the data come from the LiNGAM

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3,$$

with $\epsilon_1, \epsilon_2, \epsilon_3$ mutually independent non-Gaussian errors.

Step	Independence observed	Choice
1	X_1 is independent of the residuals of X_2, X_3 after removing X_1	choose X_1 first
2	after removing X_1's effect, X_2 is independent of the residual of X_3	choose X_2
3	the last remaining variable	choose X_3

Result: $X_1 < X_2 < X_3$. (For candidate X_2 in step 1, the residual of X_1 on X_2 shares ϵ_1, ϵ_2 with X_2, so $T(2) > T(1)$.)

4.4 After the order: estimate direct edges and effects

Suppose DirectLiNGAM found $X_1 < X_2 < X_3$. Each variable uses only the variables before it as candidate parents:

$$X_2 = \beta_{21} X_1 + \epsilon_2, \qquad X_3 = \beta_{31} X_1 + \beta_{32} X_2 + \epsilon_3.$$

In the example $\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{32} \approx 0.7$, $\hat{\beta}_{31} \approx 0$, so the estimated graph is

$$X_1 \Rightarrow X_2 \Rightarrow X_3.$$

Coefficients are estimated by least squares within the order, and small ones are pruned by sparse regression (adaptive Lasso) or tests; the order guarantees acyclicity.

5 Variants for Different Data Structures

Data or problem	Method	Caveat
independent cross-sectional data	DirectLiNGAM, ICA-LiNGAM	linear, acyclic structure, no hidden confounding
repeated measurements over time	VAR-LiNGAM	distinguish lagged and contemporaneous relations
feedback cycles	cyclic LiNGAM family	the acyclicity assumption of basic LiNGAM cannot be used
hidden common causes possible	ParceLiNGAM, RCD	estimate the identifiable part of the relations rather than the whole DAG

The LiNGAM variants are not applied one after another to patch a basic result; they are **chosen to match the data-generating structure**.

Remark: when LiNGAM is fragile

Near-Gaussian errors give weak signals, so large samples are needed. Measurement error, nonlinear effects, hidden confounders or selection bias break the independence of residuals and can make the wrong direction look better. Check the robustness of the order with bootstrap resampling, and test the residual independence of the final model.

Summary

- Conditional independence leaves directions inside the Markov equivalence class undecided; non-Gaussian independent errors break the tie.
- In the true direction the cause is independent of the regression residual; in the reverse direction residual and regressor are uncorrelated but dependent (Darmois–Skitovich). With Gaussian errors both directions look the same.
- ICA-LiNGAM recovers B from the ICA unmixing matrix; DirectLiNGAM finds exogenous variables one by one through residual independence, then estimates coefficients within the order.

6 Selected exercises

Motivation. The exercises prove two steps used without proof by the estimation algorithms (the unique row permutation in ICA-LiNGAM and the recursion of DirectLiNGAM) and compute exactly the dependence visible in the simulation figure.

Exercise 12.1 (The row permutation in ICA-LiNGAM is unique)

Let $M = I - B$, where the variables admit a causal order κ (a bijection onto $\{1, \dots, p\}$) with $B_{jk} \neq 0 \Rightarrow \kappa(k) < \kappa(j)$.
(a) Show that the identity is the only permutation σ with $M_{\sigma(i),i} \neq 0$ for all i.
(b) Let $W = \Lambda P M$ with Λ diagonal invertible and P a permutation matrix (the ICA indeterminacy). Show that exactly one row permutation of W has no zero on the diagonal, and that it recovers M after dividing each row by its diagonal entry.

Solution [Supplement].

(a)

Step 1. $M_{jk} \neq 0$ requires $j = k$ or $B_{jk} \neq 0$, hence $\kappa(k) \leq \kappa(j)$.

Step 2. If $M_{\sigma(i),i} \neq 0$ for all i, then $\kappa(i) \leq \kappa(\sigma(i))$ for all i. The permutation $\tau = \kappa \circ \sigma \circ \kappa^{-1}$ of $\{1, \ldots, p\}$ then satisfies $\tau(m) \geq m$ for all m. Since $\sum_m \tau(m) = \sum_m m$, the non-negative terms $\tau(m) - m$ sum to 0, so τ and hence σ is the identity.

(b)

Step 3. Row i of W is λ_i times row $\pi(i)$ of M, where π is the permutation encoded by P. Reordering the rows of W by a permutation q puts $\lambda_{q(i)} M_{\pi(q(i)),\cdot}$ in row i, whose diagonal entry is $\lambda_{q(i)} M_{\pi(q(i)),i}$.

Step 4. All diagonal entries are nonzero iff $\sigma = \pi \circ q$ satisfies $M_{\sigma(i),i} \neq 0$ for all i, i.e. by (a) iff $q = \pi^{-1}$. That reordering gives $\Lambda' M$ for a diagonal Λ', and since $M_{ii} = 1$, dividing each row by its diagonal entry returns M, hence $\hat{B} = I - M = B$. ■

Exercise 12.2 (The dependence in the simulation figure)

Let $X = \epsilon_X$ and $Y = X + \epsilon_Y$ with $\epsilon_X, \epsilon_Y \overset{\text{iid}}{\sim} \text{Unif}(-1, 1)$.
(a) Show that in the true direction $\text{Cov}(X^2, r_Y^2) = 0$.
(b) For the reverse regression $r_X = X - cY$, show that $c = \frac{1}{2}$, $\text{Cov}(r_X, Y) = 0$ and $\text{Corr}(r_X^2, Y^2) = -\frac{3}{7}$.

Solution [Supplement]. Write $u = \epsilon_X$, $v = \epsilon_Y$; then $\mathbb{E}u = \mathbb{E}u^3 = 0$, $\mathbb{E}u^2 = \frac{1}{3}$, $\mathbb{E}u^4 = \frac{1}{5}$, and likewise for v.

(a) $r_Y = Y - X = v$ is independent of $X = u$, so X^2 and r_Y^2 are independent and uncorrelated.

(b)

Step 1. $c = \text{Cov}(X, Y)/\text{Var}Y = \frac{1}{3}/\frac{2}{3} = \frac{1}{2}$, so $r_X = \frac{1}{2}(u - v)$ and $Y = u + v$. Then $\text{Cov}(r_X, Y) = \frac{1}{2}(\mathbb{E}u^2 - \mathbb{E}v^2) = 0$.

Step 2. Since $(u - v)^2(u + v)^2 = (u^2 - v^2)^2$,

$$\mathbb{E}(r_X^2 Y^2) = \tfrac{1}{4}\mathbb{E}(u^4 - 2u^2v^2 + v^4) = \tfrac{1}{4}(\tfrac{2}{5} - \tfrac{2}{9}) = \tfrac{2}{45},$$
$$\mathbb{E}r_X^2\, \mathbb{E}Y^2 = \tfrac{1}{4} \cdot \tfrac{2}{3} \cdot \tfrac{2}{3} = \tfrac{1}{9}, \qquad \text{Cov}(r_X^2, Y^2) = \tfrac{2}{45} - \tfrac{5}{45} = -\tfrac{1}{15}.$$

Step 3. Odd moments vanish, so $\mathbb{E}(u \pm v)^4 = \mathbb{E}u^4 + 6\mathbb{E}u^2\mathbb{E}v^2 + \mathbb{E}v^4 = \frac{16}{15}$ and $\text{Var}\{(u \pm v)^2\} = \frac{16}{15} - \frac{4}{9} = \frac{28}{45}$. Hence $\text{Var}(Y^2) = \frac{28}{45}$ and $\text{Var}(r_X^2) = \frac{1}{16} \cdot \frac{28}{45} = \frac{7}{180}$, and

$$\text{Corr}(r_X^2, Y^2) = \frac{-1/15}{\sqrt{(7/180)(28/45)}} = \frac{-1/15}{14/90} = -\frac{3}{7} \approx -0.43.$$

This matches the simulated value -0.42 in the figure: the residual of the reverse regression is uncorrelated with Y but its spread shrinks when $|Y|$ is large. ■

Exercise 12.3 (The DirectLiNGAM recursion)

In a LiNGAM $X = BX + \epsilon$ with $A = (I - B)^{-1}$, let X_j be exogenous and $r_i = X_i - a_{ij}X_j$ for $i \neq j$. Show that

$$r_i = \sum_{k \neq j} b_{ik} r_k + \epsilon_i \qquad (i \neq j),$$

i.e. $(r_i)_{i \neq j}$ is a LiNGAM with coefficient matrix B with row and column j deleted, the same errors $(\epsilon_i)_{i \neq j}$, and the induced causal order.

Solution [Supplement].

Step 1. $(I - B)A = I$ gives $A = I + BA$, i.e. $a_{ij} = \delta_{ij} + \sum_k b_{ik} a_{kj}$. Row j of B is zero (X_j is exogenous), so $X_j = \epsilon_j$ and $a_{jj} = 1$. Hence for $i \neq j$, $a_{ij} = b_{ij} + \sum_{k \neq j} b_{ik} a_{kj}$.

Step 2. Substitute $X_j = \epsilon_j$ and $X_k = r_k + a_{kj}\epsilon_j$ $(k \neq j)$ into the equation of X_i:

$$X_i = b_{ij}\epsilon_j + \sum_{k \neq j} b_{ik}(r_k + a_{kj}\epsilon_j) + \epsilon_i = \sum_{k \neq j} b_{ik} r_k + a_{ij}\epsilon_j + \epsilon_i.$$

Subtracting $a_{ij} X_j = a_{ij}\epsilon_j$ gives the claim.

Step 3. Deleting a row and the matching column keeps a matrix strictly lower triangular in the induced order, and $(\epsilon_i)_{i \neq j}$ are still mutually independent and non-Gaussian. So the assumptions of LiNGAM hold for the residuals, with the same direct effects b_{ik} among the remaining variables, and the characterization of exogenous variables can be applied again. ■

Chapter 13. Identifiable Gaussian Linear SEMs

Graphical models, Lecture 6 (equal error variances, conditional-variance ordering, ANM, the US algorithm)

1 In General Gaussian SEMs, Directions Are Not Identified

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

The three structures can produce the same conditional independences and the same Gaussian distribution (Ch. 9).

General Gaussian SEM: $\epsilon_j \sim N(0, \sigma_j^2)$	Equal-variance Gaussian SEM: $\epsilon_j \sim N(0, \sigma^2)$
if each variable may have its own error variance, several directions are compatible with the same distribution result: CPDAG	if all structural errors are restricted to the same variance, the residual-variance structure differs by direction result: DAG

LiNGAM broke the tie with non-Gaussianity (Ch. 12). This chapter shows that, even for Gaussian data, an assumption on the *error variances* can identify the full DAG.

2 The Equal-Variance Gaussian SEM

Definition: equal-variance Gaussian SEM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad \epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

$$X = BX + \epsilon, \qquad \Sigma = \sigma^2 (I - B)^{-1} (I - B)^{-\top}.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering, B is strictly lower triangular
independent errors	no hidden common cause among the analyzed variables
common error variance	all structural errors have the same σ^2
nonzero edges	coefficients of the edges in the graph are not 0

Under these assumptions B and the DAG are identified from the observational distribution.

3 Two Variables: The Equal-Variance Assumption Picks the Direction

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

so the two structural errors have $\mathrm{Var}(\epsilon_X) = \mathrm{Var}(\epsilon_Y) = \sigma^2$.

Reverse direction $Y \to X$: rewrite the same distribution as $Y = \epsilon'_Y$, $X = cY + \eta$. Then

$$\mathrm{Var}(\epsilon'_Y) = (1 + b^2)\sigma^2, \qquad \mathrm{Var}(\eta) = \frac{\sigma^2}{1 + b^2}.$$

Proof [Supplement]. $\mathrm{Var}Y = b^2\sigma^2 + \sigma^2$. The reverse regression has $c = \mathrm{Cov}(X, Y)/\mathrm{Var}Y = b/(1 + b^2)$ and residual variance $\mathrm{Var}(X \mid Y) = \sigma^2 - \dfrac{b^2\sigma^4}{(1 + b^2)\sigma^2} = \dfrac{\sigma^2}{1 + b^2}$. ■

The two reverse error variances are equal only if $(1 + b^2)^2 = 1$, i.e. $b = 0$. **Both directions explain the Gaussian distribution, but only the true direction satisfies the common error variance.**

4 Conditional Variances Reveal the Causal Order

Motivation. If all noise variances are equal, a source variable has the smallest variance, and after conditioning on it the next source has the smallest conditional variance. Repeating this recovers the causal order of a Gaussian SEM.

4.1 The key lemma

Theorem: conditional-variance ordering

In an equal-variance Gaussian SEM, let S be an *ancestral* set (it contains all ancestors of each of its members, e.g. the first few variables of a causal order) and $j \notin S$. Then

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \quad \text{if } \mathrm{pa}(j) \subseteq S, \qquad \mathrm{Var}(X_j \mid X_S) \geq \sigma^2(1 + b_{jk}^2) > \sigma^2 \quad \text{otherwise,}$$

for a suitable parent $k \notin S$ of j.

Proof [Supplement]. Write $X = A\epsilon$ with $A = (I - B)^{-1}$; a_{jl} is the total effect of l on j (sum over directed paths of products of coefficients), $a_{jj} = 1$, and $a_{jl} = 0$ unless l is an ancestor of j or $l = j$. Because S is ancestral, every X_s, $s \in S$, is a function of $(\epsilon_l)_{l \in S}$ only, and conversely $(\epsilon_l)_{l \in S}$ is recovered from X_S (invert the triangular system). Hence, with $X_j = \sum_{l \in S} a_{jl}\epsilon_l + \sum_{l \notin S} a_{jl}\epsilon_l$, the second sum is independent of X_S and the first is determined by it:

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \sum_{l \notin S} a_{jl}^2 = \sigma^2\Big(1 + \sum_{l \notin S,\, l \neq j} a_{jl}^2\Big).$$

If $\mathrm{pa}(j) \subseteq S$, every ancestor of j is in S (ancestral), so the sum is empty. Otherwise let k be the parent of j outside S that is *latest* in a causal order. Any directed path from k to j other than the edge $k \to j$ ends with an edge $m \to j$ where m is a parent of j and a descendant of k; such an m is not in S (S is ancestral and $k \notin S$) and is later than k, contradicting the choice of k. So $a_{jk} = b_{jk} \neq 0$, giving the bound. ■

So, given the already ordered variables, a *source* of the remaining graph has the smallest possible conditional variance σ^2, and every non-source has strictly more. This proves:

Theorem: identifiability (Peters and Bühlmann, 2014)

The DAG and B of an equal-variance Gaussian SEM are identified from Σ. A causal order is obtained top-down: $\pi_1 = \arg\min_j \mathrm{Var}(X_j)$, and $\pi_m = \arg\min_{j \notin S} \mathrm{Var}(X_j \mid X_S)$ with

$S = \{\pi_1, \ldots, \pi_{m-1}\}$.

4.2 A worked example

Example: three variables

Consider the Gaussian SEM

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3, \qquad \epsilon_1, \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0,1).$$

Step	Variance comparison	Choice
1	$\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = 1.64$, $\mathrm{Var}(X_3) = 1.8036$	X_1
2	$\mathrm{Var}(X_2 \mid X_1) = 1$, $\mathrm{Var}(X_3 \mid X_1) = 1.49$	X_2
3	the last remaining variable	X_3

Here $\mathrm{Var}(X_2) = 0.64 + 1$, $\mathrm{Var}(X_3) = 0.49 \times 1.64 + 1$, and $\mathrm{Var}(X_3 \mid X_1) = 0.49\,\mathrm{Var}(X_2 \mid X_1) + 1 = 1.49$. Result: $X_1 < X_2 < X_3$.

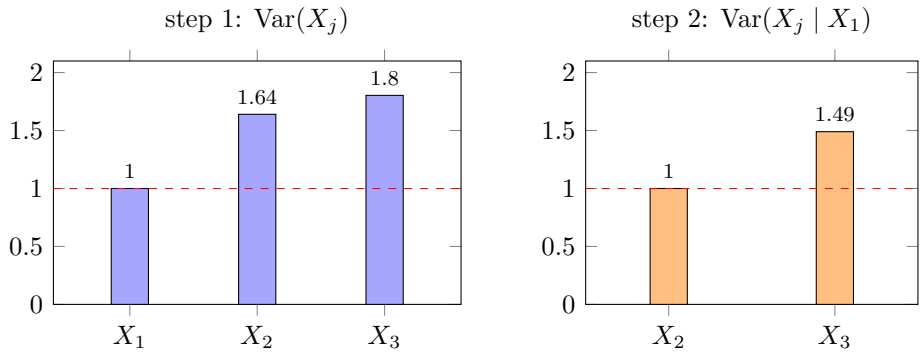


Figure 14: Top-down ordering in the equal-variance example ($\sigma^2 = 1$, red dashed line). At each step the variable whose (conditional) variance equals the common error variance — a source of the remaining graph — is selected: first X_1, then X_2.

4.3 After the order: select parents and edges

Let the estimated order be $X_1 < X_2 < X_3$. Each variable uses only earlier variables as candidate parents:

$$X_2 = \beta_{21}X_1 + \epsilon_2, \qquad X_3 = \beta_{31}X_1 + \beta_{32}X_2 + \epsilon_3.$$

Regression or sparse regression gives $\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{31} \approx 0$, $\hat{\beta}_{32} \approx 0.7$, hence

$$X_1 \to X_2 \to X_3.$$

Conditional variances find the order of the variables; regression coefficients select the direct edges within that order.

5 Equal Variance Does Not Mean Equal Observed Variances

Misconception	Correct meaning
all X_j must have the same variance	variation transmitted from the parents makes observed variances differ (as in the example)
standardizing all variables gives equal variance	standardization rescales the structural errors too, so it does not guarantee the equal-variance assumption — it typically destroys it
similar fitted residual variances prove the assumption	only a diagnostic compatible with the chosen DAG; it does not prove the true direction

When the equal-variance assumption is relatively natural: variables of the same kind and unit, whose measurement errors are also of similar size (e.g. expression levels measured on a common scale).

Remark: scale dependence and varsortability

The assumption is not scale-invariant: multiplying one variable by a constant changes its error variance. In many simulated benchmarks marginal variances simply increase along the causal order ("varsortability"), so methods exploiting variances can look excellent there; on real data with arbitrary units this signal may be absent or misleading. Check the variable units and measurement error, and the plausibility of the assumption, before trusting the order.

6 Additive Noise Models and the US Algorithm

6.1 The equal-variance Gaussian SEM is a special ANM

In an additive noise model each variable is

$$X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j, \qquad \epsilon_1, \dots, \epsilon_p \text{ independent.}$$

With S the set of already selected variables, define the expected conditional variance

$$U_j(S) = \mathbb{E}\{\mathrm{Var}(X_j \mid X_S)\}.$$

If S is ancestral and $j \notin S$, then $\epsilon_j \perp\!\!\!\perp X_S$ and

$$U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}].$$

Conditioning set S	Conditional variance
all parents are in S	$U_j(S) = \sigma_j^2$
some parent not yet in S	uncertainty created by the parents is added

6.2 ANMs are identified by an ordering of conditional variances

Let the true causal order be $\pi = (\pi_1, \dots, \pi_p)$ and $S_{m-1} = \{\pi_1, \dots, \pi_{m-1}\}$. If at each step the true next variable has a smaller conditional variance than the other candidates,

$$U_{\pi_m}(S_{m-1}) < U_k(S_{m-1}), \qquad k \notin S_{m-1} \cup \{\pi_m\},$$

then the variables can be identified one by one.

Equal-variance Gaussian SEM	General ANM
$\sigma_1^2 = \cdots = \sigma_p^2$ a variable whose parents are all included has no extra uncertainty, hence the smallest conditional variance (lemma above)	$\sigma_1^2, \ldots, \sigma_p^2$ may differ identified as long as the order of conditional variances, combining error variance and parent-induced uncertainty, is maintained

The identifiability of ANMs needs neither linearity nor Gaussianity, but the *correct conditional-variance order*. (For general ANMs with nonlinear f_j there are other identifiability results, based on residual independence as in LiNGAM; the conditional-variance route is the one used here.)

6.3 The US algorithm

Algorithm: US — order by conditional variances, then parents

Step 1: estimate the causal order. Let S be the selected set (initially $\varnothing$).

1. Regress each X_j, $j \notin S$, on X_S (flexibly, if f_j may be nonlinear).
2. Compute the residual variance $\widehat{U}_j(S)$.
3. Select the variable with the smallest value, $\hat{\pi}_m = \arg\min_{j \notin S} \widehat{U}_j(S)$.

Add the selected variable to S and repeat.
Step 2: select the parents. The candidate parents of each variable are the variables before it in the order. For each candidate, test conditional dependence given the other preceding variables:

$$X_{\pi_m} \not\perp\!\!\!\perp X_{\pi_j} \mid X_{\{\pi_1,\ldots,\pi_{m-1}\}\setminus\{\pi_j\}}.$$

If conditional dependence remains, add the edge $\pi_j \to \pi_m$.

US finds the order of directions with conditional variances and selects the actual edges with conditional independence. Within a correct order, the preceding variables other than the parents are non-descendants, so the Markov property guarantees that non-parents are conditionally independent given the rest (and faithfulness that parents are not).

7 Summary

Model	Extra assumption	Clue for the direction
LiNGAM	linear relations with independent non-Gaussian errors	independence of the cause and the regression residual
equal-variance Gaussian SEM	common variance of independent Gaussian errors	order of conditional variances starting from the source
ANM and US	an additive noise model that keeps the conditional-variance order	conditional variances find the order; conditional independence selects the parents

ANM and US allow nonlinear functions and different error variances, but still need acyclicity, no hidden confounding and causal minimality.

Key points.

- A general Gaussian linear SEM cannot distinguish Markov-equivalent directions from observational data.

- If all structural errors have the same variance, the full DAG is identified from the observational distribution.
- The source has the smallest variance; repeatedly comparing conditional variances yields the causal order.
- After the order, regression or sparse regression estimates the parents and edge coefficients.
- Check variable scales, measurement error, hidden confounding and violations of the equal-variance assumption.
- The complete DAG is obtained not from Gaussianity alone but from the additional assumption of a common structural error variance.

8 Selected exercises

Motivation. The exercises give a bottom-up version of the variance ordering based on the precision matrix, show concretely how a change of units can reverse the inferred direction, and prove the conditional-variance decomposition behind the US algorithm.

Exercise 13.1 (Bottom-up ordering from the precision matrix)

In an equal-variance Gaussian SEM with nonzero edge coefficients, let $\Omega = \Sigma^{-1}$.
(a) Show that $\Omega_{jj} = \{1 + \sum_{l \in \mathrm{ch}(j)} b_{lj}^2\}/\sigma^2$, where $\mathrm{ch}(j)$ is the set of children of j. Deduce that $\mathrm{Var}(X_j \mid X_{-j}) = \sigma^2$ if j is a sink (has no children) and $< \sigma^2$ otherwise.
(b) Show that after deleting a sink, the remaining variables again follow an equal-variance Gaussian SEM on the induced subgraph, so that repeatedly removing $\arg\min_j \Omega_{jj}$ yields a reversed causal order.

Solution [Supplement].

(a)

Step 1. As in Ch. 9, $\Omega = (I - B)^\top (I - B)/\sigma^2$, so $\Omega_{jj} = \sum_l (I - B)_{lj}^2/\sigma^2$. The term $l = j$ is 1 (no self-loops), and for $l \neq j$, $(I - B)_{lj} = -b_{lj}$, which is nonzero exactly when $j \to l$.

Step 2. By the node-wise theorem of Ch. 8, $\mathrm{Var}(X_j \mid X_{-j}) = 1/\Omega_{jj} = \sigma^2/\{1 + \sum_{l \in \mathrm{ch}(j)} b_{lj}^2\}$, which equals σ^2 iff $\mathrm{ch}(j) = \varnothing$.

(b)

Step 3. A sink j is nobody's parent, so it does not appear on the right side of any other structural equation. The equations of the other variables, with their independent $N(0, \sigma^2)$ errors, form an equal-variance Gaussian SEM on the DAG with j removed, and they determine the joint law of X_{-j}.

Step 4. In the current set, every sink has $\Omega_{jj} = 1/\sigma^2$ and every non-sink has $\Omega_{jj} > 1/\sigma^2$, so the argmin is a sink. Removing it and repeating lists the variables so that each one has no children among the variables still remaining, i.e. in reverse causal order. ■

Exercise 13.2 (A change of units can reverse the direction)

Let $X = \epsilon_X$, $Y = bX + \epsilon_Y$, $\epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2)$, $b \neq 0$, and $\tilde{Y} = sY$ with $s > 0$.
(a) Show that $(X, \tilde{Y})$ follows the Gaussian SEM $X \to \tilde{Y}$ with error variances σ^2 and $s^2\sigma^2$, which is equal-variance only for $s = 1$.
(b) Show that the top-down rule applied to $(X, \tilde{Y})$ selects $\tilde{Y}$ first iff $s^2 < 1/(1 + b^2)$.

(c) Show that for $s = 1/(1+b^2)$ the pair $(X, \tilde{Y})$ is *exactly* an equal-variance Gaussian SEM with $\tilde{Y} \to X$.

Solution [Supplement].

(a) $\tilde{Y} = sbX + s\epsilon_Y$ with $s\epsilon_Y \sim N(0, s^2\sigma^2)$ independent of X.

(b) $\mathrm{Var}X = \sigma^2$ and $\mathrm{Var}\tilde{Y} = s^2(1+b^2)\sigma^2$; the rule picks the smaller variance first, which is $\tilde{Y}$ iff $s^2(1+b^2) < 1$.

(c)

Step 1. By the two-variable computation of this chapter, the reverse factorization of (X, Y) is $Y = \epsilon'_Y$, $X = cY + \eta$ with $\mathrm{Var}\eta = \sigma^2/(1+b^2)$ and η independent of Y. Since $\tilde{Y}$ is a rescaling of Y, $X = (c/s)\tilde{Y} + \eta$ with the same η, independent of $\tilde{Y}$.

Step 2. The errors of the reverse SEM have variances $\mathrm{Var}\tilde{Y} = s^2(1+b^2)\sigma^2$ and $\sigma^2/(1+b^2)$. For $s = 1/(1+b^2)$ both equal $\sigma^2/(1+b^2)$, and $(X, \tilde{Y})$ is Gaussian, so this is an equal-variance Gaussian SEM with $\tilde{Y} \to X$.

The method then outputs the wrong direction with full confidence; nothing in the data reveals the error. Whether the equal-variance assumption holds depends on the chosen units, which is the scale dependence warned about above. ■

Exercise 13.3 (Conditional variances in an additive noise model)

In an ANM $X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j$ with independent errors, $\mathbb{E}\epsilon_j = 0$, $\mathrm{Var}\epsilon_j = \sigma_j^2$ and $\mathbb{E}f_j(X_{\mathrm{pa}(j)})^2 < \infty$, let S be an ancestral set and $j \notin S$.
(a) Show that ϵ_j is independent of $(X_S, X_{\mathrm{pa}(j)})$.
(b) Show that $\mathrm{Var}(X_j \mid X_S) = \sigma_j^2 + \mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}$ almost surely; hence $U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}]$, which equals σ_j^2 when $\mathrm{pa}(j) \subseteq S$.

Solution [Supplement].

(a)

Step 1. By induction along a causal order, each X_k is a measurable function of $(\epsilon_l)_{l \in \mathrm{an}(k)\cup\{k\}}$.

Step 2. For $k \in S$, $\mathrm{an}(k)\cup\{k\} \subseteq S$ because S is ancestral, and $j \notin S$. For $k \in \mathrm{pa}(j)$, $\mathrm{an}(k)\cup\{k\} \subseteq \mathrm{an}(j)$, which does not contain j by acyclicity. So $(X_S, X_{\mathrm{pa}(j)})$ is a function of $(\epsilon_l)_{l \neq j}$, which is independent of ϵ_j.

(b)

Step 3. Write $F = f_j(X_{\mathrm{pa}(j)})$ and $\bar{F} = \mathbb{E}(F \mid X_S)$. By (a), $\mathbb{E}(X_j \mid X_S) = \bar{F} + \mathbb{E}\epsilon_j = \bar{F}$, so

$$\mathrm{Var}(X_j \mid X_S) = \mathbb{E}\{(F-\bar{F})^2 \mid X_S\} + 2\,\mathbb{E}\{(F-\bar{F})\epsilon_j \mid X_S\} + \mathbb{E}(\epsilon_j^2 \mid X_S).$$

Step 4. Since ϵ_j is independent of (F, X_S), $\mathbb{E}\{(F-\bar{F})\epsilon_j \mid X_S\} = \mathbb{E}\epsilon_j \cdot \mathbb{E}(F-\bar{F} \mid X_S) = 0$ and $\mathbb{E}(\epsilon_j^2 \mid X_S) = \sigma_j^2$. The first term is $\mathrm{Var}(F \mid X_S)$. Taking expectations gives $U_j(S)$.

Step 5. If $\mathrm{pa}(j) \subseteq S$, then F is a function of X_S, so $\mathrm{Var}(F \mid X_S) = 0$ and $U_j(S) = \sigma_j^2$. ■